

DELOCALIZATION OF TWO-DIMENSIONAL RANDOM BAND MATRICES

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We study a random band matrix $H = (H_{xy})_{x,y}$ of dimension $N \times N$ with mean-zero complex Gaussian entries, where x, y belong to the discrete torus $(\mathbb{Z}/\sqrt{N}\mathbb{Z})^2$. The variance profile $\mathbb{E}|H_{xy}|^2 = S_{xy}$ vanishes when $|x - y| \geq W$ for some band-width parameter W depending on N .

We show that if the band-width satisfies $W \geq N^c$ for some $c > 0$, then in the large- N limit, we have the following results. The first result is a local semicircle law in the bulk down to scales $N^{-1+\varepsilon}$. The second is delocalization of bulk eigenvectors. The third is a quantum unique ergodicity for bulk eigenvectors. The fourth is universality of local bulk eigenvalue statistics. The fifth is a quantum diffusion profile for the associated T matrix.

Our method is based on embedding H inside a matrix Brownian motion H_t as done in [18, 50] for band matrices on the one-dimensional torus. In this paper, the key additional ingredient in our analysis of H_t is a new CLT-type estimate for polynomials in the entries of the resolvent of H_t .

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1. Introduction. The delocalization-localization transition for random operators has been a central question in probability theory and mathematical physics since the introduction of the Anderson model. Precisely, this model is a discrete random Schrödinger operator on $\mathbb{Z}^d$ of the form $-\Delta + \lambda V$, where Δ is the graph Laplacian on $\mathbb{Z}^d$, where V is a random potential with i.i.d. entries, and where $\lambda \in \mathbb{R}$. There has been a lot of work [7, 8, 13, 14, 29, 30, 33] showing the existence of localized states, but the existence of delocalized states has been addressed in only one work [49] in high dimension, to our knowledge.

A popular model [3, 41] that has received a lot of attention and is conjectured to exhibit similar behavior to the random Schrödinger operator is the d -dimensional *random band matrix*. This is a random Hermitian matrix $H = (H_{xy})_{x,y}$ of dimension $N \times N$ whose entries are independent mean-zero complex random variables (up to the Hermitian constraint). The indices x, y belong to the torus $(\mathbb{Z}/\sqrt[d]{N}\mathbb{Z})^d$, and the variance profile $S_{xy} = \mathbb{E}|H_{xy}|^2$ vanishes if the distance between x, y exceeds a band-width parameter W depending on N .

In the case of dimension $d = 1$, it was conjectured that [9, 10, 28] bulk eigenvector and eigenvalue statistics of H match those of the *Gaussian unitary ensemble* (GUE) if $W \gg N^{1/2}$. This was shown in [50] for Gaussian band matrices and a block variance profile. The work [50] was also extended in [21] to non-Gaussian entries and more general variance profiles. See also [2, 4–6, 18, 22–24, 31, 48], which place stronger assumptions on the band width W . On the other hand, it is conjectured that if $W \ll N^{1/2}$, the eigenvectors of H are localized, which differs from GUE eigenvector statistics. This has been shown for a general class of band matrices in [16], building on previous results [11, 12, 35, 36] under stronger assumptions on W .

The focus of this paper is dimension $d = 2$. Random band matrices are expected to behave similarly to the Anderson model for $\lambda \sim W^{-1}$. Based on this connection and non-rigorous work [1] for the Anderson model, it is conjectured that bulk eigenvectors of H are delocalized (and more generally that bulk eigenvalue and eigenvector statistics of H match those of the GUE) when $W \gg \sqrt{\log N}$. See [3, 42–44] for a further discussion.

The purpose of this paper is to establish a proof of the conjecture in dimension $d = 2$, in the setting of complex Gaussian entries and under the slightly stronger assumption that $W \gg N^\varepsilon$ for some fixed $\varepsilon > 0$. To explain this result, we recall the notion of the localization length of a band matrix ensemble, defined as the minimal scale R such that the overwhelming majority of the mass of a typical bulk eigenvector is concentrated within a ball of radius R . In dimension $d = 1$, [50] establishes delocalization at the conjectured threshold $W \gg \sqrt{N}$, so that the localization length reaches the conjectured scale $R \asymp W^2$. Our result in dimension $d = 2$ implies that for any fixed D , one has $R \geq W^D$ provided ε is taken small enough. In comparison, all previous results established only $R \geq W^{1+c}$ for some $c < 1$ in dimension $d = 2$ (see, e.g., [22, 23, 27]). The precise model and the main results are presented in the next section.

Let us now briefly discuss some of the technical contributions of this paper. The analysis of band matrices in many of the aforementioned works is built on the following T -matrix associated to the band matrix H :

$$T_{ab} = \sum_x S_{ax} |G_{xb}|^2, \quad \text{where } S_{ax} := \mathbb{E} |H_{ax}|^2.$$

Above, $G := (H - z)^{-1}$ denotes the resolvent of H at a spectral parameter z in the upper-half plane. In [18], the analysis of the T -matrix was carried out by embedding H inside a matrix Brownian motion and taking a time-dependent spectral parameter. This analysis was extended substantially in [50], which recognized the importance of a more general hierarchy of dynamical “loop” objects. These loops are essentially polynomials in the (now time-dependent) resolvent G_t ; see (31) for the exact definition.

Key to the analysis of the loop hierarchy in [50] for dimension $d = 1$ is controlling quantities such as

$$\sum_{\mathbf{b}} \mathcal{U}_{\mathbf{ab}} \cdot \mathcal{P}(G_t)_{\mathbf{b}}.$$

Above, the indices are $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{b} = (b_1, \dots, b_n)$ with a_i, b_i in the discrete torus. Also, $\mathcal{U}$ is a deterministic tensor with some smoothness properties in $\mathbf{a}, \mathbf{b}$, and $\mathcal{P}(G_t)_{\mathbf{b}}$ is a polynomial in the entries of G_t with indices close to those in $\mathbf{b}$. In [50], the control of such sums is done using the so-called “sum-zero property”, which ultimately lets one replace $\mathcal{U}$ with its discrete gradient and take advantage of its regularity.

However, in $d = 2$, the sum-zero property alone is insufficient to control these sums, because the number of terms in said sums is substantially larger than in $d = 1$. This is one of the key challenges in $d = 2$.

One of the novel contributions of this paper, which helps address this difficulty, is a set of new CLT-type estimates for $\mathcal{P}(G_t)$. (That is, we control $\mathcal{P}(G_t)_{\mathbf{b}}$ in addition to using regularity of $\mathcal{U}$.) The proof of these estimates is based on carefully analyzing correlations between entries of G_t in terms of the band width of the matrix H . This yields a new tool for the analysis of random band matrices, and the estimates themselves are perhaps of independent interest as well.

Let us illustrate the CLT-type estimates for the following simple example of $\mathcal{P}(G_t)$, in which $\ell \gg 1$ and x is fixed:

$$\sum_{|y| \leq \ell} (G_t)_{xy}.$$

For $d = 2$, the triangle inequality shows that this sum scales as $O(\ell^2)$ with respect to ℓ . In order to improve substantially on this, we will show cancellation between different G_t entries in the above sum and obtain an estimate of $O(\ell \cdot \ell_t)$ in dimension $d = 2$. Here, $\ell_t \ll \ell$ is a dynamical length-scale that captures the correlation length of G_t entries.

Roughly speaking, we will prove in $d = 2$ that there is a scale ℓ_t such that

$$\|x - y\| \gg \ell_t \implies G_{t,xy} \approx 0, \quad x, y \in \mathbb{Z}_{WL}^2.$$

We then show the ℓ_t -independence of G_t , i.e.,

$$\text{dist}(\{x, y\}, \{x', y'\}) \gg \ell_t \implies G_{t,xy} \text{ and } G_{t,x'y'} \text{ are almost independent.}$$

This independence will be used in establishing the fluctuation cancellation mentioned above. This effect will be a key step in proving the quantum diffusion. Without this cancellation, errors would have accumulated along the time evolution. This phenomenon arises only for $d \geq 2$ and it is a key ingredient in the proof of delocalization of band matrices in $d \geq 3$ [17].

We conclude this introduction with a discussion of related results. The proof of the full delocalized phase, quantum diffusion, and universality of local eigenvalue statistics for dimension $d \geq 7$ was shown in [45–47], which uses an expansion method for the resolvent of H . In dimension $d = 3$, for special variance profiles of H , supersymmetric methods have been used [15] to estimate the density of states for H . Similar methods have also been able to detect the transition between $W \gg N^{1/2}$ and $W \ll N^{1/2}$ at the level of two-point functions in $d = 1$ [38]. See [2, 19, 37, 39–41] for further discussion on the supersymmetric method.

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2. The model and main results.

2.1. Band matrix model. We will study the following two-dimensional band block matrix model. Fix positive integers $W > 0$ and $L \geq 3$, and set $N := W^2 L^2$. Let $\mathbb{Z}_L := \mathbb{Z}/L\mathbb{Z} \simeq \{1, \dots, L\}$.

The matrix of interest is $H = (H_{xy})_{x,y \in \mathbb{Z}_{WL}^2}$, where $H_{xy} = \overline{H}_{yx}$, and up to this Hermitian constraint, the entries $H_{xy} \sim \mathcal{CN}(0, S_{xy})$ are independent centered complex Gaussian random variables of variance $\mathbb{E}|H_{xy}|^2 = S_{xy}$, where $S = (S_{xy})_{x,y \in \mathbb{Z}_{WL}^2}$ is a matrix with the following block structure. We define

$$S_{ab}^{(B)} = \frac{1}{5} \mathbf{1}(|a - b|_L \leq 1), \quad (S_W)_{ij} = W^{-2}, \quad a, b \in \mathbb{Z}_L^2, \quad i, j \in \{1, \dots, W\}^2.$$

Note that $S^{(B)}$ is $L^2 \times L^2$, and that S_W is $W^2 \times W^2$. Above, $|a - b|_L$ is the periodic L^1 distance on $\mathbb{Z}_L^2$. The standing assumption $L \geq 3$ costs nothing, since all of our results are asymptotic as $N \rightarrow \infty$ and hence as $L \rightarrow \infty$; we record it because the displayed formula for $S^{(B)}$ does require it. For $L \leq 2$ the five residues $0, \pm e_1, \pm e_2$ of $\mathbb{Z}_L^2$ are not distinct, so the row sums of $S^{(B)}$ are $1/5$ when $L = 1$ and $3/5$ when $L = 2$, and $S^{(B)}$ is not stochastic. We now define S to be the following real-symmetric, stochastic matrix:

$$S = S^{(B)} \otimes S_W.$$

Note that the entries of H and S are indexed by $\mathbb{Z}_{WL}^2$.

We now introduce standard notation for the setting of this paper. We start by letting $\lambda_1 \leq \dots \leq \lambda_N$ be eigenvalues of H with corresponding unit eigenvectors $\psi_1, \dots, \psi_N$, so that

$$H\psi_k = \lambda_k\psi_k.$$

Our analysis of these quantities is based on the resolvent $G(z) := (H - z)^{-1}$ of H defined for any z with $\text{Im}z > 0$.

It is well known that the empirical spectral measure of H converges almost surely to the Wigner semicircle distribution as $N \rightarrow \infty$ if $W \rightarrow \infty$. The Stieltjes transform of the semicircle distribution is given by

$$m(z) := \int_{-2}^2 \frac{1}{2\pi} \frac{\sqrt{(4-x^2)_+}}{x-z} dx = \frac{-z + \sqrt{z^2 - 4}}{2}.$$

Finally, we will use a standard notion of stochastic domination as used in [20], defined below.

DEFINITION 2.1. (i) Consider the following two families of non-negative random variables parameterized by a set $U^{(N)}$:

$$\xi = \left(\xi^{(N)}(u) : N \in \mathbb{N}, u \in U^{(N)} \right), \quad \zeta = \left(\zeta^{(N)}(u) : N \in \mathbb{N}, u \in U^{(N)} \right).$$

We say that ξ is stochastically dominated by ζ (uniformly in u) if for any fixed (small) $\tau > 0$ and (large) $D > 0$, we have the following for $N \geq N_0(\tau, D)$:

$$\mathbb{P}\left(\bigcup_{u \in U^{(N)}} \left\{ \xi^{(N)}(u) > N^\tau \zeta^{(N)}(u) \right\}\right) \leq N^{-D}$$

We will use the notation $\xi \prec \zeta$ or $\xi = O_{\prec}(\zeta)$ to mean that ξ is stochastically dominated by ζ . If ξ is a family of complex numbers, then $\xi \prec \zeta$ and $\xi = O_{\prec}(\zeta)$ mean that $|\xi| \prec \zeta$.

(ii) As a convention, for deterministic $\xi, \zeta \geq 0$, we write $\xi \prec \zeta$ if and only if $\xi \leq N^\tau \zeta$ for any fixed $\tau > 0$.

(iii) Let A be a family of random matrices and ζ be a family of non-negative random variables. In this case, we use $A = O_{\prec}(\zeta)$ to mean that $\|A\| \prec \zeta$, where $\|\cdot\|$ is the operator norm.

(iv) We say that an event Ξ holds with high probability (w.h.p.) if for any $D > 0$, we have $\mathbb{P}(\Xi) \geq 1 - N^{-D}$ for large enough N . We say that an event Ω holds *w.h.p.* in Ξ if for any $D > 0$, we have $\mathbb{P}(\Xi \setminus \Omega) \leq N^{-D}$ for large enough N .

2.2. Main results. Our first result states a strong delocalization estimate for bulk eigenvectors of H .

THEOREM 2.2 (Delocalization). *Suppose that the following is satisfied for some fixed constant $\mathfrak{c} > 0$:*

$$(1) \quad W \geq N^{\mathfrak{c}}.$$

Then for any $\kappa, \tau, D > 0$, there exists N_0 such that for all $N \geq N_0$ we have

$$\mathbb{P}\left(\max_k \|\psi_k\|_{\infty}^2 \cdot \mathbf{1}(\lambda_k \in [-2 + \kappa, 2 - \kappa]) \leq N^{-1+\tau}\right) \geq 1 - N^{-D}$$

The proof of Theorem 2.2 is based on estimates for the Green's function $G(z)$, which we state below. First, however, we need some more notation. We start with a “diffusive” length-scale with respect to a parameter z in the upper half plane:

$$(2) \quad \ell := \ell(z) := \min\left(\eta^{-1/2}, L\right) + 1, \quad \eta := \text{Im}(z).$$

We will also introduce the fundamental control parameter to be

$$(3) \quad M_{\eta} := W^2 \ell(z)^2 \eta, \quad \eta = \text{Im}(z).$$

Next, we introduce the following index set given any $a = (a(1), a(2)) \in \mathbb{Z}_L^2$:

$$\mathcal{I}_a^{(2)} := \mathcal{I}_{a(1)} \times \mathcal{I}_{a(2)}, \quad \mathcal{I}_{a(i)} := [(a(i) - 1)W + 1, a(i)W].$$

THEOREM 2.3 (Local semicircle law). *Suppose that the assumption of Theorem 2.2 holds. For any fixed $\kappa, \tau, D > 0$, and for any $z \in \mathbb{C}$ of the form*

$$z = E + i\eta, \quad |E| \leq 2 - \kappa, \quad 1 \geq \eta \geq N^{-1+\tau},$$

there exists N_0 such that for all $N \geq N_0$ we have the following estimates. First, we have the local law:

$$(4) \quad \mathbb{P}\left(\max_{x,y} \left| (G(z) - m(z))_{xy} \right| \leq \frac{W^{\tau}}{\sqrt{M_{\eta}}} \right) \geq 1 - N^{-D}, \quad \ell = \ell(z).$$

Second, we have the partial tracial local law

$$(5) \quad \mathbb{P} \left(\max_a \left| W^{-2} \sum_{x \in \mathcal{I}_a^{(2)}} G_{xx}(z) - m(z) \right| \leq \frac{W^\tau}{M_\eta} \right) \geq 1 - N^{-D}, \quad \ell = \ell(z).$$

We now state our next two results. The first of these is quantum unique ergodicity, and the second is a quantum diffusion profile for the resolvent.

THEOREM 2.4 (Generalized quantum unique ergodicity). *Suppose the assumptions of Theorem 2.2 hold and*

$$(6) \quad 0 < \tau < \frac{\mathfrak{c}}{2},$$

where $\mathfrak{c}$ was defined in (1). Next, let E_a denote the following block identity matrix:

$$(7) \quad (E_a)_{xy} = \delta_{xy} \cdot W^{-2} \cdot \mathbf{1}(x \in \mathcal{I}_a^{(2)}), \quad x, y \in \mathbb{Z}_{WL}^2.$$

Then for any $\kappa > 0$, there exists N_0 such that for all $N \geq N_0$, we have

$$(8) \quad \max_{E: |E| < 2-\kappa} \max_{a \in \mathbb{Z}_L^2} \mathbb{P} \left(\max_{k, k' \in \mathcal{J}_E} |N(\psi_k^*(E_a - N^{-1})\psi_{k'})|^2 \geq N^{-\tau/6} \right) \leq N^{-\tau/6}.$$

Here

$$\mathcal{J}_E := \left\{ k : |\lambda_k - E| \leq N^{-1-\tau} W^{2/3} \right\}.$$

Moreover, for any subset $A \subset \mathbb{Z}_L^2$,

$$(9) \quad \max_{E: |E| < 2-\kappa} \max_{A \subset \mathbb{Z}_L^2} \mathbb{P} \left(\max_{k \in \mathcal{J}_E} \left| \sum_{a \in A} \sum_{x \in \mathcal{I}_a^{(2)}} |\psi_k(x)|^2 - \frac{|A| \cdot W^2}{N} \right| > \frac{|A| \cdot W^2}{N^{1+\tau/6}} \right) \leq N^{-\tau/6}$$

THEOREM 2.5 (Quantum diffusion). *Under the assumptions of Theorem 2.3, we have (for $m := m(z)$)*

$$(10) \quad \mathbb{P} \left(\max_{a,b} \left| \text{Tr} G E_a G^\dagger E_b - W^{-2} \left(\frac{|m|^2}{1 - |m|^2 S^{(B)}} \right)_{ab} \right| \leq \frac{W^\tau}{M_\eta^2} \right) \geq 1 - N^{-D},$$

$$(11) \quad \mathbb{P} \left(\max_{a,b} \left| \text{Tr} G E_a G E_b - W^{-2} \left(\frac{m^2}{1 - m^2 S^{(B)}} \right)_{ab} \right| \leq \frac{W^\tau}{M_\eta^2} \right) \geq 1 - N^{-D}.$$

We also have the following improved estimate at the level of expectation:

$$(12) \quad \max_{a,b} \left| \mathbb{E} \text{Tr} G E_a G^\dagger E_b - W^{-2} \left(\frac{|m|^2}{1 - |m|^2 S^{(B)}} \right)_{ab} \right| \leq M_\eta^{-3} \cdot W^\tau,$$

$$(13) \quad \max_{a,b} \left| \mathbb{E} \text{Tr} G E_a G E_b - W^{-2} \left(\frac{m^2}{1 - m^2 S^{(B)}} \right)_{ab} \right| \leq M_\eta^{-3} \cdot W^\tau.$$

Theorems 2.2 and 2.4 are simple consequences of Theorems 2.3 and 2.5. The proof of Theorem 2.2 is identical to that in [50], so we omit it. The proof of Theorem 2.4 is very similar to the argument in [50], but since there are slight dependencies on the dimension, we give the details below.

PROOF OF THEOREM 2.4. We prove (8); the proof of (9) is the same after replacing E_a by $|A|^{-1} \sum_{a \in A} E_a$ in the argument below and taking a slightly larger η , as explained at the end of the proof. Choose $z = E + \eta i$ and

$$\eta = N^{-1-\tau} W^{2/3}.$$

We first have

$$\begin{aligned} (14) \quad & \mathbb{E} \left(\sum_{k,k'} \mathbf{1}(\lambda_k, \lambda_{k'} \in \mathcal{J}_E) \cdot |N(\psi_k^*(E_a - N^{-1}) \psi_{k'})|^2 \right) \\ & \leq C \eta^4 \mathbb{E} \left(\sum_{k,k'} \frac{|N(\psi_k^*(E_a - N^{-1}) \psi_{k'})|^2}{|\lambda_k - z|^2 |\lambda_{k'} - z|^2} \right) \\ & \leq C N^2 \eta^2 \mathbb{E} \operatorname{Tr} (\operatorname{Im} G(z) (E_a - N^{-1}) \operatorname{Im} G(z) (E_a - N^{-1})) \end{aligned}$$

For the expectation of the last term above, we have

$$\begin{aligned} (15) \quad & \mathbb{E} \operatorname{Tr} (\operatorname{Im} G(z) (E_a - N^{-1}) \operatorname{Im} G(z) (E_a - N^{-1})) \\ & = \frac{1}{L^4} \sum_{b,b'} \mathbb{E} \operatorname{Tr} (\operatorname{Im} G(z) (E_a - E_b) \operatorname{Im} G(z) (E_a - E_{b'})) \\ & \leq C \max_{a,b,a',b'} \left| \mathbb{E} \operatorname{Tr} (\operatorname{Im} G(z) E_a \operatorname{Im} G(z) E_b) - \mathbb{E} \operatorname{Tr} (\operatorname{Im} G(z) E_{a'} \operatorname{Im} G(z) E_{b'}) \right| \end{aligned}$$

Recall that $N = W^2 L^2$. By (6), we have $\eta^{-1/2} \geq L$, so that $\ell(z) = L + 1$ by (2). We can bound the last line using (12)-(13) in Theorem 2.5. Recall that $M_\eta := W^2 \ell^2 \eta$, and note that $W^2 \ell^2 \geq W^2 L^2 = N$. We ultimately get

$$\begin{aligned} & \eta^2 \mathbb{E} \operatorname{Tr} (\operatorname{Im} G(z) (E_a - N^{-1}) \operatorname{Im} G(z) (E_a - N^{-1})) \\ & \leq \eta^2 M_\eta^{-3} \cdot W^\delta + \eta^2 W^{-2} \max_{a,b,a',b'} \max_{\xi=|m|^2 \text{ or } \xi=m^2} \left| \left(\frac{1}{1 - \xi \cdot S^{(B)}} \right)_{ab} - \left(\frac{1}{1 - \xi \cdot S^{(B)}} \right)_{a'b'} \right| \end{aligned}$$

for any $\delta > 0$ small. Next, fix any $a, b, a', b' \in \mathbb{Z}_L^2$, and for $\xi = |m(z)|^2, m(z)^2$ write

$$\begin{aligned} & \left| (1 - \xi \cdot S^{(B)})_{ab}^{-1} - (1 - \xi \cdot S^{(B)})_{a'b'}^{-1} \right| = \left| (1 - \xi \cdot S^{(B)})_{0(a-b)}^{-1} - (1 - \xi \cdot S^{(B)})_{0(a'-b')}^{-1} \right| \\ & \leq \left| (1 - \xi \cdot S^{(B)})_{00}^{-1} - (1 - \xi \cdot S^{(B)})_{0(a-b)}^{-1} \right| \\ & \quad + \left| (1 - \xi \cdot S^{(B)})_{00}^{-1} - (1 - \xi \cdot S^{(B)})_{0(a'-b')}^{-1} \right|. \end{aligned}$$

First let $\xi = |m|^2$, so that $|1 - \xi| \sim \eta$ and hence $\hat{\ell}(\xi) \sim L$, since $\eta^{-1/2} \geq L$. Choose a path $0 = c_0 \mapsto c_1 \mapsto \dots \mapsto a - b$ in $\mathbb{Z}_L^2$ of length $O(L)$ such that for each step in this path, we have $1/(|c_k|_L + 1) \leq C/(k+1)$ for $C = O(1)$ and all k . By the first-derivative bound in Lemma 2.14, we have

$$\left| (1 - \xi \cdot S^{(B)})_{00}^{-1} - (1 - \xi \cdot S^{(B)})_{0(a-b)}^{-1} \right| \prec \sum_{k=1}^{O(L)} \left(\frac{C}{k+1} + \eta^{-\frac{1}{2}} L^{-2} \right) \prec C' + \eta^{-\frac{1}{2}} L^{-1},$$

with $C' = O(1)$. Next let $\xi = m^2$. Since $|E| < 2 - \kappa$, we have $|1 - \xi| \geq c$ for some $c > 0$, so that $\hat{\ell}(\xi) = O(1)$ and $|1 - \xi| \cdot [\hat{\ell}(\xi)]^2 \geq c'$. (For this choice of ξ the last term in (43) is of order 1, so summing (43) along the path would only give $O(L)$.) Instead, the exponential

decay bound (42) shows that every entry of $(1 - \xi \cdot S^{(B)})^{-1}$ is $O_{\prec}(1)$, so the left-hand side of the previous display is $O_{\prec}(1)$ for $\xi = m^2$ as well. (Note that $|1 - \xi| \geq c\eta$ for some $c > 0$ and either choice of $\xi = |m|^2, m^2$, so the bounds in Lemma 2.14 can be applied.) The same holds if we replace $a - b$ by $a' - b'$. Thus, we have

$$\max_{a,b,a',b'} \max_{\xi=|m|^2 \text{ or } \xi=m^2} \left| \left(\frac{1}{1 - \xi \cdot S^{(B)}} \right)_{ab} - \left(\frac{1}{1 - \xi \cdot S^{(B)}} \right)_{a'b'} \right| \prec 1 + \eta^{-\frac{1}{2}} L^{-1}.$$

Recalling that $\eta^{-1/2} \geq L$, the dominant term on the right-hand side is $O(\eta^{-1/2} L^{-1})$. We deduce that

$$\begin{aligned} & \eta^2 \mathbb{E} \operatorname{Tr} \left(\operatorname{Im} G(z) (E_a - N^{-1}) \operatorname{Im} G(z) (E_a - N^{-1}) \right) \\ & \leq W^\delta W^{-6} L^{-6} \eta^{-1} + \eta^{\frac{3}{2}} W^{-2+\delta} L^{-1} \\ & \leq N^{-2} [W^\delta N^2 W^{-6} L^{-6} \eta^{-1} + N^2 W^{-2+\delta} \eta^{\frac{3}{2}} L^{-1}] \leq W^\delta N^{-2} [N^{-\tau/3-\delta} + N^{-3\tau/2}] < N^{-2} N^{-\tau/3}. \end{aligned}$$

Above, we used $N = W^2 L^2$ and $\eta = N^{-1-\tau} W^{2/3}$ and $W \geq N^c \geq N^{2\tau+\delta}$ to deduce the last two inequalities. Combining the above display with (14) and (15) gives

$$(16) \quad \mathbb{E} \left(\sum_{k,k'} \mathbf{1}(\lambda_k, \lambda_{k'} \in \mathcal{J}_E) \cdot |N(\psi_k^*(E_a - N^{-1}) \psi_{k'})|^2 \right) \leq N^{-\tau/3}.$$

The bound (8) now follows by (16) and the Markov inequality.

Finally, we explain the modification needed for (9). Fix $A \subset \mathbb{Z}_L^2$ with $A \neq \emptyset$ (for $A = \emptyset$ the event in (9) is empty), and set $E_A := |A|^{-1} \sum_{a \in A} E_a$. Since

$$\sum_{a \in A} \sum_{x \in \mathcal{I}_a^{(2)}} |\psi_k(x)|^2 - \frac{|A| \cdot W^2}{N} = |A| \cdot W^2 \cdot \psi_k^*(E_A - N^{-1}) \psi_k,$$

the event in (9) is the event that $|N\psi_k^*(E_A - N^{-1})\psi_k|^2 > N^{-\tau/3}$ for some $k \in \mathcal{J}_E$. Since the threshold here is $N^{-\tau/3}$ rather than $N^{-\tau/6}$, by the Markov inequality it suffices to prove (16) with E_a replaced by E_A and with $N^{-\tau/3}$ replaced by $N^{-\tau/2}$. To this end, we repeat the argument above with E_a replaced by E_A (this changes nothing in (14), (15) and the bounds that follow, since $E_A - E_b$ is an average of $E_a - E_b$ over $a \in A$), but now with

$$\eta = N^{-1-\tau/2} W^{2/3}.$$

This η is still at least the width $N^{-1-\tau} W^{2/3}$ of the window defining $\mathcal{J}_E$, so (14) remains valid; we still have $\eta^{-1/2} \geq L$, so $\ell(z) = L + 1$; and $\eta \geq N^{-1+c/3}$, so Theorem 2.5 applies. The last display before (16) then becomes

$$\begin{aligned} & \eta^2 \mathbb{E} \operatorname{Tr} \left(\operatorname{Im} G(z) (E_A - N^{-1}) \operatorname{Im} G(z) (E_A - N^{-1}) \right) \\ & \leq W^\delta W^{-6} L^{-6} \eta^{-1} + \eta^{\frac{3}{2}} W^{-2+\delta} L^{-1} = W^\delta N^{-2} [N^{\tau/2} W^{-2/3} + N^{-3\tau/4}] \leq N^{-2} N^{-2\tau/3}, \end{aligned}$$

where we used $W \geq N^c > N^{2\tau}$, so that $N^{\tau/2} W^{-2/3} \leq N^{-5\tau/6}$, and chose $\delta > 0$ small. Combining this with (14) and (15) gives (16) with E_A in place of E_a and with $N^{-\tau/2}$ in place of $N^{-\tau/3}$, and (9) follows by the Markov inequality, since $N^{\tau/3} \cdot N^{-\tau/2} = N^{-\tau/6}$. $\square$

2.3. Universality. For any $k = 1, \dots, N$, the k -point correlation function of H is the following marginal, where $\rho_H^{(N)}(\alpha_1, \dots, \alpha_N)$ denotes the joint density for the (unordered) eigenvalues of H :

$$\rho_H^{(k)}(\alpha_1, \alpha_2, \dots, \alpha_k) := \int_{\mathbb{R}^{N-k}} \rho_H^{(N)}(\alpha_1, \alpha_2, \dots, \alpha_N) d\alpha_{k+1} \cdots d\alpha_N.$$

THEOREM 2.6 (Bulk universality). *Suppose that the assumptions of Theorem 2.2 are satisfied. Fix any integer $k \geq 1$ that is independent of N . Fix any $|E| \leq 2 - \kappa$ and test function $\mathcal{O} \in \mathcal{C}_c^\infty(\mathbb{R}^k)$. We have*

$$(17) \quad \lim_{N \rightarrow \infty} \int_{\mathbb{R}^k} d\alpha \mathcal{O}(\alpha) \left\{ \left(\rho_H^{(k)} - \rho_{\text{GUE}}^{(k)} \right) \left(E + \frac{\alpha}{N} \right) \right\} = 0.$$

PROOF OF THEOREM 2.6. This argument largely follows the proof of Theorem 2.6 in [50]. Let us explain this below.

The starting point is the following matrix OU process:

$$d\mathbf{H}_t = -\frac{1}{2}\mathbf{H}_t dt + \frac{1}{\sqrt{N}} dB_t.$$

Above, the initial data is given by the band matrix in question, i.e. $\mathbf{H}_0 = H$; we use bold notation in order to not conflict with a forthcoming time-dependent random band matrix. Also, B_t is a standard Brownian motion in the space of complex Hermitian matrices.

Note that $\mathbf{H}_\infty := \lim_{t \rightarrow \infty} \mathbf{H}_t$ has the law of a GUE random matrix. In particular, to complete the proof, it suffices to show the following two limits, in which $t_* = N^{-1+\tau_U}$ with a fixed (small) $\tau_U > 0$:

$$(18) \quad \lim_{N \rightarrow \infty} \int_{\mathbb{R}^k} d\alpha \mathcal{O}(\alpha) \left\{ \left(\rho_{\mathbf{H}_{t_*}}^{(k)} - \rho_{\mathbf{H}_\infty}^{(k)} \right) \left(E + \frac{\alpha}{N} \right) \right\} = 0$$

$$(19) \quad \lim_{N \rightarrow \infty} \int_{\mathbb{R}^k} d\alpha \mathcal{O}(\alpha) \left\{ \left(\rho_{\mathbf{H}_{t_*}}^{(k)} - \rho_{\mathbf{H}_0}^{(k)} \right) \left(E + \frac{\alpha}{N} \right) \right\} = 0.$$

The first limit (18) follows by Theorem 2.3 (the local law for H) and Theorem 2.2 of [32] (in the form stated in arXiv:1609.09011v4, where the Remark following Theorem 2.2 covers the complex Hermitian case), together with the following standard steps. Theorem 2.2 of [32] is formulated for energies near the origin, so we apply it conditionally on H , with initial data given by the eigenvalues of $e^{-t_*}/2 H - E$ and with time $1 - e^{-t_*} \sim t_*$; by Theorem 2.3, this initial data is regular in the sense of Definition 2.1 of [32] (with $g = N^{-1+\tau_U}/2$ and $G = N^{-\tau_U}/8$, say) with probability $1 - O(N^{-D})$. This compares the correlation functions of $\mathbf{H}_{t_*}$ near E , rescaled by the density of the free convolution at E , with those of the GUE near 0, rescaled by $\rho_{sc}(0)$. To reach (18), in which both sides are evaluated near E , we also need to move the GUE from energy 0 to energy E . Since the GUE is invariant under the matrix OU process, $\mathbf{H}_\infty$ has the same law as $\mathbf{H}_{t_*}$ with $\mathbf{H}_0$ replaced by an independent GUE matrix; the same argument, now using the local law for the GUE (see e.g. [26]) to verify the regularity assumption, compares the rescaled correlation functions of $\mathbf{H}_\infty$ near E with those of the GUE near 0 as well. Finally, by these local laws, both free convolution densities at E are $\rho_{sc}(E) + O(N^{-c})$ for some $c > 0$ with probability $1 - O(N^{-D})$, and replacing them by $\rho_{sc}(E)$ changes the rescaled correlation functions only by $o(1)$, since $\mathcal{O}$ is Lipschitz and since, by Theorem 2.2 of [32] applied to a nonnegative test function, the integrals $\int_{[-C,C]^k} \rho^{(k)}(E + \alpha/N) d\alpha$ with $\rho^{(k)} = \rho_{\mathbf{H}_{t_*}}^{(k)}$ and $\rho^{(k)} = \rho_{\mathbf{H}_\infty}^{(k)}$ are $O(1)$ for any fixed $C > 0$. This gives (18). We spend the rest of this argument showing (19). Standard calculations for comparison of correlation functions (see Theorem 15.3 in [25], Proposition 4.17 in [45], and (2.23) in [50]) reduce the proof of (19) to showing the following claim.

Claim: Fix any positive integer n . Consider $z_j = E_j + i\eta_j$ for $j = 1, \dots, n$, where $|E_j - E| = O(N^{-1})$ and $N^{-1-\tau_U} \leq \eta_j \leq N^{-1+\tau_U}$ are deterministic. Under the assumptions of Theorem 2.6, there exist $c', C_n > 0$ such that

$$(20) \quad \sup_{0 \leq t \leq t_*} \left| \mathbb{E} \prod_{i=1}^n \text{Im } m_t(z_i) - \mathbb{E} \prod_{i=1}^n \text{Im } m_{t_*}(z_i) \right| \leq N^{-c' + C_n \tau_U}$$

where $m_t(z_i) = N^{-1} \text{Tr}(\mathbf{H}_t - z_i)^{-1}$ for any $t \geq 0$.

We now also claim the following resolvent estimates hold, in which $\mathbf{R}_t(z) := (\mathbf{H}_t - z)^{-1}$. First, we claim that for $\tau_U > 0$ sufficiently small, Theorem 2.4 holds for $\mathbf{H}_t$ with $\tau = \mathfrak{c}/3$ therein and $t \in [0, N^{-1+\tau_U}]$. We also claim that for $t \in [0, N^{-1+\tau_U}]$ and for $z = E + i\eta$ with $|E| < 2 - \kappa$ and $\eta = N^{-1+2\tau_U}$, we have $\max_x |(\mathbf{R}_t(z))_{xx}| \prec 1$. The proof of these two claims follows by treating $\mathbf{H}_t$ as a perturbation of $\mathbf{H}_0 = H$; the details are identical to the argument given in Section 7.2 of [50], so we omit them here. (While we will not explicitly cite these claims in the argument below, we clarify that they are important in following the arguments in Section 2.3 of [50].)

We now argue as in Step 3 in the proof of Theorem 2.6 in [50] to obtain the following estimate (which appears as (2.28) in [50]):

$$(21) \quad \sup_{0 \leq t \leq t_*} \left| \mathbb{E} \prod_{i=1}^n \text{Im } m_t(z_i) - \mathbb{E} \prod_{i=1}^n \text{Im } m_{t_*}(z_i) \right| \prec N^{-1+C_n \tau_U} \cdot \max_{0 \leq t \leq t_*} \left(\max_u \mathbb{E} L_{1,t}(z_u) + \max_{u \neq v} \mathbb{E} L_{2,t}(z_u, z_v) \right) + N^{-D}.$$

We have used the notation below, where $S_{xy}^\circ := S_{xy} - N^{-1}$ and $\mathbf{R}_t(z) = (\mathbf{H}_t - z)^{-1}$:

$$L_{1,t}(z) := \sum_{\mathbf{G}_1, \mathbf{G}_2 \in \{\mathbf{R}_t, \mathbf{R}_t^*\}} \left| \frac{1}{N} \sum_{a,b} (\mathbf{G}_1^2(z))_{aa} S_{ab}^\circ (\mathbf{G}_2(z))_{bb} \right|,$$

$$L_{2,t}(z_1, z_2) := \sum_{\mathbf{G}_1, \mathbf{G}_2 \in \{\mathbf{R}_t, \mathbf{R}_t^*\}} \left| \frac{1}{N^2} \sum_{a,b} (\mathbf{G}_1^2(z_1))_{ab} S_{ab}^\circ (\mathbf{G}_2^2(z_2))_{ba} \right|.$$

(For $n = 1$ the maximum over $u \neq v$ is absent. In (21), $D > 0$ is arbitrary. The additive term N^{-D} arises as follows: in the time derivative of $\mathbb{E} \prod_i \text{Im } m_t(z_i)$, the factors $\text{Im } m_t(z_j)$ that are not differentiated multiply the quantities in $L_{1,t}$ and $L_{2,t}$. They are bounded by $N^{C\tau_U}$ outside an event of probability $O(N^{-D})$, by the bound on $\mathbf{R}_t$ stated above and the monotonicity of $\eta \mapsto \eta \text{Im } m_t(E + i\eta)$. On this event all quantities involved are bounded by a fixed power of N .) In particular, in order to prove (20), it suffices to use (21) and show the following two estimates for any $\mathbf{G}_1(z), \mathbf{G}_2(z) \in \{\mathbf{R}_t, \mathbf{R}_t^*\}$, in which $C = O(1)$ and $\delta_0 > 0$ is small, and where $\mathfrak{c} > 0$ is from (1):

$$(22) \quad \max_i \max_y \mathbb{E} \left| \sum_x (\mathbf{G}_1^2(z_i))_{xx} S_{xy}^\circ (\mathbf{G}_2(z_i))_{yy} \right| \leq N^{1-\mathfrak{c}/36+C\tau_U}$$

$$(23) \quad \max_{i \neq j} \max_y \mathbb{E} \left| \sum_x (\mathbf{G}_1^2(z_i))_{xy} S_{xy}^\circ (\mathbf{G}_2^2(z_j))_{yx} \right| \prec N^{2-\mathfrak{c}/36+C\tau_U}.$$

The rest of this proof now proceeds with showing (22) and (23).

Proof of (22). We cite Lemma 4.20 in [45] (see also (2.31) in [50]); by the eigen-decomposition for $\mathbf{R}_t$, we have the following:

$$(24) \quad \left| \sum_x (\mathbf{G}_1^2(z))_{xx} S_{xy}^\circ (\mathbf{G}_2(z))_{yy} \right| \prec N^{-2} \sum_{\alpha, \beta} |p_\alpha(z)|^2 \cdot |p_\beta(z)| \cdot |M_{y,\alpha}|.$$

Above, $p_\alpha(z) := (\lambda_\alpha - z)^{-1}$, and for any y, α , we have the following (in which a_0 is such that $y \in \mathcal{I}_{a_0}^{(2)}$):

$$|M_{y,\alpha}| := \left| N \sum_x |\psi_\alpha(x)|^2 S_{xy}^\circ \right| = O(N) \left| \sum_{a: |a-a_0|_L \leq 1} \langle \psi_\alpha^* (E_a - N^{-1}I) \psi_\alpha \rangle \right|.$$

At this point, we can directly follow the proof of (2.29) in [50]. Indeed, (24) expresses the left-hand side of (22) in terms of eigenvalues and eigenvectors, which we control using Theorems 2.2, 2.3, and 2.4. So, instead of providing the details verbatim, let us briefly describe this argument.

Recall the fixed energy E in the statement of Theorem 2.6. Let $\mathcal{B}$ be the bad event where there exists α with $|\lambda_\alpha - E| \leq N^{-1+\epsilon/6}$ and $|M_{y,\alpha}| \geq N^{-\epsilon/36}$. The probability of this bad event is $O(N^{-\epsilon/18})$ by (8) for $\tau = \epsilon/3$. Indeed, $N^{-1+\epsilon/6} \leq N^{-1-\epsilon/3} W^{2/3}$ by (1), so such α belong to $\mathcal{J}_E$; moreover $|M_{y,\alpha}| \leq \max_{a: |a-a_0|_L \leq 1} |N \psi_\alpha^* (E_a - N^{-1}I) \psi_\alpha|$, and (8) with $\tau = \epsilon/3$ bounds, for each of the five blocks a , the probability that the square $|N \psi_\alpha^* (E_a - N^{-1}I) \psi_\alpha|^2$ exceeds $N^{-\epsilon/18}$, i.e. that $|N \psi_\alpha^* (E_a - N^{-1}I) \psi_\alpha|$ exceeds $N^{-\epsilon/36}$, by $N^{-\epsilon/18}$. If we restrict (24) to this bad event $\mathcal{B}$, the expectation of this quantity is $O(N^{1-\epsilon/18+C\delta_0})$, since the local law and delocalization (Theorems 2.2 and 2.3) imply that the right-hand side of (24) is $O_\prec(N^{1+C\delta_0})$; here, $C = O(1)$ and $\delta_0 > 0$ is small. On the good event $\mathcal{B}^c$, we can improve our estimate on the right-hand side of (24) using $|M_{y,\alpha}| < N^{-\epsilon/36}$ for all α with $|\lambda_\alpha - E| \leq N^{-1+\epsilon/6}$ and using the local law in Theorem 2.3. In particular, if we restrict (24) to $\mathcal{B}^c$ and take expectation, the resulting quantity is $O(N^{1-\epsilon/36+C\delta_0})$. Ultimately, (22) follows.

Proof of (23). We again use Lemma 4.20 in [45] to get the following:

$$(25) \quad \left| \sum_x (\mathbf{G}_1^2(z_i))_{xy} S_{xy}^\circ (\mathbf{G}_2^2(z_j))_{yx} \right| \prec N^{-2} \sum_{\alpha, \beta} |p_\alpha(z_i)|^2 |p_\beta(z_j)|^2 |M_{y,\alpha,\beta}|,$$

where, if we again let a_0 be such that $y \in \mathcal{I}_{a_0}^{(2)}$, we have

$$|M_{y,\alpha,\beta}| := N \left| \sum_x \psi_\alpha(x) \overline{\psi_\beta(x)} \cdot S_{xy}^\circ \right| = O(N) \left| \sum_{a: |a-a_0|_L \leq 1} \langle \psi_\alpha^* (E_a - N^{-1}I) \psi_\beta \rangle \right|.$$

At this point, we can again directly follow the proof of (2.30) in [50], with the threshold $N^{-\epsilon/36}$ for $|M_{y,\alpha,\beta}|$ in the bad event, as above. In particular, we have reduced the estimation of the left-hand side of (23) to estimation of eigenvalues and eigenvectors, which we do using Theorems 2.2, 2.3, and 2.4. So, we do not reproduce the details here. (Roughly speaking, the additional factor of $|p_\beta(z_j)|$ appearing in (25) compared to (24) is responsible for the extra factor of N on the right-hand side of (23) compared to (22); this is by the local law in Theorem 2.3). $\square$

2.4. Stochastic flow and the loop hierarchy. As in [50], the key to the proofs of Theorems 2.3 and 2.5 are estimates on a hierarchy of “ G -loops”. Besides a dimension-dependent scaling, the structure of these loops and the hierarchy are the same as in [50]. Thus, much of the rest of Section 2 follows [50] almost identically. We start with the matrix Brownian motion

$$dH_{t,xy} = \sqrt{S_{xy}} dB_{t,xy}, \quad H_0 = 0,$$

where $B_{t,xy}$ are independent standard complex Brownian motions for all $x, y \in \mathbb{Z}_{WL}^2$ up to the constraint $B_{yx} = \overline{B_{xy}}$. Along this flow, we will consider a time-dependent spectral parameter z_t as defined below.

DEFINITION 2.7. For any $E \in \mathbb{R}$, define $m^{(E)} := \lim_{\epsilon \rightarrow 0+} m_{sc}(E + i\epsilon)$, and consider the following objects:

$$\begin{aligned} z_t^{(E)} &= E + (1-t)m^{(E)}, \quad 0 \leq t \leq 1 \\ G_t^{(E)} &:= (H_t - z_t^{(E)})^{-1}. \end{aligned}$$

Note that the imaginary part of $z_t^{(E)}$ is given by the following formula:

$$(26) \quad \eta_t := \text{Im} z_t^{(E)} = (1-t)\text{Im} m^{(E)}.$$

By the Itô formula, the resolvent $G_t := G_t^{(E)}$ solves the following matrix SDE:

$$dG_t = -G_t dH_t G_t + G_t \{ \mathcal{S}[G_t] - m^{(E)} \} G_t dt.$$

Above, the linear operator $\mathcal{S} : M_N(\mathbb{C}) \rightarrow M_N(\mathbb{C})$ is defined as

$$\mathcal{S}[X]_{xy} := \delta_{xy} \sum_{v=1}^N S_{xv} X_{vv}.$$

The following will help us recover $G(z)$ from $G_t^{(E)}$; its proof is the same as Lemma 2.7 in [50].

LEMMA 2.8. Suppose that $z \in \mathbb{C}$ satisfies $0 < \text{Im} z \leq 1$ and $|\text{Re} z| \leq 2 - \kappa$ for some $\kappa > 0$. Then there exist an $E \in \mathbb{R}$ and a t with $|E| \leq 2 - \kappa$ and $0 \leq t < 1$ such that

$$(27) \quad z = t^{-1/2} \cdot z_t^{(E)}.$$

Moreover, there exists a fixed constant $c_\kappa > 0$ such that

$$(28) \quad c_\kappa \leq t \quad \text{and} \quad c_\kappa \text{Im} z \leq \text{Im} z_t \leq c_\kappa^{-1} \text{Im} z.$$

Finally, if z, E and t satisfy the relation (27), then we have

$$(29) \quad m_{sc}(z) = t^{1/2} \cdot m^{(E)}$$

as well as the distributional identity

$$(30) \quad G(z) \sim t^{1/2} \cdot G_t^{(E)}$$

For the rest of the paper, we will consider only $G_t^{(E)}$. We now define the aforementioned G -loops.

DEFINITION 2.9. For any fixed $E \in \mathbb{R}$, consider the following resolvent as a function of $+, -$ signs:

$$G_t^{(E)}(\alpha) := G_{t,\alpha}^{(E)} = \begin{cases} (H_t - z_t)^{-1}, & \alpha = + \\ (H_t - \bar{z}_t)^{-1}, & \alpha = - \end{cases}.$$

By definition, $G_{t,+}^{(E)} = \left(G_{t,-}^{(E)} \right)^\dagger$. For simplicity, we sometimes write $G_t = G_{t,+}^{(E)}$. Now, fix indices

$$\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \dots, \sigma_n), \quad \mathbf{a} = (a_1, a_2, \dots, a_n), \quad \sigma_i \in \{+, -\}, \quad a_i \in \mathbb{Z}_L^2, \quad 1 \leq i \leq n.$$

Next, recall E_a defined in (7). We define the corresponding “ n - G loop” to be the following complex number:

$$(31) \quad \mathcal{L}_{t,\sigma,\mathbf{a}} = \left\langle \prod_{i=1}^n G_t(\sigma_i) E_{a_i} \right\rangle, \quad \text{where} \quad \langle A \rangle = \text{Tr } A.$$

We also define

$$(32) \quad m(\sigma) := m^{(E)}(\sigma) = \begin{cases} m^{(E)}, & \sigma = + \\ \overline{m^{(E)}}, & \sigma = - \end{cases},$$

and

$$(33) \quad \tilde{G}_t(\sigma_k) := G_t(\sigma_k) - m(\sigma_k).$$

With these notations, we can rewrite the quantity $\text{Tr } G E_a G^\dagger E_b$ in Theorem 2.5 as follows:

$$\text{Tr } G E_a G^\dagger E_b = t \cdot \text{Tr} [G_{t,+} E_a G_{t,-} E_b] = t \cdot \mathcal{L}_{t,(+,-),(a,b)}.$$

We now define some operations on loops; these will be important for constructing a hierarchy of loops.

DEFINITION 2.10. Take a loop of the following form:

$$(34) \quad \mathcal{L}_{t,\sigma,\mathbf{a}} = \left\langle \prod_{i=1}^n G_t(\sigma_i) E_{a_i} \right\rangle, \quad \sigma_i \in \{+, -\}, \quad 1 \leq i \leq n$$

1. For $1 \leq k \leq n$, we define a “cut-and-glue” operator $\mathcal{G}_k^{(a)}$ as follows. Its action on a loop $\mathcal{L}_{t,\sigma,\mathbf{a}}$ is denoted by $\mathcal{G}_k^{(a)} \circ \mathcal{L}_{t,\sigma,\mathbf{a}}$, and it is defined to be the loop obtained by replacing $G_t(\sigma_k)$ by $G_t(\sigma_k) E_a G_t(\sigma_k)$.

In words, $\mathcal{G}_k^{(a)}$ cuts the k -th edge $G_t(\sigma_k)$ of the loop $\mathcal{L}_{t,\sigma,\mathbf{a}}$ and glues the two new ends with E_a . The new loop is a unit longer than the input loop. Since $\mathcal{G}_k^{(a)}$ turns a loop into another loop, it is an operator on the index space of $\sigma, \mathbf{a}$. Thus, we sometimes write $\mathcal{L}_{t,\mathcal{G}_k^{(a)}(\sigma,\mathbf{a})} := \mathcal{G}_k^{(a)} \circ \mathcal{L}_{t,\sigma,\mathbf{a}}$.

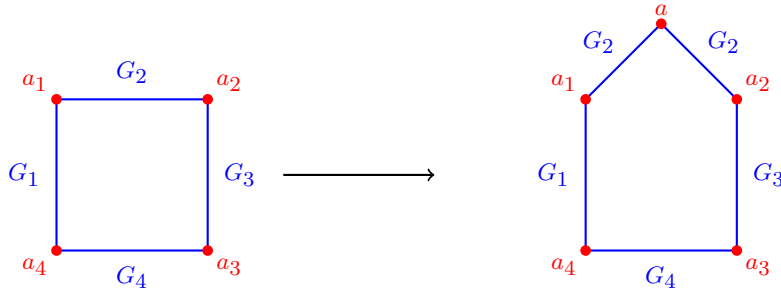


FIG 1. Illustration of $\mathcal{G}_k^{(a)}$ for $n = 4$. Taken from [50].

2. For $1 \leq k < l \leq n$, we define a “left” cut-and-glue operator $\mathcal{G}_{k,l}^{(a),L}$. Let $\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{L}_{t,\sigma,\mathbf{a}}$ be the following loop. Cut the k -th and l -th edges $G_t(\sigma_k)$ and $G_t(\sigma_l)$ (creating four end points and two “chains”). Glue the two new ends of the chain that contains E_{a_n} and insert a new E_a at the gluing point.

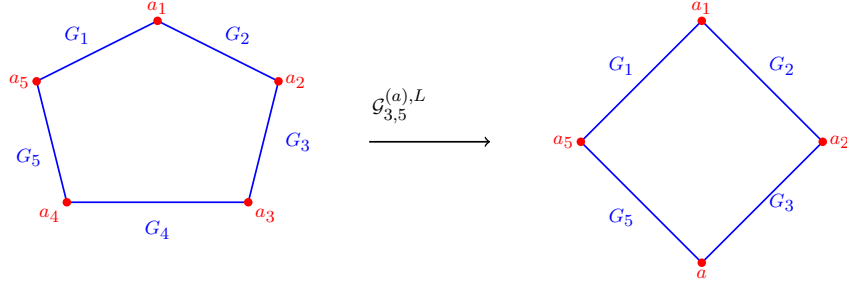


FIG 2. Illustration of operator $\mathcal{G}_{k,l}^{(a),L}$. Taken from [50].

3. For $1 \leq k < l \leq n$, we define a “right” cut-and-glue operator $\mathcal{G}_{k,l}^{(a),R}$ the same way, but we instead glue the two new ends of the chain that does not contain E_{a_n} . The length of the new loop is $l - k + 1$.

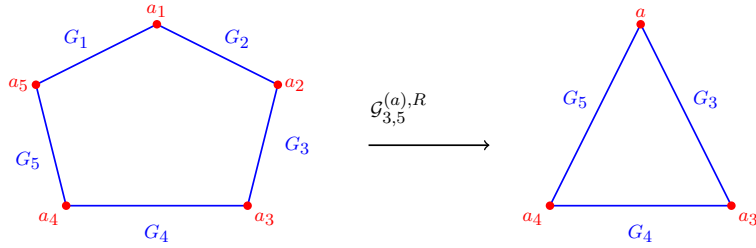


FIG 3. Illustration of operator $\mathcal{G}_{k,l}^{(a),R}$. Taken from [50].

The following result gives the evolution for a loop along the flow of $G_t^{(E)}$. It is derived via the Itô formula. Throughout this paper, we let $\mathbf{a} = (a_1, \dots, a_n)$ be a vector of indices in $\mathbb{Z}_L^2$, and we let $a \in \mathbb{Z}_L^2$ be a single index that is separate from $\mathbf{a}$. Finally, for convenience, we will use the notation $\partial_{(x,y)} := \partial_{H_t(x,y)}$.

LEMMA 2.11 (The loop hierarchy). *We have the following “loop hierarchy” SDE:*

$$(35) \quad d\mathcal{L}_{t,\sigma,a} = \mathcal{E}_{t,\sigma,a}^{(M)} + \mathcal{E}_{t,\sigma,a}^{(\tilde{G})} + W^2 \cdot \sum_{1 \leq k < l \leq n} \sum_{a,b} \left(\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{L}_{t,\sigma,a} \right) S_{ab}^{(B)} \left(\mathcal{G}_{k,l}^{(b),R} \circ \mathcal{L}_{t,\sigma,a} \right) dt.$$

Above, we used the notation

$$(36) \quad \mathcal{E}_{t,\sigma,a}^{(M)} := \sum_{\alpha=(x,y)} (\partial_\alpha \mathcal{L}_{t,\sigma,a}) \cdot (S_\alpha)^{1/2} \cdot d(B_t)_\alpha$$

$$(37) \quad \mathcal{E}_{t,\sigma,a}^{(\tilde{G})} := W^2 \cdot \sum_{1 \leq k \leq n} \sum_{a,b} \left\langle \tilde{G}_t(\sigma_k) E_a \right\rangle \cdot S_{ab}^{(B)} \cdot \left(\mathcal{G}_k^{(b)} \circ \mathcal{L}_{t,\sigma,a} \right) dt, \quad \tilde{G}_t = G_t - m.$$

The factor of W^2 in $\mathcal{E}^{(\tilde{G})}$ comes from the normalization that E_a has a factor of W^{-2} . In order to solve this hierarchy of SDEs, we introduce the notion of a convolution tree $\mathcal{K}$.

DEFINITION 2.12. For

$$\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n), \quad \mathbf{a} = (a_1, a_2, \dots, a_n), \quad \sigma_i \in \{+, -\}, \quad a_i \in \mathbb{Z}_L^2, \quad 1 \leq i \leq n$$

and $m(\sigma)$ defined in (32), we define $\mathcal{K}_{t,\sigma,\mathbf{a}}$ to be the unique solution to the equation

$$(38) \quad \frac{d}{dt} \mathcal{K}_{t,\sigma,\mathbf{a}} = W^2 \cdot \sum_{1 \leq k < l \leq n} \sum_{a,b} \left(\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{K}_{t,\sigma,\mathbf{a}} \right) S_{ab}^{(B)} \left(\mathcal{G}_{k,l}^{(b),R} \circ \mathcal{K}_{t,\sigma,\mathbf{a}} \right)$$

with initial data

$$\mathcal{K}_{0,\sigma,\mathbf{a}} = W^{-2(n-1)} \cdot \prod_{k=1}^n m(\sigma_k) \cdot \mathbf{1}(a_1 = a_2 = \dots = a_n).$$

Here, we define the $\mathcal{G}$ operator acting on $\mathcal{K}$ in the same way as it acts on $\mathcal{L}$, i.e.,

$$(39) \quad \mathcal{G}_{k,l}^{(a),L} \circ \mathcal{K}_{t,\sigma,\mathbf{a}} = \mathcal{K}_{t,\mathcal{G}_{k,l}^{(a),L}(\sigma,\mathbf{a})} \quad \forall t, k, \sigma, \mathbf{a},$$

and similarly for $\mathcal{G}_{k,l}^{(b),R}$. For the special case $n = 1$, we define $\mathcal{K}_{t,+,a} = m$ and $\mathcal{K}_{t,-,a} = \overline{m}$ for $0 \leq t \leq 1$.

Note that the lengths of $\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{K}_{t,\sigma,\mathbf{a}}$ and $\mathcal{G}_{k,l}^{(a),R} \circ \mathcal{K}_{t,\sigma,\mathbf{a}}$ are no greater than the length of $\mathcal{K}_{t,\sigma,\mathbf{a}}$. Thus, the system of equations for $\mathcal{K}$ can be solved inductively in the length of $\mathcal{K}$; we illustrate this in the next section.

2.5. *The propagator $\Theta_\xi^{(B)}$ and an estimate on $\mathcal{K}$.* The key object that we need to analyze $\mathcal{K}$ is the following propagator.

DEFINITION 2.13. We define the propagator $\Theta_\xi^{(B)}$ to be the following $L^2 \times L^2$ matrix:

$$(40) \quad \Theta_\xi^{(B)} := \frac{1}{1 - \xi \cdot S^{(B)}}, \quad \xi \in \mathbb{C}, \quad |\xi| < 1.$$

(The superscript B indicates a block-level matrix, i.e. one whose indices are parameterized by $\mathbb{Z}_L^2$.)

For convenience, we often omit the superscript (B) . (We remind the readers that Θ was often used to denote the $N \times N$ matrix $(1 - \xi S)^{-1}$ in the literature. In this paper, $\Theta_\xi = \Theta_\xi^{(B)}$.) We note that

$$(41) \quad \partial_\xi \Theta_\xi^{(B)} = \Theta_\xi^{(B)} \cdot S^{(B)} \cdot \Theta_\xi^{(B)}.$$

The following three special cases are often used in this paper (note that $|m| = |m^{(E)}| = 1$ in our setting):

$$\Theta_{tm^2}^{(B)}, \quad \Theta_{t\overline{m}^2}^{(B)}, \quad \text{and} \quad \Theta_{t|m|^2}^{(B)} = \Theta_t^{(B)}.$$

We now collect the following important properties of $\Theta^{(B)}$.

LEMMA 2.14. Suppose that $\xi \in \mathbb{C}$ satisfies $|\xi| < 1$. Define

$$\hat{\ell}(\xi) := \min \left((|1 - \xi|)^{-1/2}, L \right), \quad \xi \in \mathbb{C}$$

Then $\Theta_\xi^{(B)}$ has the following properties:

1. *Symmetry*: $(\Theta_\xi^{(B)})_{ab} = (\Theta_\xi^{(B)})_{ba}$.
2. *Translation invariance*: $(\Theta_\xi^{(B)})_{ab} = (\Theta_\xi^{(B)})_{a+s, b+s}$ for any $s \in \mathbb{Z}_L^2$.
3. *Commutativity*: $[S^{(B)}, \Theta_\xi^{(B)}] = [\Theta_{\xi'}^{(B)}, \Theta_\xi^{(B)}] = 0$ for all ξ, ξ' .
4. For any $0 \leq \xi < 1$, we have the random walk representation $\Theta_\xi^{(B)} = \sum_{k=0}^{\infty} \xi^k \cdot (S^{(B)})^k$.
5. *Exponential decay at length scale $\hat{\ell}(\xi)$* for $\xi \in \{tm^2, t\bar{m}^2, t|m|^2\}$:

$$(42) \quad (\Theta_\xi^{(B)})_{ab} \prec \frac{e^{-c|a-b|_L / \hat{\ell}(\xi)}}{|1 - \xi| \cdot [\hat{\ell}(\xi)]^2}.$$

6. *Derivative bounds* for $\xi \in \{tm^2, t\bar{m}^2, t|m|^2\}$: for any $s \in \mathbb{Z}_L^2$, we have

$$(43) \quad \left| (\Theta_\xi^{(B)})_{ab} - (\Theta_\xi^{(B)})_{a(b+s)} \right| \prec \frac{|s|_L}{|a-b|_L + 1} + \frac{|s|_L}{[\hat{\ell}(\xi)]^2 |1 - \xi|^{1/2}}$$

$$(44) \quad \left| 2(\Theta_\xi^{(B)})_{ab} - (\Theta_\xi^{(B)})_{a(b+s)} - (\Theta_\xi^{(B)})_{a(b-s)} \right| \prec \frac{|s|_L^2}{|a-b|_L^2 + 1} + \frac{|s|_L^2}{[\hat{\ell}(\xi)]^2}.$$

The proof of Lemma 2.14 is based on Fourier analysis on $\mathbb{Z}_L^2$: a contour shift in the momentum variable yields the exponential decay, and a dyadic decomposition of the momenta yields the derivative bounds. A proof is given in Section 8; the argument is elementary, so we give an intuitive explanation of Lemma 2.14 below.

The first four properties in Lemma 2.14 are standard. The exponential decay (42) for the discrete Laplacian on $\mathbb{Z}^2$ (instead of $S^{(B)} - \text{Id}$) can be found in Theorem 1.3 of [34] (see also the discussion following it therein). The difference between the full space $\mathbb{Z}^2$ and the torus $\mathbb{Z}_L^2$ is that if $\hat{\ell}(\xi) = L$, so that $|1 - \xi|^{-1/2} \geq L$, then the periodic boundary conditions on $\mathbb{Z}_L^2$ automatically introduce a space-cutoff of scale L (before the cutoff at scale $|1 - \xi|^{-1/2}$ that is imposed by the resolvent). One can also check that (42) implies

$$\sum_{b \in \mathbb{Z}_L^2} (\Theta_\xi^{(B)})_{ab} \prec |1 - \xi|^{-1},$$

which can be checked using the random walk representation as well.

The gradient bounds (43)-(44) are standard, in that the Green's function for the Laplacian on $\mathbb{R}^2$ is $\log|x|$. Thus, the first terms on the right-hand side of (43)-(44) agree with its derivatives. (The support-length of the gradients is also of order $\hat{\ell}(\xi)$, but we will not need this fact.) The additional corrections come from the periodic boundary of $\mathbb{Z}_L^2$. The scaling of the last term in (44) agrees with the heuristic that $\Theta^{(B)}$ is the resolvent for the diffusion of a generator, so its second-derivative should have L^1 norm of size $O_{\prec}(1)$. The last term in (43) therefore interpolates between (42) and the last term in (44).

Let us now illustrate $\mathcal{K}_{t, \sigma, (a_1, a_2)}$ for $n = 2$. In this case, the defining convolution tree equation (38) is

$$(45) \quad \frac{d}{dt} \mathcal{K}_{t, \sigma, (a_1, a_2)} = W^2 \cdot \sum_{a, b} \mathcal{K}_{t, \sigma, (a_1, a)} \cdot S_{ab}^{(B)} \cdot \mathcal{K}_{t, \sigma, (b, a_2)}$$

Throughout this paper, we will often use the following convenient notation:

$$(46) \quad m_i = m(\sigma_i).$$

Using the propagator $\Theta^{(B)}$ in (40) and, (41), a short calculation yields

$$(47) \quad \mathcal{K}_{t, \sigma, \mathbf{a}} = W^{-2} m_1 m_2 \left(\Theta_{t \cdot m_1 m_2}^{(B)} \right)_{a_1 a_2}, \quad \mathbf{a} = (a_1, a_2)$$

In particular, if we consider the two cases $\sigma = (+, -)$ and $\sigma = (+, +)$, then

$$(48) \quad \mathcal{K}_{t,\sigma,(a,b)} = W^{-2} \cdot \begin{cases} |m|^2 \left[(1 - t|m|^2 S^{(B)})^{-1} \right]_{ab} & , \quad \sigma = (+, -) \\ m^2 \left[(1 - tm^2 S^{(B)})^{-1} \right]_{ab} & , \quad \sigma = (+, +) \end{cases}$$

In the case where $n = 3$, we have the following formula (which comes from Example 2.16 in [50] and comes directly from the convolution tree equation, hence it has no dimension dependence):

$$\mathcal{K}_{t,\sigma,\mathbf{a}} = \sum_{b_1 b_2 b_3} \left(\Theta_{t \cdot m_1 m_2}^{(B)} \right)_{a_1 b_1} \left(\Theta_{t \cdot m_2 m_3}^{(B)} \right)_{a_2 b_2} \left(\Theta_{t \cdot m_3 m_1}^{(B)} \right)_{a_3 b_3} \cdot \mathcal{K}_{0,\sigma,\mathbf{b}}, \quad \mathbf{b} = (b_1, b_2, b_3).$$

LEMMA 2.15. *Recall $\hat{\ell}$ in (42). We have*

$$(49) \quad \mathcal{K}_{t,\sigma,\mathbf{a}} \prec M_t^{-n+1}, \quad \ell_t := \hat{\ell}(t) = \min \left((|1 - t|)^{-1/2}, L \right), \quad M_t := W^2 \ell_t^2 \eta_t.$$

Finally, we note that for any fixed $|E| < 2 - \kappa$, we have $\text{Im} z_t = \text{Im} z_t^{(E)} = \text{Im} m^{(E)}$. $(1 - t) \sim (1 - t)$. Thus, $\ell(z)$ from (2) and ℓ_t from (49) are the same order, in that $\ell(z_t) \sim \ell_t = \hat{\ell}(t)$. Furthermore, for z, t , and E satisfying (27) and (28), we have $\text{Im} z \sim \text{Im} z_t^{(E)}$ and $\ell(z) \sim \ell(z_t^{(E)}) \sim \ell_t$. Since these terms are the same order, in the following, we will use only ℓ_t .

2.6. *Estimates on loops.* Our main estimates for the G -loop are given below. (Recall $\prec$ in Definition 2.1, and recall ℓ_t, M_t in (49).)

LEMMA 2.16. *For any $\kappa, \tau > 0$ and $E \in [-2 + \kappa, 2 - \kappa]$, $0 \leq t \leq 1 - N^{-1+\tau}$, we have*

$$(50) \quad \max_{\sigma, \mathbf{a}} |\mathcal{L}_{t,\sigma,\mathbf{a}} - \mathcal{K}_{t,\sigma,\mathbf{a}}| \prec M_t^{-n}, \quad \eta_t := \text{Im} z_t,$$

$$(51) \quad \max_{\sigma, \mathbf{a}} |\mathcal{L}_{t,\sigma,\mathbf{a}}| \prec M_t^{-n+1}.$$

LEMMA 2.17 (2-loop estimate). *With the notations and assumptions of the previous lemma, if $\sigma = (+, -)$ and $\mathbf{a} = (a_1, a_2)$, then we have*

$$(52) \quad |\mathcal{L}_{t,\sigma,\mathbf{a}} - \mathcal{K}_{t,\sigma,\mathbf{a}}| \prec M_t^{-2} \exp \left(-\frac{|a_1 - a_2|_L^{1/2}}{\ell_t^{1/2}} \right) + W^{-D}.$$

LEMMA 2.18 (Local law for G_t). *With the notations and assumptions of the previous lemma, we have*

$$(53) \quad \|G_{t,+}^{(E)} - m^{(E)}\|_{\max} \prec M_t^{-1/2}.$$

LEMMA 2.19 (Improved expectation bound on 2-loop). *Adopt the notations and assumptions of the previous lemma. Fix any $\sigma \in \{+, -\}^2$ and $\mathbf{a} = (a_1, a_2) \in \mathbb{Z}_L^2 \times \mathbb{Z}_L^2$. We have*

$$(54) \quad |\mathbb{E} \mathcal{L}_{t,\sigma,\mathbf{a}} - \mathcal{K}_{t,\sigma,\mathbf{a}}| \prec M_t^{-3}.$$

PROOF OF THEOREMS 2.3 AND 2.5. This is the same argument as the proof of Theorems 2.3 and 2.5 in [50]. For each z in Theorems 2.3 and 2.5, Lemma 2.8 gives E and t satisfying (27), (28), and

$$(1-t) \sim \operatorname{Im} z_t \sim \operatorname{Im} z \geq N^{-1+\tau}, \quad \eta_t \sim \eta, \quad \ell_t \sim \ell(z).$$

Combining (29) and (30) gives

$$(55) \quad G(z) - m_{sc}(z) = t^{1/2} \left(G_t^{(E)} - m^{(E)} \right).$$

Thus, (4) follows from the estimate (53) on $G_t - m$. Similarly, the partial tracial local law (5) follows from (50) for $n = 1$ and the definition $\mathcal{K}_{t,+,a} = m^{(E)}$ from Definition 2.12. Next, by Definition 2.9,

$$(56) \quad \operatorname{Tr} G E_a G E_b = t \cdot \mathcal{L}_{t,(+,+),(a,b)}, \quad \operatorname{Tr} G E_a G^\dagger E_b = t \cdot \mathcal{L}_{t,(+,-),(a,b)}.$$

By using the formula (47) and recalling from (29) that $m_{sc}(z) = t^{1/2} m^{(E)}$, we get

$$t \cdot \mathcal{K}_{t,(+,+),(a,b)} = W^{-2} m_{sc}^2(z) \cdot \left[\frac{1}{1 - m_{sc}^2(z) \cdot S^{(B)}} \right]_{ab}, \quad t \cdot \mathcal{K}_{t,(+,-),(a,b)} = W^{-2} |m_{sc}(z)|^2 \cdot \left[\frac{1}{1 - |m_{sc}(z)|^2 \cdot S^{(B)}} \right]_{ab}.$$

Therefore, (10) and (11) in Theorem 2.5 follow from (50) in the case $n = 2$. Finally, (12)-(13) follow from Lemma 2.19. $\square$

2.7. *Strategy of the proofs of main lemmas.* It turns out that Lemma 2.19 will follow from Lemmas 2.16, 2.17, and 2.18. Thus, let us outline the proofs of Lemmas 2.16, 2.17 and 2.18. The general strategy is the same as in Section 2 of [50]. By Definitions 2.9 and 2.12, we have

$$G_0(+) = m \cdot I_{N \times N}, \quad \mathcal{L}_{0,\sigma,\mathbf{a}} = \mathcal{K}_{0,\sigma,\mathbf{a}}, \quad \forall \sigma, \mathbf{a},$$

where $I_{N \times N}$ is the identity matrix. By construction, we know that:

$$(57) \quad \text{Lemmas 2.16, 2.17 and 2.18 hold at } t = 0 \text{ with no error.}$$

The following result propagates the desired estimates forwards in time.

THEOREM 2.20. *Assume for some fixed E : $|E| \leq 2 - \kappa$ and $s \in [0, 1]$ that Lemmas 2.16, 2.17 and 2.18 hold at time s . That is, assume that for $1 \leq n \in \mathbb{N}$ and large $D > 0$, we have*

$$(58) \quad \max_{\sigma, \mathbf{a}} |\mathcal{L}_{s,\sigma,\mathbf{a}} - \mathcal{K}_{s,\sigma,\mathbf{a}}| \prec M_s^{-n}$$

$$(59) \quad |\mathcal{L}_{s,\sigma,\mathbf{a}} - \mathcal{K}_{s,\sigma,\mathbf{a}}| \prec M_s^{-2} \exp \left(- \frac{|a_1 - a_2|_L^{1/2}}{\ell_s^{1/2}} \right) + W^{-D}, \quad \sigma = (+, -)$$

$$(60) \quad \|G_{s,+}^{(E)} - m^{(E)}\|_{\max} \prec M_s^{-1/2}.$$

Then for any $t > s$ satisfying $t \leq 1 - N^{-1+\tau}$ for some fixed $\tau > 0$ and

$$(61) \quad M_s^{-1} \leq \left(\frac{1-t}{1-s} \right)^{30}$$

we have (58), (59) and (60) with s replaced by t .

PROOF OF LEMMAS 2.16, 2.17, AND 2.18. For any τ and $t \leq 1 - N^{-1+\tau}$, choose $\tau' > 0$ and $n_0 \in \mathbb{N}$ such that $M_t^{-1} \leq W^{-30\tau'}$ and $(1-t) = W^{-n_0\tau'}$. We also choose a sequence of time-scales given by $1 - s_k = W^{-k\tau'}$ for $k \leq n_0$, where n_0 is chosen so that $s_{n_0} = t$. Since M_s is decreasing in $s \in [0, 1]$, we have

$$M_{s_k}^{-1} \leq M_t^{-1} \leq W^{-30\tau'} = \left(\frac{1 - s_{k+1}}{1 - s_k} \right)^{30}$$

for all k such that $k + 1 \leq n_0$. We can now apply Theorem 2.20 for $s = s_k$ for each k inductively. (We note that $n_0 = O(1)$, so the number of implicit W^τ factors we pick up from the meaning of $\prec$ is still $W^{\tau'}$ for $\tau' > 0$ small.) This gives all the estimates in Lemmas 2.16, 2.17, and 2.18 except for (51), which follows from (50) combined with the deterministic $\mathcal{K}$ estimate from Lemma 2.15. $\square$

Theorem 2.20 and Lemma 2.19 will be proved in six steps. Their proofs are given in Section 5. The first five of these steps are meant to show Theorem 2.20. So, throughout the first five of these steps, we assume that the conditions of Theorem 2.20 are satisfied. Each step builds on the conclusions of the preceding steps.

On the other hand, the last of these steps is separate in the sense that it is not part of the propagation idea used in the proofs of Lemmas 2.16, 2.17, and 2.18 above.

Step 1 (A priori loop bounds): We will first show the following loop estimate:

$$(62) \quad \mathcal{L}_{u,\sigma,\mathbf{a}} \prec (\ell_u/\ell_s)^{2(n-1)} \cdot M_u^{-n+1}, \quad s \leq u \leq t.$$

We will also show the following weaker version of (60):

$$(63) \quad \|G_u - m\|_{\max} \prec M_u^{-1/4}, \quad s \leq u \leq t.$$

Step 2 (A priori 2-loop decay): Next, we show the following local law for $u \in [s, t]$:

$$(64) \quad \|G_u - m\|_{\max} \prec M_u^{-1/2}, \quad s \leq u \leq t.$$

Hence (60) in Theorem 2.20 will follow. In addition, for any $\sigma = (+, -)$, $s \leq u \leq t$, and $D > 0$, we show

$$(65) \quad |\mathcal{L}_{u,\sigma,\mathbf{a}} - \mathcal{K}_{u,\sigma,\mathbf{a}}| \prec (\eta_s/\eta_u)^4 \cdot M_u^{-2} \exp\left(-\frac{|a_1 - a_2|_L^{1/2}}{\ell_u^{1/2}}\right) + W^{-D}.$$

Step 3 (Sharp loop bounds): Next, we show the following *sharp* estimate on loops:

$$(66) \quad \max_{\sigma,\mathbf{a}} |\mathcal{L}_{u,\sigma,\mathbf{a}}| \prec M_u^{-n+1}, \quad s \leq u \leq t, \quad \forall n \in \mathbb{N}.$$

Step 4 (A sharp $\mathcal{L} - \mathcal{K}$ bound): We show a sharp bound on $\mathcal{L}_t - \mathcal{K}_t$, which gives (58) in Theorem 2.20:

$$(67) \quad \max_{\sigma,\mathbf{a}} |\mathcal{L}_{u,\sigma,\mathbf{a}} - \mathcal{K}_{u,\sigma,\mathbf{a}}| \prec M_u^{-n}, \quad s \leq u \leq t, \quad \forall n \in \mathbb{N}.$$

Step 5 (A sharp decay bound): For $\sigma = (+, -)$, we show the following, which gives (59) in Theorem 2.20:

$$(68) \quad |\mathcal{L}_{u,\sigma,\mathbf{a}} - \mathcal{K}_{u,\sigma,\mathbf{a}}| \prec M_u^{-2} \exp\left(-\frac{|a_1 - a_2|_L^{1/2}}{\ell_u^{1/2}}\right) + W^{-D}.$$

Step 6 (Showing Lemma 2.19): As we showed earlier, Theorem 2.20 implies Lemmas 2.16, 2.17, and 2.18. We then use these results to prove Lemma 2.19.

2.8. *Notations.* Here we summarize global notations used in this paper.

- W is band width, and L^2 is the number of blocks, so that $N = W^2 \cdot L^2$; throughout, $L \geq 3$, which is needed for $S^{(B)}$ to be the five-point stochastic matrix and is harmless because every statement below is asymptotic in $L \rightarrow \infty$
- $\mathcal{I}_a^{(2)}$ is the a -th block, $[x]$ is the block containing x , and E_a is the following matrix supported on $\mathcal{I}_a^{(2)}$:

$$x \in \mathcal{I}_{[x]}^{(2)}, \quad (E_a)_{xy} = \delta_{xy} \cdot W^{-2} \cdot \mathbf{1}(x \in \mathcal{I}_a^{(2)})$$

- The periodic distance on $\mathbb{Z}_L$ is denoted by $|\cdot|_L$ and the periodic L^1 distance on $\mathbb{Z}_L^2$ is defined for any $a = (a(1), a(2)) \in \mathbb{Z}_L^2$ by

$$|a| = |a|_L = |a(1)|_L + |a(2)|_L.$$

- The matrices $S \in \mathbb{R}^{N \times N}$, $S^{(B)} \in \mathbb{R}^{L^2 \times L^2}$, $S_W \in \mathbb{R}^{W^2 \times W^2}$ are all related to the variances of entries of the random matrix H , and $S = S^{(B)} \otimes S_W$.
- In the stochastic flow, we typically omit the superscript (E) for simplicity, so that

$$z_t = z_t^{(E)}, \quad G_t := (\sqrt{t}H - z_t)^{-1}, \quad m = \lim_{\varepsilon \searrow 0} m_{sc}(E + i\varepsilon).$$

Additionally, we use G_t to mean $(H_t - z_t)^{-1}$, where H_t has the same distribution as $\sqrt{t}H$. The context will make it clear which interpretation of G_t is being used.

- The length-scale ℓ_t is defined in (49) as $\ell_t := \hat{\ell}(t) = \min(|1 - t|^{-1/2}, L) \sim \ell(z_t)$.
- The notation $\mathcal{L}$ denotes G -loops, and $\mathcal{K}$ denotes its deterministic approximation (the “convolution tree”).
- The matrix $\Theta_\xi = \Theta_\xi^{(B)}$ is given in Definition 2.13.
- The quantities $\Xi^{(\mathcal{L})}$, $\Xi^{(\mathcal{L}-\mathcal{K})}$, $\Xi^{(\mathcal{C})}$ are loops that are normalized to have heuristic size of order 1:

$$(69) \quad \begin{aligned} \Xi_{t,m}^{(\mathcal{L})} &:= \max_{\sigma, \mathbf{a}} |\mathcal{L}_{t,\sigma,\mathbf{a}}| \cdot M_t^{m-1} \cdot \mathbf{1}(\sigma \in \{+, -\}^m), \\ \Xi_{t,m}^{(\mathcal{L}-\mathcal{K})} &:= \max_{\sigma, \mathbf{a}} |(\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}}| \cdot M_t^m \cdot \mathbf{1}(\sigma \in \{+, -\}^m). \end{aligned}$$

- The operators $\Theta_{t,\sigma}$ and $\mathcal{U}_{s,t,\sigma}$ in Definition 5.2 appear in an “integrated loop hierarchy” (Lemma 5.3).
- The $\mathcal{G}$ operators, such as $\mathcal{G}_k^{(a)}$, $\mathcal{G}_{k,l}^{(a),L}$, and $\mathcal{G}_{k,l}^{(a),R}$, are in Definition 2.10. These represent the cutting and gluing of loops in the loop hierarchy.
- The $\mathcal{E}$ terms represent lower-order terms in the (integrated) loop hierarchy (35) and (104). Specifically, $\mathcal{E}^{(M)}$ and $\mathcal{E}^{(\tilde{G})}$ are defined in (36) and (37), respectively. The term $\mathcal{E}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))}$ is defined in (99). Additionally, $(\mathcal{E} \otimes \mathcal{E})$ and $(\mathcal{E} \otimes \mathcal{E})^{(k)}$ are defined in Definition 5.4.
- The quantity $\mathcal{T}_t^{(\mathcal{L}-\mathcal{K})}$ is defined in (109); it captures tail behavior of $\mathcal{L} - \mathcal{K}$. The quantity $\mathcal{T}_{t,D}$ represents a deterministic rough tail function, as defined in (110).
- The terms $\mathcal{J}_{u,D}$ and $\mathcal{J}^*$ are introduced in (111) and (112), respectively, to describe the ratio between $\mathcal{T}_t^{(\mathcal{L}-\mathcal{K})}$ and $\mathcal{T}_{t,D}$.
- We often use an augmented length-scale $\ell_t^* = (\log W)^{3/2} \ell_t$. At this scale, Θ_t is exponentially small in W , but $\mathcal{T}_t$ is not.

3. Basic properties of $\mathcal{K}$. In this section, we record the exact properties about $\mathcal{K}$ that will be used in the analysis of the loop hierarchy. The results in this section are essentially taken from Section 3 of [50] with some slight modifications.

We start with an estimate for a “pure loop”.

LEMMA 3.1. *Suppose that $\sigma = (+, + \cdots +)$. We have*

$$(70) \quad |\mathcal{K}_{t,\sigma,a}| \leq C_n \exp \left(-c_n \max_{i,j} |a_i - a_j|_L \right).$$

PROOF. We follow the proof of Corollary 3.5 in [50]. First, we use equations (3.5) and (3.6) in [50]; this gives

$$\mathcal{K}_{t,\sigma,a} = \prod_{i=1}^n m(\sigma_i) \cdot W^{-2(n-1)} \sum_{\Gamma_a} \Gamma_a(t, \sigma).$$

The sum Γ_a has a number of terms depending only on n , and each Γ_a has the form

$$\Gamma_a(t, \sigma) := \sum_{\mathbf{b} \in (\mathbb{Z}_L^2)^n} \Gamma_a^{(\mathbf{b})}(t, \sigma),$$

$$|\Gamma_a^{(\mathbf{b})}(t, \sigma)| \leq C_n \exp \left(-c_n \max_{i,j} |b_i - b_j|_L - c_n \max_{i,j} |a_i - b_j|_L \right).$$

(Each $\Gamma_a^{(\mathbf{b})}(t, \sigma)$ term is a product of Θ_{tm^2} matrices. The second line above, which is shown in the proof of Corollary 3.5 in [50], relies only on the exponential decay of Θ_{tm^2} , which follows from Lemma 2.14 and the bound $|1 - tm^2| \geq c$ for some $c > 0$.) It now suffices to combine the previous two displays. $\square$

We now give a generalization of the Ward identity for both $\mathcal{L}$ and $\mathcal{K}$. This is Lemma 3.6 in [50].

LEMMA 3.2. *Fix a loop $\mathcal{L}_{t,\sigma,a}$ with $\sigma_1 = +$ and $\sigma_n = -$. We have*

$$(71) \quad \sum_{a_n} \mathcal{L}_{t,\sigma,a} = \frac{1}{2iW^2\eta_t} (\mathcal{L}_{t,\sigma+,a/a_n} - \mathcal{L}_{t,\sigma-,a/a_n})$$

$$(72) \quad \sum_{a_n} \mathcal{K}_{t,\sigma,a} = \frac{1}{2iW^2\eta_t} (\mathcal{K}_{t,\sigma+,a/a_n} - \mathcal{K}_{t,\sigma-,a/a_n})$$

where

- $\sigma_{\pm}$ is the length $n - 1$ sign vector obtained by removing σ_n from σ and replacing σ_1 with $\pm$, i.e.,

$$\sigma_{\pm} = (\pm, \sigma_2, \sigma_3, \dots, \sigma_{n-1}).$$

- a/a_n is obtained by removing a_n from a :

$$a/a_n = (a_1, a_2, a_3, \dots, a_{n-1})$$

PROOF. As in the proof of Lemma 3.6 in [50], (71) follows by the usual Ward identity for Green’s functions, i.e. $GG^\dagger = \frac{1}{2i\eta}(G - G^\dagger)$. The proof of (72) in the case $n = 2, 3$ is an algebraic computation using the explicit formulas for $\mathcal{K}$. For $n \geq 4$, it uses only (38). Both are given in the proof of Lemma 3.6 in [50] with no dimension dependence (i.e. no dependence on estimates of $\Theta^{(B)}$), so we omit them here. $\square$

The proof of Corollary 3.7 in [50] now gives the following as a consequence of the Ward identity (72).

COROLLARY 3.3. *Under the assumptions of Lemma 3.2, we have*

$$(73) \quad \sum_{a_2} \cdots \sum_{a_{n-1}} \sum_{a_n} \mathcal{K}_{t, \sigma, \mathbf{a}} = O([W^2 \eta_t]^{-n+1})$$

We now conclude with an estimate on $\mathcal{K}$. To state it, we need to recall a decomposition of the convolution tree $\mathcal{K}$ in terms of simpler objects. First, fix a length n for the $\mathcal{K}$ loop. We have the decomposition

$$(74) \quad \mathcal{K}_{t, \sigma, \mathbf{a}} = W^{-2(n-1)} \cdot m_\sigma \cdot \sum_{\pi} \mathcal{K}_{t, \sigma, \mathbf{a}}^{(\pi)}, \quad m_\sigma = \prod_i m_i.$$

where the notation is explained as follows. First, π is a collection of sets of the form $\{i, j\}$ with $i, j \in \mathbb{Z}_n$. (The exact constraints on π are given in Definition 3.9 in [50]. We do not record them here because these constraints will not be important for us to explicitly use; in [50], they are used to prove certain estimates that depend only on Lemma 2.14 for $|1 - \xi|, \hat{\ell}(\xi) > c$ for some $c > 0$, which we can therefore borrow directly.) The summands are given as follows:

$$(75) \quad \mathcal{K}^{(\pi)}(t, \sigma, \mathbf{a}) := \sum_{\mathbf{d} \in (\mathbb{Z}_L^2)^n} \sum_{\mathbf{c} \in (\mathbb{Z}_L^2)^{2(M-1)}} \left(\prod_{i=1}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} \right) \cdot \prod_{k=1}^{M-1} \left(\Theta_{t|m|^2}^{(B)} - 1 \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=1}^M \Sigma^{(\emptyset)}(t, \sigma^{(k)}, \mathbf{d}^{(k)}).$$

Above, $M = O(1)$ is a number determined only by n, π . The $\sigma^{(k)}$ and $\mathbf{d}^{(k)}$ are determined by $\sigma, \mathbf{d}, \pi, k$. However, the only properties that we will need for $\Sigma^{(\emptyset)}(t, \sigma^{(k)}, \mathbf{d}^{(k)})$ are the following facts, whose proofs use only $\Theta^{(B)}$ bounds of the form in Lemma 2.14 for $|1 - \xi|, \hat{\ell}(\xi) > c$ for some $c > 0$ (and can therefore be directly cited).

- Suppose $n \geq 4$ is even, and define the alternating sign vector $\sigma^{(alt)} = (\sigma_1^{(alt)}, \dots, \sigma_n^{(alt)})$ via

$$1 \leq k \leq n, \quad \sigma_k^{(alt)} = \begin{cases} +, & k \in 2\mathbb{Z} - 1 \\ -, & k \in 2\mathbb{Z} \end{cases}, \quad t \in [0, 1],$$

We have the following *sum zero property*, which follows from the proof of Lemma 3.10 in [50]:

$$(76) \quad \sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \sigma^{(alt)}, \mathbf{d}) \right) = O(\eta_t)$$

- We have the short-range exponential bound below (which comes from (3.43) in [50]):

$$(77) \quad \Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) \leq C_n \exp \left(-c_n \max_{ij} |d_i - d_j|_L \right).$$

(This comes from exponential decay of $\Theta^{(B)}$ in Lemma 2.14). However, in principle, all we need is that the sum of $|\Sigma^{(\emptyset)}(t, \sigma, \mathbf{d})|$ over $d_2, \dots, d_n$ is $O(1)$, which follows from $\sum_b (\Theta_\xi^{(B)})_{ab} = (1 - \xi)^{-1}$.

- Fix d_1 , and write $s_i = d_i - d_1$ for $i = 2, \dots, n$, and $g(\mathbf{s}) := \Sigma^{(\emptyset)}(t, \sigma, \mathbf{d})$. We have $g(\mathbf{s}) = g(-\mathbf{s})$; see the proof of Lemma 3.11 in [50].

The following is analogous to Lemma 3.11 in [50]. We explain the differences in the proofs of the result below and of Lemma 3.11 in [50], namely the dimension-dependent parts.

LEMMA 3.4. *Recall $M_t := W^2 \ell_t^2 \eta_t$. We have the estimate*

$$(78) \quad \mathcal{K}_{t,\sigma,\mathbf{a}} \prec M_t^{-n+1}.$$

PROOF. In the case $n = 2$, this follows by (47), so we focus on $n \geq 3$ for the rest of this argument. Moreover, we can assume that σ does not consist of just one sign, i.e. it is not pure, since the estimate at hand follows by Lemma 3.1 in this case. By (74), it suffices to show

$$(79) \quad \mathcal{K}_{t,\sigma,\mathbf{a}}^{(\pi)} \prec (\ell_t^2 \eta_t)^{-n+1}.$$

We first assume that $\pi = \emptyset$. Assume without loss of generality (after re-indexing) that $\sigma_1 \neq \sigma_2$. We have

$$\begin{aligned} \mathcal{K}_{t,\sigma,\mathbf{a}}^{(\emptyset)} &= \sum_{\mathbf{d}} \left(\Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) \right) \cdot \prod_{i=1}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} \\ &= \sum_{d_1} \left(\Theta_{t|m|^2}^{(B)} \right)_{a_1, d_1} \cdot \sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) \right) \cdot \prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} \end{aligned}$$

We claim that to show (79) for $\pi = \emptyset$, it suffices to show

$$(80) \quad \sum_{d_2, \dots, d_n} \left(\Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) \right) \cdot \prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i, d_i} \prec (\ell_t^2 \eta_t)^{-n+2} \cdot \left(\min_{2 \leq k \leq n} |a_k - d_1|_L^2 + 1 \right)^{-1} + (\ell_t^2 \eta_t)^{-n+1} \eta_t,$$

Indeed, we can bound entries of $\Theta_{t|m|^2}^{(B)}$ by $\ell_t^{-2} \eta_t^{-1}$ using Lemma 2.14, so that

$$\begin{aligned} &[\ell_t^2 \eta_t]^{-n+2} \sum_{d_1} \left(\Theta_{t|m|^2}^{(B)} \right)_{a_1, d_1} \cdot \left(\min_{2 \leq k \leq n} |a_k - d_1|_L^2 + 1 \right)^{-1} \\ &= [\ell_t^2 \eta_t]^{-n+1} \sum_{d_1} \left(\min_{2 \leq k \leq n} |a_k - d_1|_L^2 + 1 \right)^{-1} \prec [\ell_t^2 \eta_t]^{-n+1}. \end{aligned}$$

On the other hand, the column and row sums of $\Theta_{t|m|^2}^{(B)}$ are $O(\eta_t^{-1})$ by Lemma 2.14, so

$$[\ell_t^2 \eta_t]^{-n+1} \eta_t \sum_{d_1} \left(\Theta_{t|m|^2}^{(B)} \right)_{a_1, d_1} \prec [\ell_t^2 \eta_t]^{-n+1}.$$

To prove (80), we follow the proof of Lemma 3.11 in [50] and first assume that $\sigma_j = \sigma_{j+1}$ for some $j > 1$. (Here, $n+1$ means 1.) By Lemma 2.14, we have

$$\begin{aligned} &(\Theta_{tm_j m_{j+1}}^{(B)})_{a_j d_j} = (\Theta_{tm^2}^{(B)})_{a_j d_j} \prec 1, \\ &\prod_{i \neq 1, j} (\Theta_{tm_i m_{i+1}}^{(B)})_{a_i d_i} \prec [\ell_t^2 \eta_t]^{-n+2}. \end{aligned}$$

By (77), the display (80) then follows (with actually an exponential decay instead of polynomial decay).

We are left with the case $\sigma = \sigma^{(alt)}$. Following the proof of Lemma 3.11 in [50], we write

$$(81) \quad \sum_{d_2, \dots, d_n} \Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}) \cdot \prod_{i=2}^n \left(\Theta_{tm_i m_{i+1}}^{(B)} \right)_{a_i d_i} = \sum_{s_2, \dots, s_n} g(\mathbf{s}) \prod_{i=2}^n f(a_i, s_i)$$

$$(82) \quad = \sum_{s_2, \dots, s_n} g(\mathbf{s}) \prod_{i=2}^n \sum_{\xi_i=0,1,2} f_{\xi_i}(a_i, s_i),$$

where for any $d_1, \mathbf{a}$, we introduced the notation

$$\begin{aligned} \mathbf{s} &= (s_2, \dots, s_n) := (d_2 - d_1, \dots, d_n - d_1) \\ g(\mathbf{s}) &:= \Sigma^{(\emptyset)}(t, \sigma, \mathbf{d}), \\ f(a_i, s_i) &:= \left(\Theta_{t|m|^2}^{(B)} \right)_{a_i, (d_1 + s_i)}, \\ f_0(a_i, s_i) &:= f(a_i, 0) \\ f_1(a_i, s_i) &:= \frac{f(a_i, s_i) - f(a_i, -s_i)}{2} \\ f_2(a_i, s_i) &:= \frac{1}{2} f(a_i, s_i) + \frac{1}{2} f(a_i, -s_i) - f(a_i, 0). \end{aligned}$$

By Lemma 2.14, we have the regularity bounds

$$\begin{aligned} f_0(a_i, s_i) &\prec \ell_t^{-2} \eta_t^{-1} \\ f_1(a_i, s_i) &\prec \frac{1}{|a_i - d_1|_L + 1} + \frac{1}{\ell_t^2 \eta_t^{1/2}} \\ f_2(a_i, s_i) &\prec \frac{1}{|a_i - d_1|_L^2 + 1} + \frac{1}{\ell_t^2}. \end{aligned}$$

We now follow the same reasoning in the proof of Lemma 3.11 in [50] to deduce (80), so (79) follows for $\pi = \emptyset$. The proof for general π follows by an inductive argument given at the end of the proof of Lemma 3.11 in [50], which in dimension two should be combined with the following localized form of (80). The last term on the right-hand side of (80) does not depend on d_1 , so it should not be summed over all $d_1 \in \mathbb{Z}_L^2$ using only a uniform bound on the remaining factors: compared with a sum over a ball of radius ℓ_t , this would cost a factor $(L/\ell_t)^2$. However, for any fixed $\tau, D > 0$, the left-hand side of (80) is $O(W^{-D})$ if $|a_k - d_1|_L \geq \ell_t W^\tau$ for all $2 \leq k \leq n$ (which is only possible if $\ell_t < L$). Indeed, by (77), the contribution of the $\mathbf{d}$ with $\max_i |d_i - d_1|_L \geq \ell_t W^\tau/2$ is $O(W^{-D})$. For the remaining $\mathbf{d}$, we have $|a_i - d_i|_L \geq \ell_t W^\tau/2$ for all $i \geq 2$, so that each factor $(\Theta_{tm_i m_{i+1}}^{(B)})_{a_i d_i}$ is $O(W^{-D})$ by (42), since $\hat{\ell}(tm_i m_{i+1}) \leq C\ell_t$. Hence (80) holds with its last term multiplied by $\mathbf{1}(\min_{2 \leq k \leq n} |a_k - d_1|_L \leq \ell_t W^\tau)$, up to an additive error $O(W^{-D})$. In this form, the last term is supported on $O(W^{2\tau} \ell_t^2)$ vertices d_1 , and the counting in the inductive argument of [50] goes through in dimension two. $\square$

4. Estimates for entries of G . We now collect a set of estimates for G_t ; the proof of these estimates follows that of Lemma 4.1 in [50], which is dimension-independent and based on standard calculations for resolvents as in [26, 27].

LEMMA 4.1. Recall z_t and G_t from Definitions 2.7 and 2.9. Fix any $|E| < 2 - \kappa$ and $0 \leq t < 1$. For any $c > 0$, consider the event

$$(83) \quad \Omega(t, c) := \{\|G_t - m\|_{\max} \leq W^{-c}\}.$$

Then we have the following estimate for G_t entries in terms of loops of length 2:

$$(84) \quad \mathbf{1}_{\Omega(t, c)} \cdot \max_{\substack{x \in \mathcal{I}_a^{(2)}, y \in \mathcal{I}_b^{(2)} \\ x \neq y}} |(G_t)_{xy}|^2 \prec \sum_{a': |a' - a|_L \leq 1} \sum_{b': |b' - b|_L \leq 1} \mathcal{L}_{t, (+, -), (b', a')} + W^{-2} \cdot \mathbf{1}(|a - b|_L \leq 1),$$

$$(85) \quad \mathbf{1}_{\Omega(t, c)} \cdot \max_x |(G_t)_{xx} - m|^2 \prec \max_{a, b} \mathcal{L}_{t, (+, -), (a, b)}.$$

Moreover, suppose that the following is true for some $c > 0$:

$$(86) \quad \|G_t - m\|_{\max} \prec W^{-c}.$$

Then (84) and (85) hold without the indicator $\mathbf{1}_{\Omega(t, c)}$, and we also have

$$(87) \quad \max_a |\langle (G_t - m) E_a \rangle| \prec \max_{a, b} \mathcal{L}_{t, (+, -), (a, b)}.$$

5. Analysis of loop hierarchy. In this section, we prove Theorem 2.20. The argument largely follows that in Section 5 of [50], except for Section 5.5, which requires the main new inputs of this paper. However, even for the estimates before Section 5.5, there are a lot of technical power-countings that are dimension-dependent. (For example, the E_a matrices in (7) have a factor of W^2 with exponent matching the dimension. Also, the ball of radius ℓ in $\mathbb{Z}_L^2$ has size $O(\ell^2)$. These changes are reflected in the definition of M_t from (49), and it is important to keep track of them carefully.) For this reason, unless the argument is identical to the corresponding argument in [50] at the level of power-counting, we will provide details for proofs.

5.1. Proof of Theorem 2.20: Step 1.

PROOF. We recall that the goal of this first step is to prove (62) and (63). First, we use the assumption $t \leq 1 - N^{-1+\tau}$ in Theorem 2.20 and the definitions $\ell_t = \ell(z_t)$ and $\eta_t = \text{Im} z_t$. This gives

$$1 - u \gg N^{-1}, \quad \eta_u \gg N^{-1}, \quad u \leq t.$$

Given that $\partial_z(H - z)^{-1} = (H - z)^{-2}$ and that the entries of H_t are jointly Gaussian, we deduce that for any $C > 0$, there exists C' such that the following is true outside of events whose probabilities are exponentially small in N :

$$(88) \quad \max_{u: 1-u \geq N^{-1}} \max_{|u-u'| \leq N^{-C'}} \|G_u - G_{u'}\|_{\max} \leq N^{-C}.$$

Hence, by a standard net argument, it suffices to prove (62) and (63) for $u = t$. This continuity argument will be used repeatedly in this paper.

5.1.0.1. Case 1: $s < t \leq 1/2$. We have $\|G_t\|_{op} \leq 1/\eta_t = O(1)$. We also have $\|E_a\|_{op} \leq W^{-2}$ for all $a \in \mathbb{Z}_L^2$. Thus,

$$(89) \quad \mathcal{L}_{t, \sigma, \mathbf{a}} = \left\langle \prod_{i=1}^n G_t(\sigma_i) E_{a_i} \right\rangle = O(W^{-2(n-1)}), \quad t \leq 1/2.$$

Since $M_t = W^2 \ell_t^2 \eta_t = O(W^2)$ for $t \leq 1/2$, the bound (89) implies (62). Next, we combine (89) with (84) and (85), and we note that ℓ_t, η_t are order 1 for $t \leq 1/2$. This gives

$$\mathbf{1} \left(\|G_t - m\|_{\max} \leq W^{-1/10} \right) \cdot \|G_t - m\|_{\max} \prec W^{-1}, \quad t \leq 1/2.$$

By (88) and the previous display, we deduce that $\|G_u - m\|_{\max} \prec W^{-1}$ implies $\|G_{u'} - m\|_{\max} \prec W^{-1}$ for any $|u' - u| \leq N^{-C'}$ if $C' > 0$ is large enough. Thus, as soon as we can find a single time $u_0 \in [s, t]$, where $t \leq 1/2$, such that $\|G_{u_0} - m\|_{\max} \prec W^{-1}$, we would deduce this bound for all $u \in [s, t]$. By (60), we can choose $u_0 = s$, and thus (63) holds, i.e.

$$(90) \quad \|G_u - m\|_{\max} \prec W^{-1}, \quad u \leq 1/2.$$

5.1.0.2. *Case 2: $t > s \geq 1/2$.* By (49) from Lemma 2.15, we have $\mathcal{K}_{s, \sigma, \mathbf{a}} \prec M_s^{-n+1}$, where we recall $M_s = W^2 \ell_s^2 \eta_s$. If we combine this with the assumption (58), then we get that for any fixed n and s ,

$$(91) \quad \mathcal{L}_{s, \sigma, \mathbf{a}} \prec M_s^{-n+1}.$$

We now record the following estimate, whose proof follows from the same argument as in Section 6 of [50]. An outline of the proof is given in Section 6.

LEMMA 5.1. *Suppose that $c < t_1 \leq t_2 \leq 1$ for some constant $c > 0$. Assume that for any fixed $n \geq 1$, the following bounds hold at time t_1 :*

$$(92) \quad \max_{\sigma, \mathbf{a}} \mathcal{L}_{t_1, \sigma, \mathbf{a}} \prec M_{t_1}^{-n+1}, \quad M_t := W^2 \ell_t^2 \eta_t.$$

Next, we let Ω denote the following event:

$$(93) \quad \Omega := \{\|G_{t_2}\|_{\max} \leq 2\}.$$

Then, for any fixed $n \geq 1$, we have

$$(94) \quad \mathbf{1}_\Omega \cdot \max_{\sigma, \mathbf{a}} \mathcal{L}_{t_2, \sigma, \mathbf{a}} \prec (W^2 \ell_{t_1}^2 \eta_{t_1})^{-n+1} = \left(\frac{\ell_{t_2}}{\ell_{t_1}} \right)^{2(n-1)} \cdot M_{t_2}^{-n+1}.$$

By (89) and (91), the a priori bound (92) for $\mathcal{L}_s$ holds. Thus, we can use Lemma 5.1 to obtain

$$(95) \quad \mathbf{1}(\|G_u\|_{\max} \leq 2) \cdot \max_{\sigma, \mathbf{a}} \mathcal{L}_{u, \sigma, \mathbf{a}} \prec \left(\frac{\ell_u}{\ell_s} \right)^{2(n-1)} \cdot M_u^{-n+1}, \quad s \leq u \leq t.$$

Now, we use (95) for $n = 2$ with Lemma 4.1 gives us

$$\mathbf{1} \left(\|G_u - m\|_{\max} \leq M_u^{-1/6} \right) \cdot \|G_u - m\|_{\max} \prec \frac{\ell_u}{\ell_s} \cdot M_u^{-1/2}, \quad s \leq u \leq t.$$

By (61), we know $\ell_u \ell_s^{-1} \leq M_s^{1/60}$. Combining this with the previous display, we deduce that for any $D > 0$, we have the following for N large enough:

$$(96) \quad \mathbb{P} \left(M_u^{-1/4} \leq \|G_u - m\|_{\max} \leq M_u^{-1/6} \right) \leq N^{-D}, \quad s \leq u \leq t.$$

Another standard continuity argument (e.g. (88)) allows us to insert either a maximum or minimum over $u \in [s, t]$ in the event above.

Now, by assumption, we have the initial data bound $\|G_s - m\|_{\max} \prec M_s^{-1/2}$. Together with the previous display and continuity of G_u in u , we deduce $\|G_t - m\|_{\max} \prec M_t^{-1/4}$. Combining this bound with (95) gives (62) and (63) for $t > s \geq 1/2$.

5.1.0.3. *Case 3: $t > 1/2 > s$.* From Case 1, we know that (89) and (90) hold for $t = 1/2$. The proof of Case 2 uses only these two conditions. Thus, the current case follows as a consequence of Case 2 with $s = 1/2$. $\square$

5.2. *Dynamics of $\mathcal{L} - \mathcal{K}$.* Recall the loop hierarchy from Lemma 2.11. We now introduce the notation

$$\mathcal{L}_{t, \mathcal{G}_{k,l}^{(a),L}(\sigma, \mathbf{a})} := \left(\mathcal{G}_{k,l}^{(a),L} \circ \mathcal{L}_{t, \sigma, \mathbf{a}} \right), \quad \mathcal{L}_{t, \mathcal{G}_{k,l}^{(b),R}(\sigma, \mathbf{a})} := \left(\mathcal{G}_{k,l}^{(b),R} \circ \mathcal{L}_{t, \sigma, \mathbf{a}} \right),$$

which allows us write the hierarchy in a more convenient form that we describe below. By (5.15) in [50], we have the following equation, which uses notation to be clarified afterwards:

$$(97) \quad d(\mathcal{L} - \mathcal{K})_{t, \sigma, \mathbf{a}} = \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}=2} dt + \sum_{l_{\mathcal{K}} > 2} \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}} dt + \mathcal{E}_{t, \sigma, \mathbf{a}}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} + \mathcal{E}_{t, \sigma, \mathbf{a}}^{(M)} + \mathcal{E}_{t, \sigma, \mathbf{a}}^{(\tilde{G})}.$$

Above, the last two terms are introduced in Lemma 2.11. The first two terms are defined by the formula

$$(98) \quad W^2 \cdot \sum_{1 \leq k < l \leq n} \sum_{a,b} \left((\mathcal{L} - \mathcal{K})_{t, \mathcal{G}_{k,l}^{(a),L}(\sigma, \mathbf{a})} \cdot S_{ab}^{(B)} \cdot \mathcal{K}_{t, \mathcal{G}_{k,l}^{(b),R}(\sigma, \mathbf{a})} \cdot \mathbf{1}(\text{length of } \mathcal{K} \text{ equals } l_{\mathcal{K}}) \right. \\ \left. + (\mathcal{K} \iff \mathcal{L} - \mathcal{K}) \right) := \sum_{l_{\mathcal{K}}=2}^n \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}},$$

here, $\mathcal{K} \iff \mathcal{L} - \mathcal{K}$ denotes the terms obtained by swapping $\mathcal{K}$ and $\mathcal{L} - \mathcal{K}$. Next, we introduce

$$(99) \quad \mathcal{E}_{t, \sigma, \mathbf{a}}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} := W^2 \cdot \sum_{1 \leq k < l \leq n} \sum_{a,b} (\mathcal{L} - \mathcal{K})_{t, \mathcal{G}_{k,l}^{(a),L}(\sigma, \mathbf{a})} \cdot S_{ab}^{(B)} \cdot (\mathcal{L} - \mathcal{K})_{t, \mathcal{G}_{k,l}^{(b),R}(\sigma, \mathbf{a})} dt.$$

We will turn (97) into an integral equation. First, we introduce the main objects in this integral equation construction.

DEFINITION 5.2. Define the linear map $\Theta_{t, \sigma}$ via the following action on a tensor $\mathcal{A}$: $\mathbb{Z}_L^n \rightarrow \mathbb{C}$:

$$(100) \quad (\Theta_{t, \sigma} \circ \mathcal{A})_{\mathbf{a}} = \sum_{i=1}^n \sum_{b_i} \left(\frac{m_i m_{i+1} S^{(B)}}{1 - t m_i m_{i+1} S^{(B)}} \right)_{a_i b_i} \cdot \mathcal{A}_{\mathbf{a}^{(i)}}, \quad m_i = m(\sigma_i) \\ \mathbf{a}^{(i)} = (a_1, \dots, a_{i-1}, b_i, a_{i+1}, \dots, a_n).$$

We also define the semigroup operator $\mathcal{U}_{s,t, \sigma}$ associated to $\Theta_{t, \sigma}$ as follows

$$(101) \quad (\mathcal{U}_{s,t, \sigma} \circ \mathcal{A})_{\mathbf{a}} = \sum_{b_1, \dots, b_n} \prod_{i=1}^n \left(\frac{1 - s \cdot m_i m_{i+1} S^{(B)}}{1 - t \cdot m_i m_{i+1} S^{(B)}} \right)_{a_i b_i} \cdot \mathcal{A}_{\mathbf{b}}, \quad \mathbf{b} = (b_1, \dots, b_n), \quad m_i = m(\sigma_i)$$

Note that for any $\xi \in \mathbb{C}$, we have the following identity:

$$(102) \quad \left(\frac{1 - s \xi S^{(B)}}{1 - t \xi S^{(B)}} \right) = I - (s - t) \xi \cdot \Theta_{t\xi}^{(B)} \cdot S^{(B)}.$$

By the explicit formula (47) for $\mathcal{K}$ of rank 2, the first term on the right-hand side of (97) can be rewritten as $(\Theta_{t, \sigma} \circ (\mathcal{L} - \mathcal{K}))_{\mathbf{a}} = [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{t, \sigma, \mathbf{a}}^{l_{\mathcal{K}}=2}$. Using this and the Duhamel principle, we have the following.

LEMMA 5.3 (Integrated loop hierarchy). *For any $t \geq 0$, we have the SDE*

(103)

$$d(\mathcal{L} - \mathcal{K})_{t,\sigma,a} = (\Theta_{t,\sigma} \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\cdot})_a dt + \sum_{l_{\mathcal{K}} > 2} \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{t,\sigma,a}^{l_{\mathcal{K}}} dt + \mathcal{E}_{t,\sigma,a}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} + \mathcal{E}_{t,\sigma,a}^{(M)} + \mathcal{E}_{t,\sigma,a}^{(\tilde{G})}.$$

For any $s < t$, we have the following “integrated” loop hierarchy:

$$\begin{aligned} (\mathcal{L} - \mathcal{K})_{t,\sigma,a} &= (\mathcal{U}_{s,t,\sigma} \circ (\mathcal{L} - \mathcal{K})_{s,\sigma})_a \\ &\quad + \sum_{l_{\mathcal{K}} > 2} \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{u,\sigma}^{l_{\mathcal{K}}} \right)_a du \\ &\quad + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \right)_a du \\ &\quad + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_a du + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_a du \end{aligned} \quad (104)$$

Furthermore, let $T \geq s$ be a stopping time with respect to the matrix Brownian motion H_t and set $\tau := T \wedge t$. We have the stopped integrated loop hierarchy below for any $t \geq s$:

$$\begin{aligned} (\mathcal{L} - \mathcal{K})_{\tau,\sigma,a} &= (\mathcal{U}_{s,\tau,\sigma} \circ (\mathcal{L} - \mathcal{K})_{s,\sigma})_a \\ &\quad + \sum_{l_{\mathcal{K}} > 2} \int_s^{\tau} \left(\mathcal{U}_{u,\tau,\sigma} \circ \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{u,\sigma}^{l_{\mathcal{K}}} \right)_a du \\ &\quad + \int_s^{\tau} \left(\mathcal{U}_{u,\tau,\sigma} \circ \mathcal{E}_{u,\sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \right)_a du \\ &\quad + \int_s^{\tau} \left(\mathcal{U}_{u,\tau,\sigma} \circ \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_a du + \int_s^{\tau} \left(\mathcal{U}_{u,\tau,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_a du. \end{aligned} \quad (105)$$

The last term in (105) has the following meaning. Since $\mathcal{U}$ is deterministic, the object obtained by replacing $\mathcal{U}_{u,\tau,\sigma}$ by $\mathcal{U}_{u,t,\sigma}$ is well-defined, and this object is continuous in t , so we can evaluate it at time $t = \tau$. Of course, one must be careful in using Itô calculus to estimate it.

We now introduce notation that is necessary to compute the quadratic variation of the martingale.

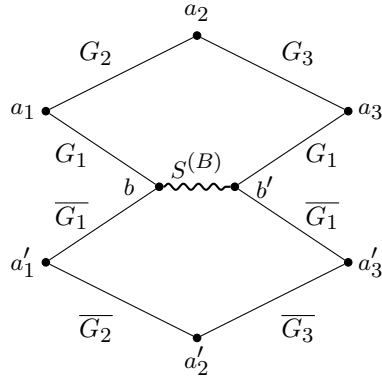
DEFINITION 5.4 (Definition of $\mathcal{E} \otimes \mathcal{E}$). Denote

$$\begin{aligned} (\mathcal{E} \otimes \mathcal{E})_{t,\sigma,\mathbf{a},\mathbf{a}'} &:= \sum_{k=1}^n (\mathcal{E} \otimes \mathcal{E})_{t,\sigma,\mathbf{a},\mathbf{a}'}^{(k)} \\ (106) \quad (\mathcal{E} \otimes \mathcal{E})_{t,\sigma,\mathbf{a},\mathbf{a}'}^{(k)} &:= W^2 \sum_{b,b'} S_{b,b'}^{(B)} \mathcal{L}_{t,\sigma^{(k)},\mathbf{a}^{(k)}} \quad \sigma^{(k)} \in \{+, -\}^{2n+2}, \end{aligned}$$

where $\mathcal{L}_{t,\sigma^{(k)},\mathbf{a}^{(k)}}$ is obtained by cutting the k -th edge of $\mathcal{L}_{t,\sigma,\mathbf{a}}$ and then attaching itself (with indices $\mathbf{a}$) to its complex conjugate loop (with indices $\mathbf{a}'$) into a bigger loop that is labeled by indices b and b' , so that

$$\begin{aligned} \mathbf{a}^{(k)} &= (a_k, a_{k+1}, \dots, a_n, a_1, \dots, a_{k-1}, b', a'_{k-1}, \dots, a'_1, a'_n, \dots, a'_k, b) \\ (107) \quad \sigma^{(k)} &= (\sigma_k, \sigma_{k+1}, \dots, \sigma_n, \sigma_1, \dots, \sigma_k, \bar{\sigma}_k, \dots, \bar{\sigma}_1, \bar{\sigma}_n, \dots, \bar{\sigma}_k) \end{aligned}$$

The symbol $\otimes$ is used to emphasize the symmetric structure of $\mathcal{E} \otimes \mathcal{E}$; it is not a tensor product.



$$k = 1$$

$$\mathbf{a}^{(1)} = (a_1, a_2, a_3, b', a'_3, a'_2, a'_1, b);$$

$$\boldsymbol{\sigma}^{(1)} = (\sigma_1, \sigma_2, \sigma_3, \sigma_1, \overline{\sigma_1}, \overline{\sigma_3}, \overline{\sigma_2}, \overline{\sigma_1}).$$

FIG 4. Example of the 8-G loop in $\mathcal{E} \otimes \mathcal{E}$: for $\sigma \in \{+, -\}^3$. Taken from [50].

We now estimate the martingale term in (105) via Lemma 5.5 in [50], which we record below.

LEMMA 5.5. *For any stopping time T with respect to the filtration generated by H_t we have*

$$(108) \quad \mathbb{E} \left[\int_s^\tau \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_a \right]^{2p} \leq C_{n,p} \mathbb{E} \left(\int_s^\tau \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\overline{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma} \right)_{a,a} du \right)^p$$

where $\tau := t \wedge T$, where $\overline{\sigma}$ is the conjugate of σ , and where

$$[(\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\overline{\sigma}}) \circ \mathcal{A}]_{a,a'} = \sum_{b,b'} \prod_{i=1}^n \left(\frac{1-u \cdot m_i m_{i+1} S^{(B)}}{1-t \cdot m_i m_{i+1} S^{(B)}} \right)_{a_i b_i} \cdot \prod_{i=1}^n \left(\frac{1-u \cdot \overline{m_i} \overline{m_{i+1}} S^{(B)}}{1-t \cdot \overline{m_i} \overline{m_{i+1}} S^{(B)}} \right)_{a'_i b'_i} \cdot \mathcal{A}_{b,b'}.$$

5.3. *Proof of Theorem 2.20, Step 2.* In this subsection, we will fix $\sigma = (+, -)$ and thus drop it from the notation. We prove (65) first; (64) will be proved at the end of this section. We start by defining the following exponential decay profiles:

$$(109) \quad \mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell) := \max_{a,b: |a-b| \geq \ell} |(\mathcal{L}-\mathcal{K})_{u,\sigma,(a,b)}|, \quad \sigma = (+, -),$$

$$(110) \quad \mathcal{T}_{u,D}(\ell) := M_u^{-2} \exp \left(-(\ell/\ell_u)_+^{1/2} \right) + W^{-D}.$$

Both $\mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}$ and $\mathcal{T}_{u,D}$ are non-increasing in the parameter ℓ , so for any $0 \leq \ell_1 \leq \ell_2$, we have $\mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell_1) \geq \mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell_2)$ and $\mathcal{T}_{u,D}(\ell_1) \geq \mathcal{T}_{u,D}(\ell_2)$. We will denote the ratio between (109) and (110) by

$$(111) \quad \mathcal{J}_{u,D}(\ell) := \left(\mathcal{T}_u^{(\mathcal{L}-\mathcal{K})}(\ell) / \mathcal{T}_{u,D}(\ell) \right) + 1$$

Our goal is to control $\mathcal{J}_{u,D}(\ell)$ for any $u \in [s, t]$ and any large $D > 0$. In particular, we aim to show

$$(112) \quad \mathcal{J}_{u,D}^* := \max_{\ell} \mathcal{J}_{u,D}(\ell) \prec (\eta_s/\eta_u)^4$$

In this subsection, we will often use the following slightly enlarged length-scale:

$$\ell_t^* := (\log W)^{3/2} \cdot \ell_t.$$

In this scale ℓ_t^* , Θ_t is exponentially small, but $\mathcal{T}_t$ is not. More precisely, we have the following lemma, which is Lemma 5.6 in [50]. (The proof is identical to that of Lemma 5.6 in [50], so we omit it. In any case, the first and third estimates below are elementary, and the last follows by decay of $\mathcal{K}$, which follows from the explicit formula (47), decay of Θ from (42), and decay of $\mathcal{L} - \mathcal{K}$.)

LEMMA 5.6. *For any fixed $D, \delta > 0$, we have*

$$(113) \quad |b - a|_L \geq \delta \cdot \ell_t^* \implies (\Theta_t)_{ab} \leq W^{-D}, \quad (\Theta_s^{-1} \Theta_t)_{ab} \leq W^{-D}$$

$$(114) \quad |b - a|_L \geq \delta \cdot \ell_t^* \implies \mathcal{L}_{t,(+,-),(a,b)} \prec \mathcal{J}_{t,D}^* \cdot \mathcal{T}_{t,D}(|a - b|_L).$$

Moreover, for any $C = O(1)$, we have

$$(115) \quad \mathcal{T}_{u,D}(\ell - C \cdot \ell_u^*) \prec \mathcal{T}_{u,D}(\ell).$$

Our proof of (65) relies on the loop hierarchy (105) for $n = 2$. We begin by bounding terms in (105).

LEMMA 5.7. *Suppose the assumptions of Theorem 2.20 and the conclusion of Step 1, i.e., (62) and (63), hold. Assume that the following are satisfied, in which $\mathbf{a} = (a_1, a_2)$ and $\mathbf{a}' = (a'_1, a'_2)$ and $\sigma = (+, -)$:*

$$(116) \quad s \leq u \leq t, \quad D \geq D_0, \quad \max_i |a_i - a'_i|_L \leq \ell_t^*.$$

Here $D_0 = D_0(\mathbf{c})$ is a sufficiently large constant (in the proof, the term W^{-D} in $\mathcal{T}_{u,D}$ is summed over at most $N^{O(1)} \leq W^{O(1/\epsilon)}$ indices). Then we have

$$(117) \quad \mathcal{E}_{u,\sigma,\mathbf{a}}^{((L-K) \times (L-K))} / \mathcal{T}_{t,D}(|a_1 - a_2|_L) \prec (\eta_u)^{-1} \cdot M_u^{-1} \cdot (\mathcal{J}_{u,D}^*)^2$$

$$(118) \quad \mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} / \mathcal{T}_{t,D}(|a_1 - a_2|_L) \prec (\eta_u)^{-1} \cdot (\ell_u/\ell_s)^4 \cdot \mathbf{1}(|a_1 - a_2|_L \leq \ell_u^*)$$

$$+ (\eta_u)^{-1} \cdot \left(M_u^{-1/2} \cdot (\mathcal{J}_{u,D}^*)^2 + M_u^{-3/2} \cdot (\mathcal{J}_{u,D}^*)^3 \right)$$

$$(119) \quad (\mathcal{E} \otimes \mathcal{E})_{u,\sigma,\mathbf{a},\mathbf{a}'} / (\mathcal{T}_{t,D}(|a_1 - a_2|_L))^2 \prec (\eta_u)^{-1} \cdot (\ell_u/\ell_s)^{10} \cdot \mathbf{1}(|a_1 - a_2|_L \leq 4\ell_t^*)$$

$$+ (\eta_u)^{-1} \cdot (\ell_u/\ell_s)^3 \cdot M_u^{-1/2} \cdot (\mathcal{J}_{u,D}^*)^2$$

$$+ (\eta_t)^{-1} \cdot M_t^{-1} \cdot (\mathcal{J}_{u,D}^*)^3$$

Assuming Lemma 5.7, we now prove (65). We will extensively use the estimates for $\mathcal{U}$ from Section 7.

PROOF OF (65). By assumption (59) on $(\mathcal{L} - \mathcal{K})_s$, the operator norm bound on $\mathcal{U}$ in Lemma 7.1, and the tail estimate (199), we can bound the first term on the right side of (105) by

$$(120) \quad (\mathcal{U}_{s,t,\sigma} \circ (\mathcal{L} - \mathcal{K})_{s,\sigma})_{\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|_L) \prec (\ell_t/\ell_s)^4 \cdot \mathbf{1}(|a_1 - a_2|_L \leq \ell_t^*) + 1,$$

where the last term of order one comes from applying (199). The factor $(\eta_s/\eta_t)^2$ from applying Lemma 7.1 becomes $(\ell_t/\ell_s)^4$ due to the prefactors in $\mathcal{T}_{s,D}$ and $\mathcal{T}_{t,D}$. For any $\mathbf{a}$ fixed and any function f , we decompose $f = f(\mathbf{b})\mathbf{1}(\|\mathbf{b} - \mathbf{a}\| \leq \ell_t^*) + f_2$. From the decay of $\mathcal{U}_{u,t}$, we

know that $(\mathcal{U}_{u,t,\sigma} \circ f_2)_a$ is exponentially small. Hence we only have to bound $\mathcal{U}_{u,t,\sigma} \circ f_1$, for which we apply Lemma 7.1. Ultimately, for the third term of (105), we have the following estimate, in which $\|\mathbf{b} - \mathbf{a}\| := \max_i |b_i - a_i|_L$:

$$\left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{((L-K) \times (L-K))} \right)_a \prec (\eta_u/\eta_t)^2 \max_{\|\mathbf{b}-\mathbf{a}\| \leq \ell_t^*} \mathcal{E}_{u,\sigma,\mathbf{b}}^{((L-K) \times (L-K))} + W^{-D}.$$

Because we restrict to $\|\mathbf{b} - \mathbf{a}\| \leq \ell_t^*$, (115) implies $\mathcal{T}_{t,D}(|b_1 - b_2|_L) \prec \mathcal{T}_{t,D}(|a_1 - a_2|_L)$. Thus, to control the right-hand side of the previous estimate, we can use the estimate (117) and closeness of s, t in (61) to get

$$(121) \quad \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{((L-K) \times (L-K))} \right)_a / \mathcal{T}_{t,D}(|a_1 - a_2|_L) \prec \frac{1}{\eta_u} (\eta_u/\eta_t)^2 \cdot M_u^{-1} \cdot (\mathcal{J}_{u,D}^*)^2$$

Similarly, the estimate (118) on $\mathcal{E}^{(\tilde{G})}$ implies

$$(122) \quad \begin{aligned} \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_a / \mathcal{T}_{t,D}(|a_1 - a_2|_L) &\prec \mathbf{1}(|a_1 - a_2|_L \leq 3\ell_t^*) \cdot \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^2 \cdot (\ell_u/\ell_s)^4 \\ &+ \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^2 \cdot M_u^{-1/2} \cdot (\mathcal{J}_{u,D}^*)^2 \\ &+ \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^2 \cdot M_u^{-3/2} \cdot (\mathcal{J}_{u,D}^*)^3, \end{aligned}$$

and the estimate (119) on $\mathcal{E} \otimes \mathcal{E}$ implies

$$(123) \quad \begin{aligned} \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma,\bar{\sigma}} \right)_{a,a} / (\mathcal{T}_{t,D}(|a_1 - a_2|_L))^2 &\prec \mathbf{1}(|a_1 - a_2|_L \leq 6\ell_t^*) \cdot \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^4 \cdot (\ell_u/\ell_s)^{10} \\ &+ (\eta_u/\eta_t)^4 \cdot \frac{1}{\eta_u} \cdot (\ell_u/\ell_s)^3 \cdot M_u^{-1/2} \cdot (\mathcal{J}_{u,D}^*)^2 \\ &+ (\eta_u/\eta_t)^4 \cdot \frac{1}{\eta_t} \cdot M_t^{-1} \cdot (\mathcal{J}_{u,D}^*)^3. \end{aligned}$$

We now insert these bounds into the stopped equation (105) with $\tau := T \wedge t$, where, for a fixed and sufficiently small $\delta > 0$,

$$(124) \quad T := \min\{u \geq s : \mathcal{J}_{u,D}^* \geq N^\delta (\eta_s/\eta_t)^4\}$$

(If we divide each estimate in Lemma 5.7 by its right-hand side, the resulting estimates all hold with probability $1 - O(N^{-D})$ for any $D > 0$ simultaneously for all $u \in [s, t]$. This follows by another continuity argument.) With this stopping time, the quadratic variation of the martingale in (105) is bounded as follows:

$$(125) \quad \begin{aligned} &\int_s^\tau \left((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u,\sigma,\bar{\sigma}} \right)_{a,a} du / \mathcal{T}_{t,D}(|a_1 - a_2|_L)^2 \\ &\prec \int_s^\tau du \left\{ \mathbf{1}(|a_1 - a_2|_L \leq 6\ell_t^*) \cdot \frac{1}{\eta_u} \cdot (\eta_u/\eta_t)^4 \cdot (\ell_u/\ell_s)^{10} \right. \\ &\quad \left. + (\eta_u/\eta_t)^4 \cdot \frac{1}{\eta_u} \cdot (\ell_u/\ell_s)^3 \cdot M_u^{-1/2} \cdot (\mathcal{J}_{u,D}^*)^2 + (\eta_u/\eta_t)^4 \cdot \frac{1}{\eta_t} \cdot M_t^{-1} \cdot (\mathcal{J}_{u,D}^*)^3 \right\} \\ &\leq C[(\eta_s/\eta_t)^5 \cdot \mathbf{1}(|a_1 - a_2|_L \leq 6\ell_t^*) + 1] \end{aligned}$$

To see the last inequality, set $r := \eta_s/\eta_t$, and recall that $\eta_u = (1 - u)\text{Imm}$ and that M_u is comparable to $W^2 \min(1, L^2(1 - u))$. Hence $(\ell_u/\ell_s)^2 \leq \eta_s/\eta_u$ and $M_s/M_u \leq \eta_s/\eta_u \leq$

r for $u \in [s, t]$, so that (61) gives $M_u \geq M_t \geq M_s/r \geq r^{29}$. Moreover, $M_t \geq N^{c_0}$ for some $c_0 > 0$, by the assumption $t \leq 1 - N^{-1+\tau}$ and (1). For the first term, we use $\int_s^t \eta_u^{-1}(\eta_u/\eta_t)^4(\eta_s/\eta_u)^5 du = r^4(r-1)/\text{Imm}$. For the remaining terms, we use $\mathcal{J}_{u,D}^* \leq N^\delta r^4$ for $s \leq u < \tau$, which holds by definition of T , together with $\int_s^t \eta_u^{-1}(\eta_u/\eta_t)^4(\eta_s/\eta_u)^{3/2} du \leq Cr^4$ and $\int_s^t \eta_t^{-1}(\eta_u/\eta_t)^4 du \leq Cr^5$. This bounds the remaining terms by $CN^{2\delta}r^{12}M_t^{-1/2} + CN^{3\delta}r^{17}M_t^{-1} \leq CN^{3\delta}M_t^{-5/58}$, which is at most 1 if $\delta > 0$ is small enough (depending on c_0). Now, by Lemma 5.5 and the previous display, we deduce

$$\int_s^\tau \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|_L) \prec (\eta_s/\eta_t)^{5/2} \mathbf{1}(|a_1 - a_2|_L \leq 6\ell_t^*) + 1.$$

Because this bound holds at the level of stochastic domination, a union bound and a net argument lets us replace t on the left-hand side by a supremum over $t' \in [s, t]$. In particular, we can evaluate it at $t' = \tau$. (We emphasize that the stochastic domination nature of the previous estimate is crucial for this argument.) Combining this bound with (120), (121), (122) and (105), we have

$$(\mathcal{L} - \mathcal{K})_{\tau,\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|_L) \prec [(\eta_s/\eta_t)^{5/2} \cdot \mathbf{1}(|a_1 - a_2|_L \leq 6\ell_t^*) + 1],$$

where we used $\mathcal{J}_{s,D}^* \prec 1$ from (58) and (59); in particular, because of the factor N^δ in the definition of T , we have $T > s$ with probability $1 - O(N^{-D})$. (The integrals over $u \in [s, \tau]$ of the bounds (121) and (122) are estimated as in the proof of (125): the term with the indicator is $O_\prec((\eta_s/\eta_t)^2)$, since $\int_s^t \eta_u^{-1}(\eta_u/\eta_t)^2(\eta_s/\eta_u)^2 du = (\eta_s/\eta_t)^2 \log(\eta_s/\eta_t)/\text{Imm}$, and the remaining terms are bounded by $CN^{3\delta}M_t^{-5/58} \leq 1$.) By another continuity argument, the above estimate holds for all t satisfying the constraint (61) simultaneously with probability $1 - O(N^{-D})$ for any $D > 0$. In particular, it holds even at the random time $t = \tau$, which implies

$$(126) \quad \mathcal{J}_{\tau,D}^* \prec (\eta_s/\eta_t)^{5/2}$$

Since $(\eta_s/\eta_t)^{5/2} \leq N^{-\delta/2} \cdot N^\delta(\eta_s/\eta_t)^4$, we deduce $\mathbb{P}(T \leq t) \leq N^{-D}$ for any large $D > 0$, and (65) follows. $\square$

We note that the previous argument gives

$$(127) \quad (\mathcal{L} - \mathcal{K})_{t,\mathbf{a}} / \mathcal{T}_{t,D}(|a_1 - a_2|_L) \prec [(\eta_s/\eta_t)^{5/2} \cdot \mathbf{1}(|a_1 - a_2|_L \leq 6\ell_t^*) + 1].$$

PROOF OF LEMMA 5.7. This argument is essentially the same as the proof of Lemma 5.7 in [50], but we give the proof here because it is fairly long and technical, and because there are delicate power-countings that are dimension-dependent (as discussed at the beginning of this section).

Proof of (117). Recall the monotonicity property $u \leq t \Rightarrow \ell_u \leq \ell_t$, which itself implies $u \leq t \Rightarrow \mathcal{T}_{u,D} \leq \mathcal{T}_{t,D}$. By definition, we have

$$(128) \quad \mathcal{E}_{u,\sigma,\mathbf{a}}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} = W^2 \sum_{b_1, b_2} (\mathcal{L} - \mathcal{K})_{u,\sigma,(a_1,b_1)} S_{b_1,b_2}^{(B)} (\mathcal{L} - \mathcal{K})_{u,\sigma,(b_2,a_2)}$$

By definition of $S^{(B)}$, we can restrict the sum to $|b_1 - b_2|_L \leq 1$. Next, we use the elementary inequalities:

$$\int_0^a \exp\left(-\sqrt{(a-x)} - \sqrt{x} + \sqrt{a}\right) dx \leq C \approx 6.12 \quad \text{and} \quad \int_0^\infty \exp(-\sqrt{x}) dx = 2.$$

This gives $\sum_x \mathcal{T}_{u,D}(|a_1 - x|_L) \mathcal{T}_{u,D}(|a_2 - x|_L) \prec M_u^{-2} \ell_u^2 \mathcal{T}_{u,D}(|a_1 - a_2|_L)$, where the factor of M_u^{-2} comes from counting M_u factors on each side of this identity using (110), and the factor of ℓ_u^2 comes from re-scaling in ℓ of ℓ_u in (110). Therefore, we have the following, which implies (117):

$$(129) \quad \begin{aligned} \mathcal{E}_{u,\sigma,\mathbf{a}}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} &\prec W^2 (\mathcal{J}_{u,D}^*)^2 \cdot \sum_x \mathcal{T}_{u,D}(|a_1 - x|_L) \cdot \mathcal{T}_{u,D}(|a_2 - x|_L) \\ &\prec (\mathcal{J}_{u,D}^*)^2 \cdot \eta_u^{-1} \cdot M_u^{-1} \cdot \mathcal{T}_{u,D}(|a_1 - a_2|_L). \end{aligned}$$

Proof of (118). By definition, we have

$$(130) \quad \max_{\mathbf{a}} \mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} \prec W^2 \sum_{b_1, b_2} \langle \tilde{G}_u E_{b_1} \rangle \cdot S_{b_1, b_2}^{(B)} \cdot \mathcal{L}_{u,(-,+,+), (a_1, b_2, a_2)} + c.c., \quad \tilde{G} = G - m$$

Recall $\Xi_{t,m}^{(\mathcal{L})} := \max_{\sigma, \mathbf{a}} |\mathcal{L}_{t,\sigma,\mathbf{a}}| \cdot M_t^{m-1} \cdot \mathbf{1}(\sigma \in \{+, -\}^m)$. By (87), (109), (110), (111), and (112), we have

$$\langle \tilde{G}_u E_{b_1} \rangle \prec \Xi_{u,2}^{(\mathcal{L})} \cdot M_u^{-1} \prec M_u^{-1} + \mathcal{J}_{u,D}^* \cdot M_u^{-2}.$$

Therefore

$$(131) \quad \max_{\mathbf{a}} \mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} \prec \frac{1}{\ell_u^2 \eta_u} \cdot \sum_b |\mathcal{L}_{u,(-,+,+), (a_1, b, a_2)}| (1 + \mathcal{J}_{u,D}^* \cdot M_u^{-1})$$

Now, let us first restrict to the case $|a_1 - a_2|_L \leq \ell_u^*$. Define

$$\ell_u^{**} := \ell_u \cdot (\log W)^3 = \ell_u^* \cdot (\log W)^{3/2}$$

Note that $\mathcal{T}_{u,D}(\ell_u^{**}) \prec W^{-D}$. Now, consider b in (131) such that $|b - a_1|_L \leq \ell_u^{**}$; there are $\prec (\ell_u^{**})^2 \prec \ell_u^2$ many such indices since we are in dimension 2. For this restricted summation, we use (62) to bound the loops in (131). This gives

$$(132) \quad \sum_b \mathbf{1}(|b - a_1|_L \leq \ell_u^{**}) \cdot |\mathcal{L}_{u,(-,+,+), (a_1, b, a_2)}| \prec (\ell_u / \ell_s)^4 \cdot M_u^{-2} \cdot \ell_u^2.$$

We are left to handle indices $|b - a_1|_L \geq \ell_u^{**}$. For this, we use a simpler estimate

$$|\mathcal{L}_{u,(-,+,+), (a_1, b, a_2)}| \prec \max_{x_1 \in \mathcal{I}_{a_1}^{(2)}, y \in \mathcal{I}_b^{(2)}} |G_{x_1, y}|.$$

Next, we use (84) to estimate the right-hand side of the previous display:

$$\max_{x_1 \in \mathcal{I}_{a_1}^{(2)}, y \in \mathcal{I}_b^{(2)}} |G_{x_1, y}| \prec \sum_{a'_1, b'} (\mathcal{L}_{u,(+,-), (b', a'_1)})^{1/2} \mathbf{1}(|a'_1 - a_1|_L \leq 1, |b - b'|_L \leq 1).$$

Since $|b - a_1| \geq \ell_u^{**}$, by (114) and $\mathcal{T}_{u,D}(|a_1 - b|_L) \prec W^{-D}$, we have

$$\mathcal{L}_{u,(+,-), (b', a'_1)} \prec \mathcal{J}_{u,D}^* \cdot W^{-D}.$$

Therefore, the contribution from indices b in (131) satisfying $|b - a_1| \geq \ell_u^{**}$ part is negligible, i.e.

$$(133) \quad \sum_b \mathbf{1}(|b - a_1| \geq \ell_u^{**}) \cdot |\mathcal{L}_{u,(-,+,+), (a_1, b, a_2)}| \prec \mathcal{J}_{u,D}^* \cdot W^{-D}.$$

We will not track the contribution of the last term; it does not affect the argument given below. Now, since $|a_1 - a_2|_L \leq \ell_u^*$, we know that $\mathcal{T}_{u,D}(|a_1 - a_2|_L)$ is bounded away from 0; this

follows by (115). Combining this with (132) and (131) gives the following, which implies the desired estimate (118) for $|a_1 - a_2|_L \leq \ell_u^*$:

(134)

$$|a_1 - a_2| \leq \ell_u^* \implies \mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} / \mathcal{T}_{u,D}(|a_1 - a_2|_L) \prec (\eta_u)^{-1} \cdot (\ell_u/\ell_s)^4 \cdot (1 + (\mathcal{J}_{u,D}^*)^2 M_u^{-1}).$$

(Here we note that $(\ell_u/\ell_s)^4 M_u^{-1} \leq M_u^{-1/2}$, since $(\ell_u/\ell_s)^2 \leq \eta_s/\eta_u \leq M_s^{1/30}$ and $M_u \geq M_s^{29/30}$ by (61).) We now prove the desired estimate (118) under the assumption that $|a_1 - a_2| \geq \ell_u^*$. For this, we consider the following two cases:

$$(1) : \min_i |a_i - b| \leq \ell_u^*/2, \quad (2) : \min_i |a_i - b| \geq \ell_u^*/2.$$

In the first case, we assume without loss of generality that $|a_1 - b|_L \leq \ell_u^*/2$. Applying the Schwarz inequality to the two G edges in $\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)}$ connecting the block a_2 gives

$$G_{x_1 x_2} G_{x_2 y} G_{y x_1} \prec |G_{x_1 x_2}|^2 |G_{y x_1}| + |G_{x_2 y}|^2 |G_{y x_1}|.$$

Therefore, we have

(135)

$$\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec W^{-2} \left(\mathcal{L}_{u,(-,+,+),(a_1,a_2)} \max_{x_1 \in \mathcal{I}_{a_1}^{(2)}} \sum_{y \in \mathcal{I}_b^{(2)}} |G_{x_1 y}| + \mathcal{L}_{u,(-,+,+),(a_2,b)} \max_{y \in \mathcal{I}_b^{(2)}} \sum_{x_1 \in \mathcal{I}_{a_1}^{(2)}} |G_{x_1 y}| \right).$$

If we use (84), the $\mathcal{K}$ bound and $\sqrt{1+c} \leq 1+c$, (109), (110), (111), and (112), we have

$$\mathbf{1}(x_1 \neq x_2) |G_{x_1 x_2}| \prec M_u^{-1/2} (1 + \mathcal{J}_{u,D}^* M_u^{-1}).$$

Thus, for any a, b , we have

$$(136) \quad W^{-2} \max_{y \in \mathcal{I}_b^{(2)}} \sum_{x \in \mathcal{I}_a^{(2)}} |G_{xy}| \prec M_u^{-1/2} (1 + \mathcal{J}_{u,D}^* M_u^{-1})$$

Plugging this bound into (135) gives

$$(137) \quad \mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec (\mathcal{L}_{u,(-,+,+),(a_1,a_2)} + \mathcal{L}_{u,(-,+,+),(a_2,b)}) \cdot M_u^{-1/2} (1 + \mathcal{J}_{u,D}^* M_u^{-1})$$

Since $|a_2 - b| \geq \ell_u^*/2$ and $|a_1 - a_2| \geq \ell_u^*$, we can estimate $\mathcal{L}_{u,(-,+,+),(a_1,a_2)} + \mathcal{L}_{u,(-,+,+),(a_2,b)}$ with (114) to have

$$\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec \mathcal{J}_{u,D}^* \cdot (\mathcal{T}_{u,D}(|a_1 - a_2|) + \mathcal{T}_{u,D}(|b - a_2|)) \cdot M_u^{-1/2} (1 + \mathcal{J}_{u,D}^* M_u^{-1})$$

Since $|b - a_2|_L \geq |a_1 - a_2|_L - |a_1 - b|_L \geq |a_1 - a_2|_L - \ell_u^*/2$, we can use (115) to deduce

$$\mathcal{T}_{u,D}(|b - a_2|) \prec \mathcal{T}_{u,D}(|a_1 - a_2|)$$

Combining these bounds along with bound $\mathcal{J}_{u,D}^* \geq 1$ gives the following for $|a_1 - a_2|_L \geq \ell_u^*$:

$$(138) \quad \sum_b^{\text{Case 1}} |\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)}| \prec (\mathcal{J}_{u,D}^*)^2 \cdot \mathcal{T}_{u,D}(|a_1 - a_2|_L) \cdot \ell_u^2 M_u^{-1/2}.$$

We now consider case (2), defined below (134). We bound $\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)}$ as follows

(139)

$$\mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec \max_{x_1, x_2, y} |G_{x_1 y}| |G_{y x_2}| |G_{x_1 x_2}| \mathbf{1}(x_1 \in \mathcal{I}_{a_1}^{(2)}) \mathbf{1}(x_2 \in \mathcal{I}_{a_2}^{(2)}) \mathbf{1}(y \in \mathcal{I}_b^{(2)})$$

We note that since a_1, a_2 and b are all different, x_1, x_2 and y must be all different as well.

Next we use (84) to bound each G entry above, and then we apply (114). Next, recall that the distance between any two of a_1, a_2, b is at least $\ell_u^*/2$. Thus, because of the indicators on the right-hand side of (139), we deduce that $|x_1 - y|_L \geq |a_1 - b|_L - c\ell_u^*$. Doing this for the other pairs of indices above, namely (y, x_2) and (x_1, x_2) , we ultimately have

$$(140) \quad |G_{yx_2}| \cdot |G_{x_1x_2}| \cdot |G_{x_1y}| \prec (\mathcal{J}_{u,D}^*)^{3/2} \left(\mathcal{T}_{u,D}(|a_1 - a_2|_L) \mathcal{T}_{u,D}(|a_1 - b|_L) \mathcal{T}_{u,D}(|a_2 - b|_L) \right)^{1/2}$$

By combining (139) and (140), we arrive at

$$(141) \quad \begin{aligned} \text{Case 2} \\ \sum_b \mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} &\prec (\mathcal{J}_{u,D}^*)^{3/2} \left(\mathcal{T}_{u,D}(|a_1 - a_2|_L) \right)^{1/2} \sum_b \left(\mathcal{T}_{u,D}(|a_1 - b|_L) \mathcal{T}_{u,D}(|a_2 - b|_L) \right)^{1/2} \\ &\prec (\mathcal{J}_{u,D}^*)^{3/2} \mathcal{T}_{u,D}(|a_1 - a_2|_L) \cdot \ell_u^2 \cdot M_u^{-1} \end{aligned}$$

where, to deduce the second line, we use

$$\int_0^a \exp\left(-\sqrt{(a-x)}/2 - \sqrt{x}/2 + \sqrt{a}/2\right) dx \leq C.$$

We now combine this with (138). We deduce that if $|a_1 - a_2|_L \geq \ell_u^*$ then

$$(142) \quad \sum_b \mathcal{L}_{u,(-,+,+),(a_1,b,a_2)} \prec (\mathcal{J}_{u,D}^*)^2 \mathcal{T}_{u,D}(|a_1 - a_2|_L) \cdot \ell_u^2 \cdot M_u^{-1/2}$$

Plugging this into (131) and combining it with (134) completes the proof of (118).

Proof of (119). By definition (106), $(\mathcal{E} \otimes \mathcal{E})$ is the sum of $(\mathcal{E} \otimes \mathcal{E})^{(k)}$, which can be written in terms of $\mathcal{L}^{(k)} := \mathcal{L}_{u,\sigma^{(k)},\mathbf{a}^{(k)}}$. Here $\sigma^{(1)} = (+, -, +, -, +, -)$, $\mathbf{a}^{(1)} = (a_1, a_2, b', a'_2, a'_1, b)$ and $\sigma^{(2)} = (-, +, -, +, -, +)$, $\mathbf{a}^{(2)} = (a_2, a_1, b', a'_1, a'_2, b)$. By symmetry, we only need to prove (119) for $(\mathcal{E} \otimes \mathcal{E})^{(1)}$. Note that $|b - b'|_L \leq 1$ in this case, so we can treat $b = b'$ for all practical purposes in the following argument.

Case 1: $|a_1 - a_2|_L \leq 4\ell_t^*$. We split the sum $\sum_{b,b'}$ into the following two regions:

$$|b - a_1|_L \leq \ell_u^{**} := (\log W)^3 \ell_u \quad \text{and} \quad |b - a_1|_L \geq \ell_u^{**}$$

Note that $\mathcal{T}_{u,D}(\ell_u^{**})$ is exponentially small in W . This, and arguments similar to those used in (133), gives

$$\sum_{b,b'} \mathbf{1}(|b - a_1|_L \geq \ell_u^{**}) \mathcal{L}^{(1)} \prec \mathcal{J}_{u,D}^* \cdot W^{-D}.$$

For $|b - a_1|_L \leq \ell_u^{**}$, we use (62) for $n = 6$. This gives

$$(143) \quad \sum_{b,b'} \mathbf{1}(|b - a_1|_L \leq \ell_u^{**}) \mathcal{L}^{(1)} \prec (\ell_u/\ell_s)^{10} \cdot M_u^{-5} \cdot \ell_u^2.$$

If $|a_1 - a_2|_L \leq 4\ell_t^*$, then $\mathcal{T}_{u,D}(|a_1 - a_2|_L) \prec 1$ by (115). By this and the last two displays, we get (119).

Case 2: $|a_1 - a_2|_L \geq 4\ell_t^*$. Recall the assumption $|a'_i - a_i|_L \leq \ell_t^*$ (116) and the fact that we can treat $b = b'$ in the following proof. We again split the sum $\sum_{b,b'}$ into two regions:

$$(1) : |b - a_1|_L \leq |b - a_2|_L, \quad (2) : |b - a_1|_L \geq |b - a_2|_L.$$

Case (2) is treated in the same way as case (1) after interchanging the roles of (a_1, a'_1, b) and (a_2, a'_2, b') , i.e. by applying the Cauchy-Schwarz inequality below to the factor $(GE_{b'}G^\dagger)_{x_2x'_2}$

instead of $(G^\dagger E_b G)_{x_1 x'_1}$. So it suffices to consider case (1). Similar to (139) (but without controlling averages over x_1, x'_1 by maxima), we bound $\mathcal{L}^{(1)}$ by the product of four G 's and $G^\dagger E_b G$ as follows (recall that $\mathcal{L}^{(1)}$ is a six- G loop, hence six factors of G):

$$\mathcal{L}^{(1)} \leq W^{-4} \max_{x_2 \in \mathcal{I}_{a_2}^{(2)}} \max_{y' \in \mathcal{I}_{b'}^{(2)}} \max_{x'_2 \in \mathcal{I}_{a'_2}^{(2)}} \sum_{x_1 \in \mathcal{I}_{a_1}^{(2)}} \sum_{x'_1 \in \mathcal{I}_{a'_1}^{(2)}} |G_{x_1 x_2}| |G_{x_2 y'}| |G_{y' x'_2}| |G_{x'_2 x'_1}| \left| \left(G^\dagger E_b G \right)_{x_1 x'_1} \right|.$$

Now, we use the Cauchy-Schwarz inequality with respect to the x_1, x'_1 summations to obtain

$$\mathcal{L}^{(1)} \leq \max_{x_1, x_2, y', x'_1, x'_2}^* |G_{x_1 x_2}| |G_{x_2 y'}| |G_{y' x'_2}| |G_{x'_2 x'_1}| \cdot \left(W^{-4} \sum_{x_1 \in \mathcal{I}_{a_1}^{(2)}} \sum_{x'_1 \in \mathcal{I}_{a'_1}^{(2)}} (G^\dagger E_b G)_{x_1 x'_1} (G^\dagger E_b G)_{x'_1 x_1} \right)^{\frac{1}{2}}$$

(144)

$$\leq \max_{x_1, x_2, y', x'_1, x'_2}^* |G_{x_1 x_2}| |G_{x_2 y'}| |G_{y' x'_2}| |G_{x'_2 x'_1}| \cdot \left(\max_{\mathbf{a}} \max_{\sigma \in \{+, -\}^4} \mathcal{L}_{\sigma, \mathbf{a}} \right)^{\frac{1}{2}},$$

where the second line follows by definition of $\mathcal{L}$. Above, the $*$ indicates that the maximum is over all indices satisfying the condition

$$\mathbf{1}(x_1 \in \mathcal{I}_{a_1}^{(2)}) \mathbf{1}(x_2 \in \mathcal{I}_{a_2}^{(2)}) \mathbf{1}(y' \in \mathcal{I}_{b'}^{(2)}) \mathbf{1}(x'_1 \in \mathcal{I}_{a'_1}^{(2)}) \mathbf{1}(x'_2 \in \mathcal{I}_{a'_2}^{(2)}) = 1.$$

We now claim that

$$(145) \quad |G_{y' x_2}| \cdot |G_{y' x'_2}| \cdot |G_{x_1 x_2}| \cdot |G_{x'_1 x'_2}| \prec (\mathcal{J}_{u,D}^*)^2 \cdot \mathcal{T}_{t,D}(|a_1 - a_2|_L) \cdot \mathcal{T}_{t,D}(|b - a_2|_L)$$

Let us prove this claim. Since we assumed $|a_1 - a_2|_L \geq 4\ell_t^*$, we have, similar to (140), that

$$(146) \quad |G_{x_1 x_2}|^2 \leq \mathcal{J}_{u,D}^* \cdot \mathcal{T}_{u,D}(|a_1 - a_2|_L),$$

$$(147) \quad |G_{x'_1 x'_2}|^2 \leq \mathcal{J}_{u,D}^* \cdot \mathcal{T}_{u,D}(|a'_1 - a'_2|_L) \prec \mathcal{J}_{u,D}^* \cdot \mathcal{T}_{t,D}(|a_1 - a_2|_L),$$

where we used $\mathcal{T}_u \leq \mathcal{T}_t$, the assumption of $|a'_i - a_i|_L \leq \ell_t^*$ and (115) to get the second line. Thus,

$$|G_{x_1 x_2}| \cdot |G_{x'_1 x'_2}| \prec \mathcal{J}_{u,D}^* \cdot \mathcal{T}_{t,D}(|a_1 - a_2|_L).$$

For the indices $y' \in \mathcal{I}_b^{(2)}$, we similarly have

$$|G_{y' x_2}| \cdot |G_{y' x'_2}| \prec \mathcal{J}_{u,D}^* \cdot \mathcal{T}_{u,D}^{1/2}(|b - a_2|_L) \cdot \mathcal{T}_{u,D}^{1/2}(|b - a'_2|_L).$$

Using $\mathcal{T}_u \leq \mathcal{T}_t$ and $|a_2 - a'_2|_L \leq \ell_t^*$ then gives

$$|G_{y' x_2}| \cdot |G_{y' x'_2}| \leq \mathcal{J}_{u,D}^* \cdot \mathcal{T}_{t,D}(|b - a_2|_L).$$

By combining these bounds, we get (145). Next, we combine (145), (144), and the a priori estimate (62) on the $\mathcal{L}$ factor in (144). This gives

$$\mathcal{L}^{(1)} \prec (\ell_u/\ell_s)^3 M_u^{-3/2} \cdot (\mathcal{J}_{u,D}^*)^2 \cdot \mathcal{T}_{t,D}(|a_1 - a_2|) \cdot \mathcal{T}_{t,D}(|b - a_2|).$$

We will now consider the following two cases:

$$(148) \quad \begin{aligned} (1a) : & |a_1 - b|_L \leq \ell_u^*, \quad \text{or} \quad |a'_1 - b|_L \leq \ell_u^* \\ (1b) : & |a_1 - b|_L \geq \ell_u^*, \quad \text{and} \quad |a'_1 - b|_L \geq \ell_u^* \end{aligned}$$

Consider the case (1a), so $|a_2 - b|_L \geq |a_1 - a_2|_L - \ell_t^*$. We have $\mathcal{T}_{t,D}(|b - a_2|_L) \prec \mathcal{T}_{t,D}(|a_1 - a_2|_L)$ by (115). Combining these bounds with (145), we have the following in which we bound $\mathcal{L}^{(1)}$ pointwise and sum over the $O(\ell_u^*)^2$ -many indices b in case (1a):

$$(149) \quad \sum_{b,b'}^{(1a)} \mathcal{L}^{(1)} \prec (\ell_u/\ell_s)^3 \cdot \ell_u^2 \cdot M_u^{-3/2} \cdot (\mathcal{J}_{u,D}^*)^2 \cdot \mathcal{T}_{t,D}(|a_1 - a_2|_L)^2.$$

We now consider the case (1b). Here, we have the estimate

$$(G^\dagger E_b G)_{x_1 x'_1} \leq \max_{y \in \mathcal{I}_b^{(2)}} |G_{x_1 y}| |G_{x'_1 y}|.$$

Since $|a_1 - b|_L$ and $|a'_1 - b|_L \geq \ell_u^*$ and $|a_1 - a'_1|_L \leq \ell_t^*$, we have, similar to (140),

$$(G^\dagger E_b G)_{x_1 x'_1} \prec \mathcal{J}_{u,D}^* \cdot \mathcal{T}_{t,D}(|b - a_1|_L).$$

Combining this with (145) gives the following, in which the second line is bounded as in the proof of (129):

$$(150) \quad \begin{aligned} \sum_{b,b'}^{(1b)} \mathcal{L}^{(1)} &\prec (\mathcal{J}_{u,D}^*)^3 \cdot \mathcal{T}_{t,D}(|a_1 - a_2|_L) \cdot \sum_b \mathcal{T}_{t,D}(|b - a_1|_L) \mathcal{T}_{t,D}(|b - a_2|_L) \\ &\prec (\mathcal{J}_{u,D}^*)^3 \cdot \ell_t^2 \cdot M_t^{-2} \cdot (\mathcal{T}_{t,D}(|a_1 - a_2|_L))^2. \end{aligned}$$

The bounds for cases (1a) and (1b), multiplied by the factor W^2 in (106) (recall $W^2 \ell_u^2 = M_u/\eta_u$), give the last two terms in (119) and thus complete the proof of Lemma 5.7. $\square$

PROOF OF (64). Combining the $\mathcal{L} - \mathcal{K}$ estimate (65) and the $\mathcal{K}$ bound in Lemma 2.15 gives

$$(151) \quad \max_{a,b} \mathcal{L}_{u,(+,-),(a,b)} \prec (\eta_s/\eta_u)^4 \cdot M_u^{-2} + M_u^{-1} \prec M_u^{-1}, \quad u \in [s, t]$$

where we used the inductive hypothesis (61) to control $(\eta_s/\eta_u)^4$. Now, we use the weak local law (63); this lets us apply (85) and (84). Using these bounds with (151) gives

$$\|G_u - m\|_{\max}^2 \prec \max_{a,b} \mathcal{L}_{u,(+,-),(a,b)} \prec M_u^{-1}.$$

This completes the proof of (64). $\square$

5.4. Fast decay property. In the second step of the proof for Theorem 2.20, we established a strong decay property (65) for 2-loops. We now extend this decay to general loops $\mathcal{L}$ and $\mathcal{L} - \mathcal{K}$. The idea is to use (84) to control entries of G in terms of 2-loops, and then use this to control general loops

We now define the desired ℓ_u -decay property for general loops. We then state the decay property for $\mathcal{L}_u$, whose proof is identical to that of Lemma 5.9 in [50] and is thus omitted.

DEFINITION 5.8. Let $\mathcal{A}$ be a tensor $\mathbb{Z}_L^n \rightarrow \mathbb{R}$, and fix $\tau, D > 0$. We say $\mathcal{A}$ has (u, τ, D) decay at time u if

$$(152) \quad \max_{i,j} |a_i - a_j|_L \geq \ell_u W^\tau \implies \mathcal{A}_{\mathbf{a}} = O(W^{-D}), \quad \mathbf{a} = (a_1, a_2, \dots, a_n)$$

LEMMA 5.9. Assume that (64) and (65) hold. Then for any $n \geq 2$, and any $\tau, D, D' > 0$, the loop $\mathcal{L}_u$ has (u, τ, D) decay with probability $1 - O(W^{-D'})$:

$$(153) \quad \mathbb{P} \left(\max_{\sigma} \left(|\mathcal{L}_{u, \sigma, a}| + |(\mathcal{L} - \mathcal{K})_{u, \sigma, a}| \right) \cdot \mathbf{1} \left(\max_{i, j} |a_i - a_j|_L \geq \ell_u W^\tau \right) \geq W^{-D} \right) \leq W^{-D'}.$$

We now use Lemma 5.9 to provide the following estimate for $\mathcal{E}$ terms.

LEMMA 5.10. Assume that (64) and (65) hold. Recall $\Xi^{(\mathcal{L})}$ from (69), and define

$$(154) \quad \Xi_{t, m}^{(\mathcal{L}-\mathcal{K})} := \max_{\sigma, a} |(\mathcal{L} - \mathcal{K})_{t, \sigma, a}| \cdot M_t^m \cdot \mathbf{1}(\sigma \in \{+, -\}^m),$$

Then, we have

$$(155) \quad \begin{aligned} [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{u, \sigma}^{l_{\mathcal{K}}} &\prec \left(\max_{k < n} \Xi_{u, k}^{(\mathcal{L}-\mathcal{K})} \right) \cdot M_u^{-n} \cdot \frac{1}{\eta_u} \\ \mathcal{E}_{u, \sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} &\prec \left(\max_{k: 2 \leq k \leq n} \Xi_{u, k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u, n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} \right) \cdot M_u^{-n} \cdot \frac{1}{\eta_u} \\ \mathcal{E}_{u, \sigma}^{(\tilde{G})} &\prec \Xi_{u, n+1}^{(\mathcal{L})} \cdot M_u^{-n} \cdot \frac{1}{\eta_u} \\ (\mathcal{E} \otimes \mathcal{E})_{u, \sigma} &\prec \Xi_{u, 2n+2}^{(\mathcal{L})} \cdot M_u^{-2n} \cdot \frac{1}{\eta_u} \end{aligned}$$

Furthermore, for any $u \in [s, t]$, all of the $\mathcal{E}$ terms at time u on the left-hand side have (u, τ, D) decay.

PROOF OF LEMMA 5.10. The proof follows from the definitions of these terms and Lemma 5.9. Below, all ℓ_u^2 factors come from summing an index restricted to a length ℓ_u -neighborhood in $\mathbb{Z}^2$, and W^2 factors come from our choice of scaling.

- We use the formula (98) for $[\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{u, \sigma}^{l_{\mathcal{K}}}$ and bound $\mathcal{K}$ with (49). This gives

$$(156) \quad [\mathcal{K} \sim (\mathcal{L} - \mathcal{K})]_{u, \sigma}^{l_{\mathcal{K}}} \prec W^2 \cdot M_u^{-l_{\mathcal{K}}+1} \cdot \ell_u^2 \cdot \Xi_{u, (n-l_{\mathcal{K}}+2)}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-(n-l_{\mathcal{K}}+2)}$$

where we used $l_{\mathcal{K}} \geq 3$. The ℓ_u^2 factor comes from the sum over one index a that is restricted by the decay of $\mathcal{K}$ and another index b that is restricted by $S_{ab}^{(B)}$.

- By (99), $\mathcal{E}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))}$ is a product of two loops, whose total length is $n + 2$. In particular, we have

$$(157) \quad \mathcal{E}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \prec W^2 \cdot \sum_{k=2}^n \Xi_{u, k}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-k} \cdot \ell_u^2 \cdot \Xi_{u, (n-k+2)}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-(n-k+2)}$$

- By (37), $\mathcal{E}^{(\tilde{G})}$ is a product of $(\mathcal{L} - \mathcal{K})$ with length one and an $(n+1)$ - G loop. We use (87) to bound said G loop, and then we use (65) and (49), which implies $\Xi_{u, 2}^{\mathcal{L}} \prec 1$. This gives

$$(158) \quad \mathcal{E}^{(\tilde{G})} \prec W^2 \cdot \Xi_{u, 2}^{\mathcal{L}} \cdot M_u^{-1} \cdot \ell_u^2 \cdot \Xi_{u, (n+1)}^{\mathcal{L}} \cdot M_u^{-n} \prec W^2 \cdot M_u^{-n-1} \cdot \ell_u^2 \cdot \Xi_{u, (n+1)}^{\mathcal{L}}.$$

- By (107), $(\mathcal{E} \otimes \mathcal{E})$ can be written in terms of $(2n+2)$ - G -loops. Thus

$$(159) \quad (\mathcal{E} \otimes \mathcal{E}) \prec \Xi_{u, 2n+2}^{(\mathcal{L})} \cdot W^2 \cdot \ell_u^2 \cdot M_u^{-2n-1}.$$

This completes the proof. $\square$

LEMMA 5.11. *Suppose that the assumptions of Theorem 2.20, the local law (64) and the decay property for $\mathcal{L} - \mathcal{K}$ (65) hold, and that σ is not the alternating sign vector, in that*

$$(160) \quad \exists k, \text{ s.t. } \sigma_k = \sigma_{k+1}$$

. If $\max_{u \in [s, t]} \Xi_{u, 2n+2}^{(\mathcal{L})} \prec \Lambda$ for some deterministic $\Lambda \geq 1$, then

$$(161) \quad \Xi_{t, n}^{(\mathcal{L}-\mathcal{K})} \prec \Lambda^{1/2} + \max_{u \in [s, t]} \left(\max_{k < n} \Xi_{u, k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \Xi_{u, k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u, n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} + \Xi_{u, n+1}^{(\mathcal{L})} \right)$$

PROOF. By Lemma 7.3, the $\max \rightarrow \max$ norm for $\mathcal{U}_{u, t, \sigma}$ for fast decay tensors and for σ satisfying (160) is $\prec [(\ell_u^2 \eta_u)/(\ell_t^2 \eta_t)]^n$. Using this operator norm bound, Lemma 5.10, and (104), we have

$$(162) \quad \begin{aligned} M_t^n \cdot (\mathcal{L} - \mathcal{K})_{t, \sigma, \mathbf{a}} &\prec 1 + \int_s^t \left(\max_{k < n} \Xi_{u, k}^{(\mathcal{L}-\mathcal{K})} \right) \cdot \frac{1}{\eta_u} du \\ &\quad + \int_s^t \left(\max_{k: 2 \leq k \leq n} \Xi_{u, k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u, n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} \right) \cdot \frac{1}{\eta_u} du \\ &\quad + \int_s^t \Xi_{u, n+1}^{(\mathcal{L})} \cdot \frac{1}{\eta_u} du + M_t^n \cdot \int_s^t \left(\mathcal{U}_{u, t, \sigma} \circ \mathcal{E}_{u, \sigma}^{(M)} \right)_{\mathbf{a}} du. \end{aligned}$$

Above, the operator norm bound of $\prec [(\ell_u^2 \eta_u)/(\ell_t^2 \eta_t)]^n$ is used to turn the factor of M_t^n on the left-hand side into M_u^n , which is why we have $\Xi_{u, k}$ -terms on the right-hand side. Next, we use Lemma 5.5 to get

$$(163) \quad \mathbb{E} \left[M_t^n \int_s^t \left(\mathcal{U}_{u, t, \sigma} \circ \mathcal{E}_{u, \sigma}^{(M)} \right)_{\mathbf{a}} \right]^{2p} \leq C_{n, p} \mathbb{E} \left(M_t^n \int_s^t \left((\mathcal{U}_{u, t, \sigma} \otimes \mathcal{U}_{u, t, \bar{\sigma}}) \circ (\mathcal{E} \otimes \mathcal{E})_{u, \sigma} \right)_{\mathbf{a}, \mathbf{a}} du \right)^p$$

By (155), the right-hand side of (163) is $O_{\prec} (\int_s^t \Xi_{u, 2n+2}^{(\mathcal{L})} \eta_u^{-1} du)^p$, which completes the proof. $\square$

5.5. *Sum zero operator $\mathcal{Q}_t$ and mean zero operator $\mathbb{Q}$.* By Lemma 5.11, we are left to consider the alternating sign vector now. In particular, we assume $\sigma = \sigma^{(alt)}$ for the rest of this subsection. To do this, we now introduce the first of two operators that will be important for analyzing the loop hierarchy.

DEFINITION 5.12. Let $\mathcal{A}_{\mathbf{a}}$ be a tensor with $\mathbf{a} = (a_1, \dots, a_n) \in (\mathbb{Z}_L^2)^n$ and $n \geq 2$. Define

$$(\mathcal{P} \circ \mathcal{A})_{a_1} = \sum_{a_2, a_3, \dots, a_n} \mathcal{A}_{\mathbf{a}}.$$

In particular, we fix a_1 and sum over all other indices. We now define the sum-zero operator to be

$$(\mathcal{Q}_t \circ \mathcal{A})_{\mathbf{a}} = \mathcal{A}_{\mathbf{a}} - (\mathcal{P} \circ \mathcal{A})_{a_1} \cdot \vartheta_{t, \mathbf{a}}, \quad \vartheta_{t, \mathbf{a}} := (1-t)^{n-1} \prod_{i=2}^n \left(\Theta_t^{(B)} \right)_{a_1 a_i}.$$

Finally, we say that a tensor has the sum-zero property if $\mathcal{P} \circ \mathcal{A} = 0$.

By the random walk representation of $\Theta_{\xi}^{(B)}$ from Lemma 2.14, the row sums of $\Theta^{(B)}$ are equal to $(1-\xi)^{-1}$. This implies $\mathcal{P} \circ \vartheta_{t, \mathbf{a}} = 1$; thus, $\mathcal{P} \circ \mathcal{Q} = 0$.

The following lemma collects important facts about $\mathcal{Q}_t$; it is Lemma 5.13 in [50], and it relies on rapid decay of $\vartheta_{t, \mathbf{a}}$.

LEMMA 5.13. *If $\mathcal{A} : (\mathbb{Z}_L^2)^n \rightarrow \mathbb{R}$ has (t, τ, D) decay, then $\mathcal{Q}_t \circ \mathcal{A} : (\mathbb{Z}_L^2)^n \rightarrow \mathbb{R}$ has (t, τ, D) decay. Moreover, for some $C_n = O(1)$, we have*

$$(164) \quad \max_{\mathbf{a}} (\mathcal{Q}_t \circ \mathcal{A})_{\mathbf{a}} \leq W^{C_n \tau} \cdot \max_{\mathbf{a}} \mathcal{A}_{\mathbf{a}} + W^{-D+C_n}.$$

We now introduce a version of the loop hierarchy obtained after applying the sum-zero operator. First, define $\dot{\vartheta}_{t,\mathbf{a}} = \frac{d}{dt} \vartheta_{t,\mathbf{a}}$. We have the following equation, which follows from (5.90) in [50] (the proof is identical, since we only use that $\Theta_{t,\sigma}$ from (5.2) preserves the sum-zero property, which itself follows because the row and column sums of $\Theta_{\xi}^{(B)}$ are equal to $(1 - \xi)^{-1}$):

$$(165) \quad \begin{aligned} \mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &= (\mathcal{U}_{s,t,\sigma} \circ \mathcal{Q}_s \circ (\mathcal{L} - \mathcal{K})_{s,\sigma})_{\mathbf{a}} \\ &+ \sum_{l_{\mathcal{K}} > 2} \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{u,\sigma}^{l_{\mathcal{K}}} \right)_{\mathbf{a}} du \\ &+ \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \right)_{\mathbf{a}} du \\ &+ \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(\tilde{G})} \right)_{\mathbf{a}} du + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} du \\ &+ \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ ([\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}) \right)_{\mathbf{a}} du \\ &- \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u] \right)_{\mathbf{a}} du. \end{aligned}$$

Above, $[\mathcal{Q}_u, \Theta_{u,\sigma}] := \mathcal{Q}_u \circ \Theta_{u,\sigma} - \Theta_{u,\sigma} \circ \mathcal{Q}_u$ is the commutator for operators. We also record the following identity below, which is (5.88) in [50]:

$$(166) \quad \begin{aligned} \mathcal{Q}_t \circ \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{t,\sigma,\mathbf{a}}^{l_{\mathcal{K}}=2} &= \mathcal{Q}_t \circ \Theta_{t,\sigma} \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} \\ &= \Theta_{t,\sigma} \circ \mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} + [\mathcal{Q}_t, \Theta_{t,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}}, \end{aligned}$$

Not only does $\mathcal{Q}_t \circ \mathcal{A}$ satisfy the sum zero property, but so does $\Theta_{t,\sigma} \circ \mathcal{Q}_t$ by definition of $\Theta_{t,\sigma}$ from (5.2) and because the row and column sums of $\Theta_{\xi}^{(B)}$ are $1/(1 - \xi)$. Thus, the last term in (166) is also sum-zero:

$$(167) \quad \mathcal{P} \circ [\mathcal{Q}_t, \Theta_{t,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} = 0.$$

Now, note that if $\mathcal{A}$ is sum zero, then $\mathcal{A} = \mathcal{Q}_t \circ \mathcal{A}$. Using this and (167) gives

$$\int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ ([\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}) \right)_{\mathbf{a}} du = \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ ([\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}) \right)_{\mathbf{a}} du.$$

Since $\mathcal{P}\vartheta = 1$, differentiating this identity gives $\mathcal{P}\dot{\vartheta} = 0$. In particular, we deduce

$$\int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u] \right)_{\mathbf{a}} du = \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u] \right)_{\mathbf{a}} du.$$

Using the previous two identities, we can rewrite (165) into a compact form:

$$(168) \quad \mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} = (\mathcal{U}_{s,t,\sigma} \circ \mathcal{Q}_s \circ \mathcal{B}_0)_{\mathbf{a}}$$

$$\begin{aligned}
& + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \sum_{m=1}^5 \mathcal{B}_m \right)_a du \\
& + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_a du
\end{aligned}$$

where the integrands are defined as follows:

$$\begin{aligned}
(169) \quad \mathcal{B}_0 &:= (\mathcal{L} - \mathcal{K})_{s,\sigma} \\
\mathcal{B}_1 &:= \sum_{l_{\mathcal{K}} > 2} \left[\mathcal{K} \sim (\mathcal{L} - \mathcal{K}) \right]_{u,\sigma}^{l_{\mathcal{K}}} \\
\mathcal{B}_2 &:= \mathcal{E}_{u,\sigma}^{((\mathcal{L}-\mathcal{K}) \times (\mathcal{L}-\mathcal{K}))} \\
\mathcal{B}_3 &:= \mathcal{E}_{u,\sigma}^{(\tilde{G})} \\
\mathcal{B}_4 &:= [\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \\
\mathcal{B}_5 &:= -\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u
\end{aligned}$$

Before we analyze the integrals of $\mathcal{B}_m$, we provide estimates for these terms. For the rest of this subsection, we fix the length n and adopt the following assumption, in which $\Lambda \geq 1$ constants are deterministic:

$$(170) \quad \Xi_{u,m}^{(\mathcal{L})} \prec \Lambda_{u,m}^{(\mathcal{L})}, \quad \Xi_{u,m}^{(\mathcal{L}-\mathcal{K})} \prec \Lambda_{u,m}^{(\mathcal{L}-\mathcal{K})}, \quad \forall m \leq 2n+2$$

First, $\mathcal{B}_0$ is bounded using the assumption (58). For $1 \leq m \leq 3$, the $\mathcal{B}_m$ are bounded using Lemma 5.10. Next, we have

$$(171) \quad \|\mathcal{B}_5\|_{\max} \prec \max_a \left| \left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \right]_a \right| \cdot \max_{\mathbf{a}} \left| \dot{\vartheta}_{u,\mathbf{a}} \right|$$

Recall that σ is alternating. In this case, we claim the following holds for any loop of length n :

$$(172) \quad \left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \right]_a \prec (W^2 \eta_u)^{-n} \cdot \ell_u^{2(n-2)} \cdot \ell_u^{-2(n-1)} \cdot \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})}.$$

To see this, since σ is non-constant, there exists $1 < k \leq n$ such that $\sigma_k = \bar{\sigma}_{k+1}$. By Lemma 3.2, we can sum $\mathcal{L} - \mathcal{K}$ of rank n over a_k and write the resulting sum in terms of $\mathcal{L} - \mathcal{K}$ of rank $n-1$ with a multiplicative factor $(W^2 \eta_u)^{-1}$. The summation over the remaining $n-2$ many indices results in a factor of $\prec \ell_u^{2(n-2)}$ because $\mathcal{L} - \mathcal{K}$ has the (u, τ, D) decay property for any $\tau, D > 0$ by assumption (65). Finally, the additional $(W^2 \ell_u^2 \eta_u)^{-n+1} = M_u^{-n+1}$ comes from compensating the factor of M_u^{n-1} in the definition of $\Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})}$. Next,

$$(173) \quad \max_{\mathbf{a}} \left| \dot{\vartheta}_{u,\mathbf{a}} \right| \prec \ell_u^{-2n+2} \cdot \eta_u^{-1}$$

by definition of ϑ and the estimate (42) for $\Theta^{(B)}$. Inserting these two bounds into (171) and using (170) gives

$$(174) \quad \|\mathcal{B}_5\|_{\max} \prec (M_u)^{-n} \cdot \eta_u^{-1} \cdot \Lambda_{u,n-1}^{(\mathcal{L}-\mathcal{K})}$$

By a similar argument, for $\mathcal{B}_4$, we have

$$\begin{aligned}
 (175) \quad \|\mathcal{B}_4\|_{\max} &= \max_{\mathbf{a}} \left| \left[\Theta_{u,\sigma} \circ ((\mathcal{P} \circ (\mathcal{L} - \mathcal{K})) \cdot \vartheta_u) \right]_{\mathbf{a}} - [(\mathcal{P} \circ (\Theta_{u,\sigma} \circ (\mathcal{L} - \mathcal{K}))) \cdot \vartheta_u]_{\mathbf{a}} \right| \\
 &\prec \|\vartheta_u\|_{\max} \cdot \eta_u^{-1} \max_a \left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \right]_a \\
 &\prec (M_u)^{-n} \cdot \eta_u^{-1} \cdot \Lambda_{u,n-1}^{(\mathcal{L}-\mathcal{K})}
 \end{aligned}$$

In [50], the sum zero operator and the previous bounds on $\mathcal{B}_m$ terms are enough to analyze the loop hierarchy. For dimension 2, we need another “mean zero operator” $\mathbb{Q} := 1 - \mathbb{E}$. We now decompose (168) according to $\mathbb{E}$ and $\mathbb{Q}$:

$$\begin{aligned}
 (176) \quad \mathcal{Q}_t \circ \mathbb{Q} \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &= (\mathcal{U}_{s,t,\sigma} \circ \mathcal{Q}_s \circ \mathbb{Q} \circ \mathcal{B}_0)_{\mathbf{a}} \\
 &\quad + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathbb{Q} \circ \sum_{m=1}^5 \mathcal{B}_m \right)_a du \\
 &\quad + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_a du
 \end{aligned}$$

$$\begin{aligned}
 (177) \quad \mathcal{Q}_t \circ \mathbb{E} \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &= (\mathcal{U}_{s,t,\sigma} \circ \mathcal{Q}_s \circ \mathbb{E} \circ \mathcal{B}_0)_{\mathbf{a}} \\
 &\quad + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathbb{E} \circ \sum_{m=1}^5 \mathcal{B}_m \right)_a du
 \end{aligned}$$

By definition, each of $\mathcal{Q}_s \circ \mathcal{B}_0$ and $\mathcal{Q}_u \circ \mathcal{B}_m$ for $m = 1, \dots, 5$ has fast decay, the sum-zero property, and the form (205). Thus, since we have the mean-zero operator $\mathbb{Q}$, we can use the enhanced $\mathcal{U}_{u,t,\sigma}$ estimate from case 3 of Lemma 7.3. In the case of $\mathcal{B}_0$, we have $\|\mathcal{B}_0\|_{\max} \prec M_s^{-n}$, and thus

$$M_t^n \cdot (\mathcal{U}_{s,t,\sigma} \circ \mathcal{Q}_s \circ \mathbb{Q} \circ \mathcal{B}_0)_{\mathbf{a}} \prec M_t^n \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^n M_s^{-n} + W^{-D+C_n} \prec \Lambda_{s,n}^{(\mathcal{L}-\mathcal{K})} + W^{-D+C_n}$$

for $D > 0$ large and $C_n = O(1)$, where the last bound follows since $\Lambda \geq 1$. If we let $\Lambda_m \geq 1$ be deterministic numbers such that $\|\mathcal{B}_m\|_{\max} \prec \Lambda_m$, then we also get

$$M_t^n \cdot \mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathbb{Q} \circ \mathcal{B}_m \prec M_t^n \left(\frac{\ell_u^2 \eta_u}{\ell_t^2 \eta_t} \right)^n \Lambda_m + W^{-D+C_n} \prec M_u^n \|\mathcal{B}_m\|_{\max} + W^{-D+C_n}.$$

For clarity of presentation, we will not keep track of the W^{-D+C_n} errors; they will not affect the argument below in any case. For $1 \leq m \leq 3$, the $\mathcal{B}_m$ terms are bounded in Lemma 5.10, whereas $\mathcal{B}_4, \mathcal{B}_5$ are bounded in (174) and (175). Ultimately, we obtain that

$$\begin{aligned}
 (178) \quad &(\mathcal{U}_{s,t,\sigma} \circ \mathcal{Q}_s \circ \mathbb{Q} \circ \mathcal{B}_0)_{\mathbf{a}} + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathbb{Q} \circ \sum_{m=1}^5 \mathcal{B}_m \right)_a du \\
 &\prec (M_t)^{-n} \max_{u \in [s,t]} \left(\max_{k < n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Lambda_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} + \Lambda_{u,n+1}^{(\mathcal{L})} \right)
 \end{aligned}$$

We now analyze the terms in (177). First, we note that each of $\mathcal{Q}_s \circ \mathbb{E} \circ \mathcal{B}_0$ and $\mathcal{Q}_u \circ \mathbb{E} \circ \mathcal{B}_m$ for $m = 1, \dots, 5$ has fast decay, sum zero and symmetry, in the sense that we have

$$(179) \quad \mathcal{P}(\mathbb{Q} \circ \mathbb{E} \circ \mathcal{B}) = 0, \quad \mathcal{R} \circ (\mathbb{Q} \circ \mathbb{E} \circ \mathcal{B}) = \mathbb{Q} \circ \mathbb{E} \circ \mathcal{B}$$

where

$$(\mathcal{R} \circ \mathcal{A})_{a, a+s_2, \dots, a+s_n} := \mathcal{A}_{a, a-s_2 \dots a-s_n}$$

for any $a, s_2, \dots, s_n \in \mathbb{Z}_L^2$. For such tensors, we have another enhanced $\mathcal{U}_{u,t,\sigma}$ estimate from case 4 of Lemma 7.3. This gives

$$M_t^n \cdot \mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathbb{E} \circ \mathcal{B}_m \prec M_u^n \|\mathbb{E} \mathcal{B}_m\|_{\max}$$

Similar to (178), with estimates for $\mathcal{B}_m$ from (58), Lemma 5.10, (174), and (175), we deduce (180)

$$\begin{aligned} & (\mathcal{U}_{s,t,\sigma} \circ \mathcal{Q}_s \circ \mathbb{E} \circ \mathcal{B}_0)_{\mathbf{a}} + \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathbb{E} \circ \sum_{m=1}^5 \mathcal{B}_m \right)_a du \\ & \prec (M_t)^{-n} \max_{u \in [s,t]} \left(\max_{k < n} \mathbb{E} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \mathbb{E} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} + \mathbb{E} \Xi_{u,n+1}^{(\mathcal{L})} \right) \end{aligned}$$

We are left to estimate the martingale in (176). First, we introduce the notation

$$\begin{aligned} ((\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ \mathcal{A})_{\mathbf{a}, \mathbf{b}} &= \mathcal{A}_{\mathbf{a}, \mathbf{b}} - \delta_{a'_1, a_1} \sum_{a'_2 \dots a'_n} \mathcal{A}_{\mathbf{a}', \mathbf{b}} \cdot \vartheta_{u, \mathbf{a}} - \delta_{b'_1, b_1} \sum_{b'_2 \dots b'_n} \mathcal{A}_{\mathbf{a}, \mathbf{b}'} \cdot \vartheta_{u, \mathbf{b}} \\ (181) \quad &+ \delta_{a'_1, a_1} \delta_{b'_1, b_1} \sum_{a'_2 \dots a'_n} \sum_{b'_2 \dots b'_n} \mathcal{A}_{\mathbf{a}', \mathbf{b}'} \cdot \vartheta_{u, \mathbf{a}} \cdot \vartheta_{u, \mathbf{b}}, \end{aligned}$$

where we used the notation $\mathbf{a} = (a_1, \dots, a_n)$, $\mathbf{a}' = (a'_1, \dots, a'_n)$ and similarly for $\mathbf{b}$ and $\mathbf{b}'$. The definition of ϑ and an elementary calculation gives

$$\sum_{a_2, \dots, a_n} ((\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ \mathcal{A})_{\mathbf{a}, \mathbf{b}} = \sum_{b_2, \dots, b_n} ((\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ \mathcal{A})_{\mathbf{a}, \mathbf{b}} = 0.$$

By case 5 in Lemma 7.3, we have the following for any tensor $\mathcal{A}$ of rank- $2n$ that has fast decay at scale ℓ_u :

$$((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ \mathcal{A})_{\mathbf{a}, \mathbf{a}} \prec \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_u^2 \eta_u}{\ell_t^2 \eta_t} \right)^{2n}.$$

By this estimate applied to $\mathcal{E} \otimes \mathcal{E}$, we have

$$((\mathcal{U}_{u,t,\sigma} \otimes \mathcal{U}_{u,t,\bar{\sigma}}) \circ (\mathcal{Q}_u \otimes \mathcal{Q}_u) \circ (\mathcal{E} \otimes \mathcal{E})_{u, \sigma})_{\mathbf{a}, \mathbf{a}} \prec \|(\mathcal{E} \otimes \mathcal{E})\|_{\max} \cdot \left(\frac{\ell_u^2 \eta_u}{\ell_t^2 \eta_t} \right)^{2n}$$

Next, by (108), if for some deterministic quantity $\Lambda_{u,2n+2}^{(L)}$, we have $\Xi_{u,2n+2}^{(\mathcal{L})} \prec \Lambda_{u,2n+2}^{(L)}$, then

$$(182) \quad \int_s^t \left(\mathcal{U}_{u,t,\sigma} \circ \mathcal{Q}_u \circ \mathcal{E}_{u,\sigma}^{(M)} \right)_{\mathbf{a}} \prec \max_{u \in [s,t]} \left(\Lambda_{u,2n+2}^{(L)} \right)^{1/2}.$$

We now combine the estimates in this subsection to prove the following.

LEMMA 5.14. *Suppose the assumptions of Theorem 2.20, local law (64) and decay of $\mathcal{L} - \mathcal{K}$ in (65) all hold. Assume that for some deterministic quantities $\Lambda \geq 1$, we have*

$$(183) \quad \Xi_{u,m}^{(\mathcal{L})} \prec \Lambda_{u,m}^{(\mathcal{L})}, \quad \Xi_{u,m}^{(\mathcal{L}-\mathcal{K})} \prec \Lambda_{u,m}^{(\mathcal{L}-\mathcal{K})}, \quad \forall m \leq 2n+2.$$

Then, we have

(184)

$$\begin{aligned} \Xi_{t,n}^{(\mathcal{L}-\mathcal{K})} &\prec \max_{u \in [s,t]} \left(\max_{k < n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{2 \leq k \leq n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Lambda_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} + \Lambda_{u,n+1}^{(\mathcal{L})} + \left(\Lambda_{u,2n+2}^{(\mathcal{L})} \right)^{1/2} \right) \\ &\quad + \max_{u \in [s,t]} \left(\max_{k < n} \mathbb{E} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \mathbb{E} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} + \mathbb{E} \Xi_{u,n+1}^{(\mathcal{L})} \right). \end{aligned}$$

PROOF. Lemma 5.11 treats (184) for non-alternating case σ , so we assume σ is alternating, so that $\sigma_k = \bar{\sigma}_{k+1}$ for all k . We combine (176), (177), (178), (180) and (182), to get

$$\begin{aligned} \mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} &\prec M_t^{-n} \cdot \max_{u \in [s,t]} \left(\max_{k < n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{2 \leq k \leq n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Lambda_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} + \Lambda_{u,n+1}^{(\mathcal{L})} + \left(\Lambda_{u,2n+2}^{(\mathcal{L})} \right)^{1/2} \right) \\ &\quad + M_t^{-n} \cdot \max_{u \in [s,t]} \left(\max_{k < n} \mathbb{E} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{k: 2 \leq k \leq n} \mathbb{E} \Xi_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Xi_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} + \mathbb{E} \Xi_{u,n+1}^{(\mathcal{L})} \right). \end{aligned}$$

We now write the left-hand side of the previous display as follows:

(185)

$$\mathcal{Q}_t \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} = (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} - \mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} \cdot \vartheta_{\mathbf{a}} = (\mathcal{L} - \mathcal{K})_{t,\sigma,\mathbf{a}} + O_{\prec} \left(M_t^{-n} \cdot \Lambda_{t,n-1}^{(\mathcal{L}-\mathcal{K})} \right)$$

This completes the proof of Lemma 5.14. $\square$

5.6. *Proof of Theorem 2.20, Step 3.* The argument is the exact same as the proof of Step 3 in [50]. Thus, let us give a summary of the argument. For the sake of stochastic domination, we can drop the second line of (184). We will proceed in the following inductive fashion. We say that $\mathcal{S}(n, k, s, u, t)$ holds if $\Xi_{u,n}^{(\mathcal{L}-\mathcal{K})} \prec \Psi(n, k, s, u, t)$, where

$$\Psi(n, k, s, u, t) := M_s^{\frac{1}{2}} + \frac{\ell_t^{2n-2}}{\ell_s^{2n-2}} \cdot \begin{cases} M_u^{1-\frac{k}{4}} & k = 0 \\ M_s^{1-\frac{k}{4}} & k \geq 1 \end{cases}, \quad u \in [s, t].$$

Suppose that $n \geq 3$ and $k \geq 1$, and that $\mathcal{S}(m, l, s, u, t)$ holds for (m, l) satisfying $l = k$ and $m \leq n - 1$ or $l = k - 1$ and $m \leq n + 2$. Under this assumption, we will show that $\mathcal{S}(n, k, s, u, t)$ holds. Given this, we now establish the base cases that $\mathcal{S}(m, l, s, u, t)$ holds for (m, l) if $l = 0$ and $m \geq 1$ or $l \geq 0$ and $m \leq 2$. For $m = 1$, and for $m = 2$ with $\sigma \in \{(+, -), (-, +)\}$, this follows from (60), (65) and (87). For $m = 2$ and $\sigma \in \{(+, +), (-, -)\}$, the estimate (65) does not apply, and (161) (with $n = 2$) contains the self-quadratic term $(\Xi_{u,2}^{(\mathcal{L}-\mathcal{K})})^2 M_u^{-1}$. In this case, we use a standard continuity argument in $u \in [s, t]$ based on Lemma 5.11 with $n = 2$, the a priori bound (62) from Step 1, and the initial bound (58) at time s . Indeed, as long as $\Xi_{u,2}^{(\mathcal{L}-\mathcal{K})} \leq N^\delta M_s^{1/2}$ on $[s, u']$, the right-hand side of (161) at time u' is $O_{\prec}(M_s^{1/2})$, because $(N^\delta M_s^{1/2})^2 M_u^{-1} \leq N^{2\delta} M_s / M_u \leq N^{2\delta} M_s^{1/30}$ by (61), and the terms involving $\Xi^{(\mathcal{L})}$ are $O_{\prec}((\ell_t / \ell_s)^5) = O_{\prec}(M_s^{1/12})$ by (62). We then deduce that $\mathcal{S}(m, l, s, u, t)$ holds for $(3, 1)$. By continuing inductively, we deduce that it holds for $(n, 1)$ for any $n \geq 3$ as well. Then, we use these estimates to deduce that $\mathcal{S}(m, l, s, u, t)$ holds for $(n, 2)$ for any $n \geq 1$, and by continuing inductively, we deduce it for any (n, k) . This would yield $\Xi_{u,n}^{(\mathcal{L}-\mathcal{K})} \prec M_s^{1/2}$ for any $u \in [s, t]$. Moreover, we have $\Xi_{u,n}^{(\mathcal{L})} \prec 1 + M_u^{-1} \Xi_{u,n}^{(\mathcal{L}-\mathcal{K})}$ by the $\mathcal{K}$ -bound in Lemma 3.4 and the difference in scaling for $\Xi^{(\mathcal{L})}$ and $\Xi^{(\mathcal{L}-\mathcal{K})}$. Thus, the desired estimate (66) follows.

Because we assume that $\mathcal{S}(m, l, s, u, t)$ holds for (m, l) satisfying $l = k$ and $m \leq n - 1$ or $l = k - 1$ and $m \leq n + 2$, we can trade the first two terms on the right-hand side of (184) for 1, assuming we have proven the main estimate (66) of Step 3 for $k < n$. This gives us the following, in which we think of $\Lambda \geq 1$ constants as our “best bounds available” for Ξ parameters:

$$\Lambda_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec 1 + \max_{u \in [s,t]} \left(\Lambda_{u,2n+2}^{(\mathcal{L})} \right)^{\frac{1}{2}} + \max_{u \in [s,t]} \Lambda_{u,n+1}^{(\mathcal{L})}.$$

Next, we use (5.118) in [50] (whose proof is based only on dimension-independent matrix calculations); this gives the estimate

$$\Xi_{u,2n+2}^{(\mathcal{L})} \leq M_u \cdot \Xi_{u,2l_1}^{(\mathcal{L})} \Xi_{u,2l_2}^{(\mathcal{L})}$$

where $l_1 = \lfloor (n+1)/2 \rfloor$ and $l_2 = n+1 - l_1$. In particular, we can take $\Lambda_{u,2n+2}^{(\mathcal{L})} \leq M_u \cdot \Lambda_{u,2l_1}^{(\mathcal{L})} \Lambda_{u,2l_2}^{(\mathcal{L})}$. Now, if $n+1$ is even, then $2l_1 = 2l_2 = n+1$, and we arrive at

$$\Lambda_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec 1 + \max_{u \in [s,t]} M_u^{\frac{1}{2}} \cdot \Lambda_{u,n+1}^{(\mathcal{L})}$$

Because the scaling in $\Xi^{(\mathcal{L}-\mathcal{K})}$ is smaller than the scaling in $\Xi^{(\mathcal{L})}$ by M_u^{-1} , we have $\Lambda_{u,n+1}^{(\mathcal{L})} \prec 1 + M_u^{-1} \Lambda_{u,n+1}^{(\mathcal{L}-\mathcal{K})}$. If we combine this with the previous display, then we have

$$\Lambda_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec \max_{u \in [s,t]} M_u^{\frac{1}{2}} + \max_{u \in [s,t]} M_u^{-\frac{1}{2}} \Lambda_{u,n+1}^{(\mathcal{L}-\mathcal{K})}.$$

If $n+1$ is odd, then $2l_1 = n$ and $2l_2 = n+2$, so we arrive instead at

$$\Lambda_{u,2n+2}^{(\mathcal{L})} \leq M_u \Lambda_{u,n}^{(\mathcal{L})} \Lambda_{u,n+2}^{(\mathcal{L})} \leq M_u |\Lambda_{u,n}^{(\mathcal{L})}|^2 + M_u |\Lambda_{u,n+2}^{(\mathcal{L})}|^2.$$

Using the previous display, the same logic as in the case of $n+1$ being even yields

$$\Lambda_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec \max_{u \in [s,t]} M_u^{\frac{1}{2}} + \max_{u \in [s,t]} M_u^{-\frac{1}{2}} \Lambda_{u,n}^{(\mathcal{L}-\mathcal{K})} + \max_{u \in [s,t]} M_u^{-\frac{1}{2}} \Lambda_{u,n+2}^{(\mathcal{L}-\mathcal{K})}.$$

Since $M_u \geq W^\delta$ for some fixed $\delta > 0$, we can move the second term on the right-hand side to the left-hand side. Ultimately, we obtain the following regardless of the parity of n :

$$\Lambda_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec \max_{u \in [s,t]} M_u^{\frac{1}{2}} + \max_{u \in [s,t]} M_u^{-\frac{1}{2}} \Lambda_{u,n+1}^{(\mathcal{L}-\mathcal{K})} + \max_{u \in [s,t]} M_u^{-\frac{1}{2}} \Lambda_{u,n+2}^{(\mathcal{L}-\mathcal{K})}.$$

By assumption, we have $\Lambda_{u,n+1}^{(\mathcal{L}-\mathcal{K})} + \Lambda_{u,n+2}^{(\mathcal{L}-\mathcal{K})} \prec M_s^{1/2} + \ell_t^{2n+2} \ell_s^{-2n-2} M_s^{1-(k-1)/4}$, where the exponent $2n+2$ is the value of $2m-2$ at $m = n+2$. We now plug this bound into the previous display to deduce that

$$\Lambda_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec M_s^{\frac{1}{2}} + \frac{\ell_t^{2n-2}}{\ell_s^{2n-2}} M_s^{1-\frac{k}{4}}.$$

Here, we used that the leftover factor $(\ell_t/\ell_s)^4$ is absorbed by the gain in M . Indeed, $(\ell_t/\ell_s)^4 \leq (\eta_s/\eta_t)^2$, while (61) gives $(\eta_s/\eta_t)^{30} \leq M_s$, so that $(\ell_t/\ell_s)^4 \leq M_s^{1/15}$. We note that the same argument applies if we replace t by any $u' \in [s, t]$. Therefore, $\mathcal{S}(n, k, s, u, t)$ holds, and the proof of step 3 is complete.

5.7. *Proof of Theorem 2.20: Step 4 and 5.* For step 4, the argument is the same as the proof of step 4 in [50], so we give a summary. We use step 3 to control all $\Xi^{(\mathcal{L})}$ terms in (184) by $O_{\prec}(1)$. Thus, we arrive at

$$(186) \quad \begin{aligned} \Lambda_{t,n}^{(\mathcal{L}-\mathcal{K})} &\prec 1 + \max_{u \in [s,t]} \left(\max_{k < n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{2 \leq k \leq n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot \Lambda_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-1} \right) \\ &\prec 1 + \max_{u \in [s,t]} \left(\max_{k < n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} + \max_{2 \leq k \leq n} \Lambda_{u,k}^{(\mathcal{L}-\mathcal{K})} \cdot M_u^{-\frac{1}{3}} \right), \end{aligned}$$

where the second line follows from the bound $\Lambda_{u,n-k+2}^{(\mathcal{L}-\mathcal{K})} \prec M_s^{1/2}$ from Step 3, together with

$$M_s^{1/2} M_u^{-1} = M_u^{-1/2} (M_s/M_u)^{1/2} \leq M_u^{-1/2} (\eta_s/\eta_u)^{1/2} \leq M_u^{-1/2} M_s^{1/60} \leq M_u^{-1/3},$$

which holds by (61) (recall that $M_s/M_u \leq \eta_s/\eta_u$ and $M_u \geq M_s^{29/30}$). Now, we use (87) to get $\Lambda_{t,1}^{(\mathcal{L}-\mathcal{K})} \prec 1$. At this point, because $M_u \geq W^\delta$ for some $\delta > 0$, we can induct in n using (186) to deduce $\Lambda_{t,n}^{(\mathcal{L}-\mathcal{K})} \prec 1$. (For each $n \geq 2$, the term $k = n$ on the right-hand side of (186) is linear in $\Lambda_{u,n}^{(\mathcal{L}-\mathcal{K})}$ with the small factor $M_u^{-1/3}$. Since (186) holds with t replaced by any $t' \in [s, t]$, and since Step 3 gives the a priori bound $\Lambda_{u,n}^{(\mathcal{L}-\mathcal{K})} \prec M_s^{1/2}$, iterating (186) finitely many times removes this term.) This is the desired estimate for step 4.

For step 5, we use step 4 to get $(\mathcal{L} - \mathcal{K})_{t,\sigma,a} \prec M_t^{-2}$ in the case of $n = 2$. Since $\mathcal{T}_{t,D}(\ell)$ is bounded away from 0 for $\ell = O(\ell_t^*)$, this implies that $(\mathcal{L} - \mathcal{K})_{t,\sigma,a}/\mathcal{T}_{t,D}(|a_1 - a_2|_L) \prec 1$ in the case where $|a_1 - a_2|_L = O(\ell_t^*)$. This is the desired estimate. So, it suffices to assume that $|a_1 - a_2|_L > 6\ell_t^*$. In this case, the desired estimate (68) follows immediately from (127). This completes the proofs of step 4 and 5. $\square$

5.8. *Proof of Lemma 2.19: Step 6.* We recall that Lemma 2.19 asks for an improved estimate on the expectation of $\mathcal{L}$ of length $n = 2$. The first ingredient is the following improved estimate on loops of length 1, which is Lemma 5.15 in [50].

LEMMA 5.15. *Fix any $a \in \mathbb{Z}_L^2$, any $\tau, \kappa > 0$, any $E \in [-2 + \kappa, 2 - \kappa]$, and $0 \leq u \leq 1 - N^{-1+\tau}$. We have*

$$(187) \quad |\mathbb{E} \mathcal{L}_{u,(+),(a)} - \mathcal{K}_{u,(+),(a)}| = |\mathbb{E} \langle (G_u - m) E_a \rangle| \prec M_u^{-2}.$$

PROOF. We start with the identity $G_u - m = -m(H_u + um)G_u$, which follows from $z_u = E + (1 - u)m$ and $m(m + E) = -1$. This gives us the first line of the following calculation, after which we use Gaussian integration by parts (note that the variance of the entries of H_u are all scaled by u):

$$\begin{aligned} \mathbb{E} \langle (G_u - m) E_a \rangle &= m \cdot \mathbb{E} \langle (-H_u - um) G_u E_a \rangle \\ &= u \cdot m \sum_{b,b'} \mathbb{E} \left(\langle (G_u - m) E_b \rangle W^2 \cdot S_{bb'}^{(B)} \cdot \langle E_{b'} G_u E_a \rangle \right) \\ &= u \cdot m \sum_b \mathbb{E} \left(\langle (G_u - m) E_b \rangle S_{ba}^{(B)} \langle G E_a \rangle \right). \end{aligned}$$

(In the above calculation, we use the structure of $S_{ba}^{(B)} := \frac{1}{5} \mathbf{1}(|b - a|_L \leq 1)$ and of $E_{b'}$ from (7).) At this point, we decompose $\langle G_u E_a \rangle = m + \langle (G_u - m) E_a \rangle$ and plug it into the last

line above. As in (5.124) of [50], this gives us a linear equation for the vector $a \mapsto \mathbb{E}\langle (G_u - m)E_a \rangle$. Recalling that $\Theta_{um^2}^{(B)} := (1 - um^2 S^{(B)})^{-1}$, we arrive at

$$\mathbb{E}\langle (G_u - m)E_a \rangle = \sum_{a'} (\Theta_{um^2}^{(B)})_{aa'} \cdot \mathbb{E} \left(u \cdot m \sum_b \langle (G_u - m)E_b \rangle S_{ba'}^{(B)} \langle (G_u - m)E_{a'} \rangle \right).$$

By (50), we know that $\langle (G_u - m)E_a \rangle \prec M_u^{-1}$. Moreover, since $|1 - um^2|$ is bounded uniformly away from 0, the random walk representation in Lemma 2.14 gives $\sum_{a'} |(\Theta_{um^2}^{(B)})_{aa'}| = O(1)$. If we plug these two bounds into the right-hand side above, then the proof is complete. $\square$

PROOF OF LEMMA 2.19. We first assume that σ is alternating, i.e. $\sigma \in \{(+, -), (-, +)\}$; the case $\sigma \in \{(+, +), (-, -)\}$ is treated at the end of the proof. We will take expectation of the integrated loop hierarchy after applying the sum-zero operator. In particular, consider (168) with $s = 0$ and $n = 2$, and take the expectation of this identity. When we do this, the $m = 1$ term in (168) vanishes because $n = 2$, and the martingale term vanishes because we take expectation. Thus, we get

$$\begin{aligned} (188) \quad \mathcal{Q}_t \circ \mathbb{E}(\mathcal{L} - \mathcal{K})_{t, \sigma, \mathbf{a}} &= \int_0^t \left(\mathcal{U}_{u, t, \sigma} \circ \mathcal{Q}_u \circ \mathbb{E} \mathcal{E}_{u, \sigma}^{((\mathcal{L} - \mathcal{K}) \times (\mathcal{L} - \mathcal{K}))} \right)_{\mathbf{a}} du \\ &\quad + \int_0^t \left(\mathcal{U}_{u, t, \sigma} \circ \mathcal{Q}_u \circ \mathbb{E} \mathcal{E}_{u, \sigma}^{(\tilde{G})} \right)_{\mathbf{a}} du \\ &\quad + \int_0^t \left(\mathcal{U}_{u, t, \sigma} \circ \mathcal{Q}_u \circ \mathbb{E} [\mathcal{Q}_u, \Theta_{u, \sigma}] \circ (\mathcal{L} - \mathcal{K})_{u, \sigma} \right)_{\mathbf{a}} du \\ &\quad - \int_0^t \left(\mathcal{U}_{u, t, \sigma} \circ \mathcal{Q}_u \circ \mathbb{E} [\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u, \sigma} \cdot \dot{\vartheta}_u] \right)_{\mathbf{a}} du. \end{aligned}$$

(Note that there is no initial data term, since $(\mathcal{L} - \mathcal{K})$ vanishes at time 0.)

We can now control the expectation in the first term on the right-hand side, which is defined in (99), as follows. First, in (99), we use (u, τ, D) decay from Lemma 5.9 and the $O(1)$ range of $S^{(B)}$ to restrict the sum over a, b on the right-hand side of (99) to a total of $W^\tau \ell_u^2$ many indices (for $\tau > 0$ small). Then, we bound each loop on the right-hand side of (99) using (50). This gives the following analogue of (5.129) in [50]:

$$(189) \quad \mathbb{E} \mathcal{E}_{u, \sigma}^{((L - K) \times (L - K))} \prec W^2 \ell_u^2 M_u^{-4} \prec \eta_u^{-1} M_u^{-3}.$$

(We recall that $M_u = W^2 \ell_u^2 \eta_u$.) For the second term on the right-hand side of (188), we can control the expectation therein in the following way. Recall the definition of $\mathcal{E}_{u, \sigma}^{(\tilde{G})}$ from (37). Each summand in (37) is a product of a one-loop and a three-loop. In particular, we can use the same fast decay property from Lemma 5.9 to restrict the sum on the right-hand side of (37) to $O(\ell_u^2)$ -many indices. We get

$$\begin{aligned} (190) \quad \mathbb{E} \mathcal{E}_{u, \sigma, \mathbf{a}}^{(\tilde{G})} &\prec W^2 \ell_u^2 \cdot \max_a \max_{\mathbf{a}} \max_{\sigma \in \{+, -\}^3} \left| \mathbb{E} \left[\langle (G_u - m)E_a \rangle \cdot \mathcal{L}_{u, \mathbf{a}, \sigma} \right] \right| \\ &\prec W^2 \ell_u^2 \cdot \max_a \left| \mathbb{E} \langle (G_u - m)E_a \rangle \right| \cdot \max_{\mathbf{a}} \max_{\sigma \in \{+, -\}^3} |\mathcal{K}_{u, \mathbf{a}, \sigma}| \\ &\quad + W^2 \ell_u^2 \cdot \max_a |(\mathcal{L} - \mathcal{K})_{u, (+), (a)}| \cdot \max_{\mathbf{a}} \max_{\sigma \in \{+, -\}^3} |(\mathcal{L} - \mathcal{K})_{u, \mathbf{a}, \sigma}|. \end{aligned}$$

We now use Lemma 5.15 along with the estimates (49) and (50) to obtain

$$(191) \quad \mathbb{E}\mathcal{E}_{u,\sigma,\mathbf{a}}^{(\tilde{G})} \prec \eta_u^{-1} M_u^{-3}.$$

Let us now move to the last term in (188). Since the lengths of $\mathcal{L}$ and $\mathcal{K}$ are both equal to $n = 2$, we can use Lemma 3.2 (the Ward identity) to write

$$(192) \quad \mathbb{E}[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}] = \frac{1}{2iW^2\eta_u} (\mathbb{E}[\mathcal{L}_{u,(+)} - \mathcal{K}_{u,(+)}] - \mathbb{E}[\mathcal{L}_{u,(-)} - \mathcal{K}_{u,(-)}]) \prec \ell_u^2 M_u^{-3},$$

where the last bound follows from Lemma 5.15 and $M_u = W^2 \ell_u^2 \eta_u$. We now combine the above display with (173) to deduce

$$(193) \quad \|\mathbb{E}[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma} \cdot \dot{\vartheta}_u]\|_{\max} \prec \eta_u^{-1} M_u^{-3}.$$

Finally, for the third term on the right-hand side of (188), we follow (175), except we use the improved expectation bound (192). In particular, we have

$$(194) \quad \|\mathbb{E}[\mathcal{Q}_u, \Theta_{u,\sigma}] \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}\|_{\max} \prec \|\vartheta_u\|_{\max} \cdot \eta_u^{-1} \max_a |\mathbb{E}[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\sigma}]_a|$$

$$(195) \quad \prec \eta_u^{-1} M_u^{-3}.$$

We clarify that the last bound above uses also the bound $\|\vartheta_u\|_{\max} \prec \ell_u^{-2}$ for $n = 2$ which can be deduced from Definition 5.12 and Lemma 2.14. Now, we recall (179), and we recall from the sentence before (179) that each term which is acted on by $\mathcal{U}_{u,t,\sigma}$ in (188) has (u, τ, D) decay for any $\tau, D > 0$. This lets us apply the $\mathcal{U}_{u,t,\sigma}|_{\max \rightarrow \max}$ operator norm from case 4 of Lemma 7.3. If we combine this operator norm estimate with (188), (189), (191), (193), and (195), we deduce the following (for any $D > 0$ fixed):

$$\begin{aligned} \|\mathcal{Q}_t \circ \mathbb{E}(\mathcal{L} - \mathcal{K})_{t,\sigma}\|_{\max} &\prec \int_0^t \left(\frac{M_u}{M_t}\right)^2 \eta_u^{-1} M_u^{-3} du + W^{-D} \\ &\leq M_t^{-3} \int_0^t \eta_u^{-1} du + W^{-D} \prec M_t^{-3}. \end{aligned}$$

Finally, we again use (192) and the bound $\|\vartheta_t\|_{\max} \prec \ell_t^{-2}$ to deduce the estimate

$$\|\mathcal{P} \circ \mathbb{E}(\mathcal{L} - \mathcal{K})_{t,\sigma} \cdot \vartheta_t\|_{\max} \prec M_t^{-3}.$$

Combining the previous two displays and using $\mathcal{Q}_t \circ \mathbb{E}(\mathcal{L} - \mathcal{K})_{t,\sigma} + \mathcal{P} \circ \mathbb{E}(\mathcal{L} - \mathcal{K})_{t,\sigma} \cdot \vartheta_t = \mathbb{E}(\mathcal{L} - \mathcal{K})_{t,\sigma}$ completes the proof for alternating σ .

Finally, let $\sigma \in \{(+, +), (-, -)\}$. Then σ is not alternating, and the Ward identity used in (192) does not apply; however, the sum-zero operator is not needed in this case. We take the expectation of (104) with $s = 0$ and $n = 2$: the initial data term and the sum over $l_{\mathcal{K}} > 2$ vanish, and so does the expectation of the martingale term. The two remaining terms have (u, τ, D) decay, so case 1 of Lemma 7.3 applies to them, and by (189) and (191) (which hold for any $\sigma \in \{+, -\}^2$) we obtain

$$\|\mathbb{E}(\mathcal{L} - \mathcal{K})_{t,\sigma}\|_{\max} \prec \int_0^t \left(\frac{M_u}{M_t}\right)^2 \eta_u^{-1} M_u^{-3} du + W^{-D} \prec M_t^{-3}.$$

This completes the proof. $\square$

6. Proof of Lemma 5.1. As explained in the proof of Lemma 5.1 in [50], Cauchy-Schwarz allows us to reduce to loops $\mathcal{L}_{t,\sigma,\mathbf{a}}$ of even length $2m$ that are symmetric. In particular, we have the following, which is (6.4) in [50]:

$$(196) \quad \max_{\mathbf{a},\sigma} \left| \mathcal{L}_{t,\sigma,\mathbf{a}}^{(\text{length}=2m+1)} \right|^2 \leq \left| \max_{\mathbf{a},\sigma} \mathcal{L}_{t,\sigma,\mathbf{a}}^{(\text{length}=2m)} \cdot \max_{\mathbf{a},\sigma} \mathcal{L}_{t,\sigma,\mathbf{a}}^{(\text{length}=2m+2)} \right|.$$

We record this bound because it will be important shortly for another reason. Our proof of Lemma 5.1 proceeds inductively in m , where we recall that $2m$ is the length of $\mathcal{L}$. Recall Ω from (93). We claim

$$(197) \quad \mathbf{1}_\Omega \cdot \mathcal{L}_{t_2,\sigma',\mathbf{a}'} \prec (W^2 \ell_{t_1}^2 \eta_{t_2})^{-2m+1} + \mathbf{1}_\Omega \frac{\eta_{t_1}}{\eta_{t_2}} \cdot \left(\mathcal{L}_{t,\sigma_m^{(1)},\mathbf{a}_m} - \mathcal{L}_{t,\sigma_m^{(2)},\mathbf{a}_m} \right) M_{t_1}^{-1},$$

Recall that $2m$ is the length of $\mathcal{L}$. Above, we have also used the notation

$$\begin{aligned} \mathbf{a}_l &= (a_0, a_1, \dots, a_{l-1}, a_{l-1}, \dots, a_1) \\ \sigma_l^{(1)} &= (\sigma_1, \sigma_2, \dots, \sigma_{l-1}, \sigma_l, \bar{\sigma}_{l-1}, \dots, \bar{\sigma}_2, \bar{\sigma}_1), \\ \sigma_l^{(2)} &= (\sigma_1, \sigma_2, \dots, \sigma_{l-1}, \bar{\sigma}_l, \bar{\sigma}_{l-1}, \dots, \bar{\sigma}_2, \bar{\sigma}_1). \end{aligned}$$

In particular, $\mathcal{L}_{t_2,\sigma_m^{(1)},\mathbf{a}_m}$ and $\mathcal{L}_{t_2,\sigma_m^{(2)},\mathbf{a}_m}$ are loops of length $2m-1$, and $\sigma_m^{(1)}, \sigma_m^{(2)} \in \{+, -\}^{2m-1}$. The estimate (197) is (6.13) in [50]. It follows from the same argument; indeed, the only difference between (197) and (6.13) in [50] is the scaling of W^2 versus W and ℓ^2 versus ℓ . But in the proof of (6.13) in [50], the factors of W come from the scaling of E_a matrices, the form of the right-hand side of (92) when we apply induction in m , and the coefficient in the Ward identity in Lemma 3.2. In our setting, the correct factor is W^2 . The scaling of ℓ in the proof of (6.13) in [50] comes from the size of $\mathcal{I}_a$ sets therein (which are replaced by $\mathcal{I}_a^{(2)}$ here) and the form of the right-hand side of (92) when we apply induction in m . Thus, in our argument, ℓ should be replaced by ℓ^2 .

To perform the induction, we note that the event Ω in (93) implies $\mathbf{1}_\Omega \cdot \mathcal{L}_{t_2,\sigma_m^{(j)},\mathbf{a}_m} \prec 1$ when $m=1$ for $j=1, 2$. Next, we apply (196) with the left-hand side given by a loop of size $2m-1$ to bound it by a loop of length $2m-2$ and another of length $2m$. Then, we apply the inductive assumption that (94) holds for the loop of length $2m-2$. This gives the following for $j=1, 2$ on the left-hand side below:

$$\mathbf{1}_\Omega \cdot \mathcal{L}_{t_2,\sigma_m^{(j)},\mathbf{a}_m} \prec \mathbf{1}_\Omega \left[(W^2 \ell_{t_1}^2 \eta_{t_2})^{-2m+3} \max_{\sigma,\mathbf{a}} \left| \mathcal{L}_{t_2,\sigma,\mathbf{a}}^{\text{length}=2m} \right| \right]^{1/2}.$$

Plugging this into (197) gives

$$\mathbf{1}_\Omega \cdot \max_{\sigma',\mathbf{a}'} \left| \mathcal{L}_{t_2,\sigma',\mathbf{a}'}^{\text{length}=2m} \right| \prec (W^2 \ell_{t_1}^2 \eta_{t_2})^{-2m+1} + (W^2 \ell_{t_1}^2 \eta_{t_2})^{-m+1/2} \mathbf{1}_\Omega \max_{\sigma',\mathbf{a}'} \left| \mathcal{L}_{t_2,\sigma',\mathbf{a}'}^{\text{length}=2m} \right|^{1/2}$$

If $W^2 \ell_{t_1}^2 \eta_{t_2} \geq W^\tau$ for any $\tau > 0$, then we can move the last term to the left-hand side and deduce the desired estimate (94). If $W^2 \ell_{t_1}^2 \eta_{t_2} \leq W^\tau$ for $\tau > 0$ small, then the desired estimate (94) follows by the restriction to the event Ω from (93) anyway. This completes the proof. $\square$

7. Evolution kernel estimates. In this section we collect the bounds on $L_{\max \rightarrow \max}$ norm of $\mathcal{U}$ defined in (101). First, we state the trivial bound, proved in Lemma 7.1 of [50]. We note that the proof in [50] is independent of the dimension of the band matrix d .

LEMMA 7.1 ($\|\mathcal{U}\|_{\max \rightarrow \max}$ estimate). *Let $\mathcal{A}$ be a tensor $(\mathbb{Z}_L^2)^n \rightarrow \mathbb{C}$ and $n \geq 2$. Then*

$$(198) \quad \|\mathcal{U}_{s,t,\sigma} \circ \mathcal{A}\|_{\max} \prec \|\mathcal{A}\|_{\max} \cdot (\eta_s / \eta_t)^n, \quad s < t.$$

The following lemma shows how $\mathcal{U}_{s,t,(+,-)}$ affects the decay property of the 2-tensors.

LEMMA 7.2 (Tail Estimates). *Recall*

$$\mathcal{T}_t(\ell) := M_t^{-2} \exp\left(-|\ell/\ell_t|^{1/2}\right)$$

Fix $\sigma = (+, -)$. Assume that for $\mathcal{A}_a$, $a \in (\mathbb{Z}_L^2)^2$ and for some large $D > 0$, we have

$$\mathcal{A}_a \leq \mathcal{T}_s(|a_1 - a_2|_L) + W^{-D}$$

Recall $\mathcal{U}$ from Definition 101. We have

(199)

$$(\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_a \prec \mathcal{T}_t(|a_1 - a_2|_L) + W^{-D} \cdot (\eta_s/\eta_t)^2, \quad \text{if } |a_1 - a_2|_L \geq \ell_t^* := (\log W)^{3/2} \ell_t$$

The analogous statement was proved in Lemma 7.2 of [50] in case of one-dimensional band matrices $d = 1$. The proof of [50] goes through for $d = 2$ as well. The only two adjustments necessary are the infinity norm bound $\|\Theta_t^{(B)}\|_{\max} \prec \ell_t^{-2} \eta_t^{-1}$, and the summation bound $\sum_{b, |a-b|_L \leq \ell} 1 \prec \ell^2$ for any $a \in \mathbb{Z}_L^2$ and $\ell \in \mathbb{Z}_+$.

The main technical part of this section is the following improved $L_{\max \rightarrow \max}$ bounds on $\mathcal{U}$.

LEMMA 7.3 ($\mathcal{U}_{s,t}$ on fast decay tensor). *Let $\mathcal{A}$ be a tensor $(\mathbb{Z}_L^2)^n \rightarrow \mathbb{R}$, $n \geq 2$. We say that $\mathcal{A}$ has (τ, D) decay at time s if for some $\tau, D > 0$, the following bound holds:*

$$(200) \quad \max_{i,j} |a_i - a_j|_L \geq \ell_s W^\tau \implies \mathcal{A}_a = O(W^{-D}), \quad a = (a_1, a_2, \dots, a_n).$$

Now, suppose that $\mathcal{A}$ has (τ, D) decay at time s , and that $\mathcal{A}$ has rank $n \geq 2$. We have

$$(201) \quad (\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_a \leq C_n W^{C_n \tau} \cdot \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_t}{\ell_s}\right)^2 \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t}\right)^n + W^{-D+C_n}.$$

We will now consider five cases in which the bound (201) can be improved.

Case 1: Suppose that for some $1 \leq k \leq n$, we have

$$\sigma_k = \sigma_{k-1}, \quad \sigma = (\sigma_1, \dots, \sigma_n).$$

Then the following estimate holds:

$$(202) \quad (\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_a \leq W^{C_n \tau} \cdot \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t}\right)^n + W^{-D+C_n}$$

Case 2: Suppose that $\mathcal{A}_b$ has the sum zero property, i.e.

$$(203) \quad \sum_{a_2, \dots, a_n} \mathcal{A}_a = 0, \quad \forall a_1.$$

Then the following estimate holds:

$$(204) \quad (\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_a \leq W^{C_n \tau} \cdot \|\mathcal{A}\|_{\max} \cdot \frac{\ell_t}{\ell_s} \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t}\right)^n + W^{-D+C_n}$$

Case 3: Suppose that $\mathcal{A}_b$ has the sum zero property (203). Suppose also that $\mathcal{A}_b$ has the following form, which we explain afterwards:

$$(205) \quad \mathcal{A}_b = \sum_{u=0}^K \sum_{\mathbf{x}, \mathbf{y} \in (\mathbb{Z}_{W_L}^2)^u} C_{b,\mathbf{x},\mathbf{y}} \cdot \prod_{i=1}^u G_s(\sigma_i)_{x_i y_i}.$$

Here, $K = O(1)$ is fixed. The coefficients $C_{\mathbf{b}, \mathbf{x}, \mathbf{y}}$ are deterministic, and they satisfy the three conditions below. First, we have the following for some constant $C' = O(1)$:

$$\|C\|_{\max} \leq N^{C'}.$$

Second, there exists $\tau > 0$ such that

$$\max_i \min_k \left(|[x_i] - b_k|_L + |[y_i] - b_k|_L \right) \geq W^\tau \ell_s \Rightarrow C_{\mathbf{b}, \mathbf{x}, \mathbf{y}} = 0.$$

Third, let $\Lambda > 0$ be a deterministic constant such that

$$\|\mathcal{A}\|_{\max} \prec \Lambda.$$

Now, suppose $0 \leq s \leq 1 - N^{-1+\delta}$ for some fixed $\delta > 0$, and that (64) and (65) hold at time s . Then

$$(206) \quad (\mathcal{U}_{s,t,\sigma} \circ (\mathcal{A} - \mathbb{E}\mathcal{A}))_{\mathbf{a}} \prec C_n W^{C_n \tau} \cdot \Lambda \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^n + W^{-D+C_n}.$$

Case 4: Suppose that $\mathcal{A}_{\mathbf{b}}$ has the sum zero property (203) and that $\mathcal{A}$ is symmetric, i.e.:

$$\mathcal{A}_{a, a+s_2, \dots, a+s_n} = \mathcal{A}_{a, a-s_2, \dots, a-s_n}$$

for all $a, s_2, \dots, s_n \in \mathbb{Z}_L^2$. Then we have the following bound:

$$(207) \quad (\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_{\mathbf{a}} \leq W^{C_n \tau} \cdot \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^n + W^{-D+C_n}.$$

Case 5: Let $\mathcal{A}$ be a $2n$ -tensor $(\mathbb{Z}_L^2)^n \times (\mathbb{Z}_L^2)^n \rightarrow \mathbb{R}$, $n \geq 2$ with (τ, D) decay at time s . Suppose that $\mathcal{A}$ has the double sum zero property, meaning that

$$\sum_{a_2, \dots, a_n \in \mathbb{Z}_L^2} \mathcal{A}_{a, \mathbf{a}'} = 0 \quad \forall a_1, \mathbf{a}', \quad \text{and} \quad \sum_{a_2', \dots, a_n' \in \mathbb{Z}_L^2} \mathcal{A}_{a, \mathbf{a}'} = 0 \quad \forall a, \mathbf{a}'.$$

Then we have the bound

$$(208) \quad ((\mathcal{U}_{s,t,\sigma} \otimes \mathcal{U}_{s,t,\bar{\sigma}}) \circ \mathcal{A})_{\mathbf{a}, \mathbf{a}} \leq W^{C_n \tau} \cdot \|\mathcal{A}\|_{\max} \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{2n} + W^{-D+C_n}.$$

The cases 1-2 of this Lemma 7.3 are analogous to Lemma 7.3 of [50]. However, in two dimensions the additional factor $\frac{\ell_s}{\ell_t}$ coming from the sum zero property of $\mathcal{A}$ is insufficient for the analysis of the loop hierarchy in Section 5. Thus, we show that another $\frac{\ell_s}{\ell_t}$ factor can be obtained from symmetry, double sum zero property of $\mathcal{A}$ in cases 4-5, or a CLT-like cancellation in case 3.

PROOF OF LEMMA 7.3. By definition of $\mathcal{U}_{s,t}$ in (101) and (102), we have

$$(209) \quad (\mathcal{U}_{s,t,\sigma} \circ \mathcal{A})_{\mathbf{a}} = \sum_{b_1, \dots, b_n} \prod_{i=1}^n \psi_i \cdot \mathcal{A}_{\mathbf{b}}, \quad \mathbf{b} = (b_1, \dots, b_n),$$

$$(210) \quad \psi_i = \delta_{a_i b_i} + \Xi_i, \quad \Xi_i := -(s-t)\xi_i \cdot \left(S^{(B)} \cdot \Theta_{t\xi_i}^{(B)} \right)_{a_i b_i}, \quad \xi_i = m(\sigma_i)m(\sigma_{i+1}).$$

We now expand

$$\sum_{b_1, \dots, b_n} \prod_{i=1}^n \psi_i \cdot \mathcal{A}_{\mathbf{b}} = \sum_{A \subset \llbracket 1, n \rrbracket} \sum_{b_1, \dots, b_n} \left(\prod_{j \in A^c} \Xi_j \right) \left(\prod_{j \in A} \delta_{a_j b_j} \right) \cdot \mathcal{A}_{\mathbf{b}}.$$

By (210), Lemma 2.14 and $|t-s| \leq C\eta_s$, the entries of Ξ are $O_{\prec}(\eta_s \ell_t^{-2} \eta_t^{-1})$. For any $A \neq \emptyset$, there must be at least one factor of $\delta_{a_j b_j}$ on the r.h.s. above. By the (τ, D) decay property of $\mathcal{A}_{\mathbf{b}}$ at time s , this implies that all but $O(W^\tau \ell_s^2)^{|A^c|}$ many indices in the sum over $b_1, \dots, b_n$ on the r.h.s. must be $O(W^{-D})$. Ultimately, we deduce that

$$\begin{aligned} \sum_{\emptyset \neq A \subset \llbracket 1, n \rrbracket} \sum_{b_1, \dots, b_n} \prod_{i=1}^n \psi_i \cdot \mathcal{A}_{\mathbf{b}} &\leq \sum_{\emptyset \neq A \subset \llbracket 1, n \rrbracket} C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{|A^c|} \|\mathcal{A}\|_{\max} + W^{-D+C_n} \\ (211) \quad &\leq C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^n \|\mathcal{A}\|_{\max} + W^{-D+C_n}, \end{aligned}$$

where the last bound follows because $\ell_s^2 \eta_s \geq \ell_t^2 \eta_t$.

Thus we only need to prove the bounds (201), (202), (204), (207) and (206) on the main term $\sum_{\mathbf{b}} (\prod_{i=1}^n \Xi_i) \mathcal{A}_{\mathbf{b}}$. By definition of $\Theta^{(B)}$, we have

$$\Xi_i \prec \eta_s \ell_t^{-2} \eta_t^{-1}.$$

Then, by (τ, D) decay property of $\mathcal{A}$ at time s in (200), we have

$$\begin{aligned} (212) \quad \sum_{\mathbf{b}} \prod_{i=1}^n \Xi_i \cdot \mathcal{A}_{\mathbf{b}} &\leq C W^\tau \cdot \frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \max_{b_n} \cdot \left| \sum_{b_1 \dots b_{n-1}} \prod_{i=1}^{n-1} \Xi_i \cdot \mathcal{A}_{\mathbf{b}} \right| + W^{-D+C} \\ &\leq C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-1} \max_{b_2, \dots, b_n} \left| \sum_{b_1} \Xi_1 \mathcal{A}_{\mathbf{b}} \right| + W^{-D+C_n} \end{aligned}$$

By the decay of (τ, D) decay of $\Theta_t^{(B)}$ at time t , we have

$$(213) \quad \sum_{b_1} \Xi_1 |\mathcal{A}_{\mathbf{b}}| \leq C W^\tau (\eta_s / \eta_t) \|\mathcal{A}\|_{\max} + W^{-D}.$$

Together with (212) and (211), we deduce (201). For a non-alternating tensor, we can assume without loss of generality that $\sigma_1 = \sigma_2 = +$. Then $\sum_{b_1} |\Xi_1| = O(1)$, and we get an improved bound

$$\sum_{b_1} \Xi_1 |\mathcal{A}_{\mathbf{b}}| \leq C \|\mathcal{A}\|_{\max}.$$

Together with (212) and (211), this proves (202).

Now we assume that $\mathcal{A}$ has the sum zero property (203). Since we proved (202) for case 1, we can assume that $\sigma_i \neq \sigma_{i+1}$ for any $1 \leq i \leq n$, and thus $\xi_i = |m|^2 = 1$ (in (210)). For each Ξ_i with $i \geq 2$, we write

$$\Xi_i = \Xi_i^0 + \Xi_i^1 + \Xi_i^2 := \Xi^0(a_i - b_1) + \Xi^1(a_i - b_1, r_i) + \Xi^2(a_i - b_1, r_i),$$

where the shifts are defined by $r_i = b_i - b_1$ for $2 \leq i \leq n$, and

$$\begin{aligned} \Xi^0(a) &= (t-s) \left(S^{(B)} \Theta_t^{(B)} \right)_{a,0}, \\ \Xi_i^1(a, r) &= (t-s) \left[\frac{1}{2} \left(S^{(B)} \Theta_t^{(B)} \right)_{a,r} - \frac{1}{2} \left(S^{(B)} \Theta_t^{(B)} \right)_{a,-r} \right], \\ \Xi_i^2(a, r) &= (t-s) \left[\frac{1}{2} \left(S^{(B)} \Theta_t^{(B)} \right)_{a,r} + \frac{1}{2} \left(S^{(B)} \Theta_t^{(B)} \right)_{a,-r} - \left(S^{(B)} \Theta_t^{(B)} \right)_{a,0} \right]. \end{aligned}$$

By Lemma 2.14 and $t - s = O(\eta_s)$, we have

$$(214) \quad |\Xi^{0,1,2}(a)| \prec \frac{\eta_s}{\ell_t^2 \eta_t},$$

$$(215) \quad |\Xi^{1,2}(a, r)| \prec \frac{\eta_s |r|_L}{|a|_L + 1} + \frac{\eta_s |r|_L}{\ell_t^2 \eta_t^{1/2}},$$

$$(216) \quad |\Xi^2(a, r)| \prec \frac{\eta_s |r|_L^2}{(|a|_L + 1)^2} + \frac{\eta_s |r|_L^2}{\ell_t^2}.$$

We expand the main term

$$(217) \quad \sum_{\mathbf{b}} \left(\prod_{i=1}^n \Xi_i \right) \cdot \mathcal{A}_{\mathbf{b}} = \sum_{\mathbf{b}} \Xi_1 \left(\prod_{i=2}^n (\Xi_i^0 + \Xi_i^1 + \Xi_i^2) \right) \cdot \mathcal{A}_{\mathbf{b}} \\ = \sum_{\llbracket 2, n \rrbracket = A \sqcup B \sqcup C} \sum_{\mathbf{b}} \prod_{i \in A} \Xi_i^0 \prod_{i \in B} \Xi_i^1 \prod_{i \in C} \Xi_i^2 \cdot \Xi_1 \mathcal{A}_{\mathbf{b}}.$$

The leading term, with $B, C = \emptyset$, vanishes due to the sum zero property of $\mathcal{A}$. For other partitions of $\llbracket 2, n \rrbracket$, we pick an $i \in B \cup C$ and bound the corresponding $\Xi_i^{1,2}$ with (215). The rest of the $\Xi_i^{0,1,2}$ factors are bounded by (214). By the decay (200) of $\mathcal{A}$, the main contribution to the last equation comes from $|r_i|_L \leq W^\tau \ell_s$, thus the summation over each r_i contributes a factor of $W^{2\tau} \ell_s^2$. Therefore, we obtain the following bound for (212).

$$(218) \quad \sum_{\mathbf{b}} \prod_{i=1}^n \Xi_i \cdot \mathcal{A}_{\mathbf{b}} \leq C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-2} \max_{b_2, \dots, b_n} \sum_{b_1} \left[\sum_{i=2}^n \frac{\eta_s \ell_s^3}{|a_i - b_1|_L + 1} + \frac{\eta_s \ell_s^3}{\ell_t^2 \eta_t^{1/2}} \right] \Xi_1 |\mathcal{A}_{\mathbf{b}}| + W^{-D+C_n}$$

Due to the exponential decay property of Ξ_1 , we can restrict the summation over b_1 to $|a_1 - b_1|_L \leq W^\tau \ell_t$. For $i \geq 2$, we have the following summation bound

$$(219) \quad \sum_{|b_1 - a_1| \leq W^\tau \ell_t} \frac{1}{|a_i - b_1|_L + 1} \leq C W^\tau \ell_t.$$

Using this together with the entrywise bound (214) on Ξ_1 and (213), we have

$$(220) \quad \sum_{\mathbf{b}} \prod_{i=1}^n \Xi_i \cdot \mathcal{A}_{\mathbf{b}} \leq C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-2} \left[\frac{(\ell_s^2 \eta_s)^2}{\ell_t^2 \eta_t} \cdot \frac{\ell_t}{\ell_s} + \frac{(\ell_s^2 \eta_s)^2}{(\ell_t^2 \eta_t)^{3/2}} \cdot \frac{\ell_t}{\ell_s} \right] \|\mathcal{A}\|_{\max} + W^{-D+C_n}.$$

This implies (204), since $\ell_t^2 \eta_t = O(1)$. In case 4, we additionally have the symmetry property of $\mathcal{A}$, so that $\Xi^1(a, r) = -\Xi^1(a, -r)$ by definition, and thus

$$\sum_{r_2, \dots, r_n} \Xi^1(a_i - b_1, r_i) \mathcal{A}_{b_1, b_1 + r_2, \dots, b_1 + r_n} = 0.$$

Then the terms in the expansion of (217) with $|B| = 1$ and $C = \emptyset$ vanish as well. We split the remaining non-zero terms in (217) into two groups. The first group contains the partitions with $C \neq \emptyset$, where we use (216) on one of Ξ_i^2 with $i \in C$ and (214) on the other $\Xi_i^{0,1,2}$. The second group covers the partitions with $C = \emptyset$, $|B| \geq 2$, where we use (215) for two terms $\Xi_{i,j}^1$ with $i, j \in B$ and (214) on the other $\Xi_i^{0,1}$ terms. In total we get

$$\sum_{\mathbf{b}} \prod_{i=1}^n \Xi_i \cdot \mathcal{A}_{\mathbf{b}} \leq C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-2} \max_{b_2, \dots, b_n} \sum_{b_1} \left[\sum_{i=2}^n \frac{\ell_s^4 \eta_s}{(|a_i - b_1|_L + 1)^2} + \frac{\ell_s^4 \eta_s}{\ell_t^2} \right] \Xi_1 |\mathcal{A}_{\mathbf{b}}| + W^{-D+C_n}$$

(221)

$$+ \delta_{n \geq 3} C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-3} \max_{b_2, \dots, b_n} \sum_{b_1} \left[\sum_{i=2}^n \frac{\ell_s^3 \eta_s}{|a_i - b_1|_L + 1} + \frac{\ell_s^3 \eta_s}{\ell_t^2 \eta_t^{1/2}} \right]^2 \Xi_1 |\mathcal{A}_{\mathbf{b}}|$$

Similarly to the proof of case 2, we now bound Ξ_1 entrywise using (214), use the decay of Ξ_1 to restrict b_1 summation to $|b_1 - a_1|_L \leq W^\tau \ell_t$ and sum out the prefactors using (219) and

$$(222) \quad \sum_{|b_1 - a_1|_L \leq W^\tau \ell_t} \frac{1}{(|a_i - b_1|_L + 1)(|a_j - b_1|_L + 1)} \leq W^\tau$$

for $i, j \geq 2$. Then

$$\begin{aligned} \sum_{\mathbf{b}} \prod_{i=1}^n \Xi_i \cdot \mathcal{A}_{\mathbf{b}} &\leq C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-2} \frac{(\ell_s^2 \eta_s)^2}{\ell_t^2 \eta_t} \|\mathcal{A}\|_{\max} \\ &\quad + \delta_{n \geq 3} C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-3} \left[\frac{(\ell_s^2 \eta_s)^3}{\ell_t^2 \eta_t} + \frac{(\ell_s^2 \eta_s)^3}{(\ell_t^2 \eta_t)^{3/2}} + \frac{(\ell_s^2 \eta_s)^3}{(\ell_t^2 \eta_t)^2} \right] \|\mathcal{A}\|_{\max} + W^{-D+C_n}. \end{aligned}$$

This finishes the proof of (207).

We now prove (206). Because $\mathcal{A}_{\mathbf{b}}$ has the fast decay property, at the cost of an error term of the form $O(W^{-D})$ for any $D > 0$ fixed, we can restrict $\mathcal{A}_{\mathbf{b}}$ to indices $\mathbf{b} = (b_1, \dots, b_n)$ such that

$$\max_i |b_i - b_1|_L \leq W^\tau \ell_s.$$

In particular, we can assume that in the representation (205), the coefficients $C_{\mathbf{b}, \mathbf{x}, \mathbf{y}}$ satisfy

$$(223) \quad \max_i \left(|[x_i] - b_1|_L + |[y_i] - b_1|_L \right) \geq W^{2\tau} \ell_s \Rightarrow C_{\mathbf{b}, \mathbf{x}, \mathbf{y}} = 0.$$

Take (217). Because $\mathcal{A}_{\mathbf{b}}$ has the sum-zero property by assumption, the term with $B, C = \emptyset$ vanishes as explained after (217). For other partitions $\llbracket 2, n \rrbracket = A \sqcup B \sqcup C$, there must be an index $i \in B \sqcup C$ with a factor of either Ξ_i^1 or Ξ_i^2 ; as before, call this factor $\Xi_i^{1,2}$, and without loss of generality (upon relabeling indices), assume that $i = 2$. Thus, the quantity from (217) that we must control has the form

$$\sum_{\llbracket 2, n \rrbracket = A \sqcup B \sqcup C} \sum_{b_1} \Xi_1 \left(\sum_{b_2, \dots, b_n} \Xi_2^{1,2} \prod_{i \in A} \Xi_i^0 \prod_{i \in B \setminus \{2\}} \Xi_i^1 \prod_{i \in C \setminus \{2\}} \Xi_i^2 \cdot (\mathcal{A}_{\mathbf{b}} - \mathbb{E} \mathcal{A}_{\mathbf{b}}) \right)$$

By the fast decay property of $\mathcal{A}_{\mathbf{b}}$, the sum over $b_2, \dots, b_n$ can be restricted to $b_2, \dots, b_n$ such that $|b_i - b_1|_L \leq W^\tau \ell_s$ for every $i = 2, \dots, n$. Moreover, if we use the fast decay of Ξ_1 , it suffices to restrict to b_1 such that $|b_1 - a_1|_L \leq W^\tau \ell_t$ in the above display. In particular, the object we must control (in terms of the right-hand side of the desired estimate (206)) is

$$(224) \quad \Gamma := \sum_{|b_1 - a_1|_L \leq W^\tau \ell_t} \Xi_1 \left(\sum_{b_2, \dots, b_n} \Xi_2^{1,2} \prod_{i \in A} \Xi_i^0 \prod_{i \in B \setminus \{2\}} \Xi_i^1 \prod_{i \in C \setminus \{2\}} \Xi_i^2 \cdot (\mathcal{A}_{\mathbf{b}} - \mathbb{E} \mathcal{A}_{\mathbf{b}}) \right).$$

Now, recall that $\mathcal{A}_{\mathbf{b}} \prec \Lambda$. Also, by (205) and the assumption that $s \leq 1 - N^{-1+\delta}$ for some fixed $\delta > 0$, we have $\mathcal{A}_{\mathbf{b}} = O(W^C)$ for some $C = O(1)$. Thus, we deduce $\mathbb{E} \mathcal{A}_{\mathbf{b}} \prec \Lambda + W^{-D'}$ for any $D' = O(1)$. So, if we follow the derivation of (218), we deduce the following for any $D = O(1)$:

$$\sum_{b_2, \dots, b_n} \Xi_2^{1,2} \prod_{i \in A} \Xi_i^0 \prod_{i \in B \setminus \{2\}} \Xi_i^1 \prod_{i \in C \setminus \{2\}} \Xi_i^2 (\mathcal{A}_{\mathbf{b}} - \mathbb{E} \mathcal{A}_{\mathbf{b}}) \prec \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-2} \eta_s \ell_s^3 \left(\frac{1}{1 + |a_2 - b_1|_L} + \frac{1}{\ell_t^2 \eta_t^{1/2}} \right) \Lambda + W^{-D}.$$

We now use Lemma 7.4 in two ways. First, we use case 2 of Lemma 7.4 with

$$Z_{ab_1} := (1 + |a - b_1|_L)^{-1} \cdot \eta_s^{-1} \ell_t^2 \eta_t \cdot (\Xi_1)_{a_1 b_1} \cdot \mathbf{1}[|b_1 - a|_L \leq W^\tau \ell_t],$$

$$Y_{s, b_1} := \left(\frac{1}{1 + |a_2 - b_1|_L} + \frac{1}{\ell_t^2 \eta_t^{1/2}} \right)^{-1} \cdot \sum_{b_2, \dots, b_n} \Xi_2^{1,2} \prod_{i \in A} \Xi_i^0 \prod_{i \in B \setminus \{2\}} \Xi_i^1 \prod_{i \in C \setminus \{2\}} \Xi_i^2 \cdot (\mathcal{A}_b - \mathbb{E} \mathcal{A}_b).$$

(In this application of Lemma 7.4, the τ parameter there is equal to two times the τ parameter here; see the decay condition in (223), for example. Moreover, recall from (214) that $\Xi_1 \prec \eta_s \ell_t^{-2} \eta_t^{-1}$. Thus, our choice of Z_{ab_1} satisfies the assumption (231).) Recall also that

$$Y_{s, b_1} \prec \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-2} \cdot \eta_s \ell_s^3 \cdot \Lambda.$$

Thus, when we do this application of Lemma 7.4, we arrive at the following for any $D = O(1)$:

$$\begin{aligned} \eta_s \ell_t^{-2} \eta_t^{-1} \cdot \sum_{b_1} Z_{a_2 b_1} Y_{s, b_1} &\prec W^{C\tau} \ell_s \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-2} \cdot \eta_s \ell_t^{-2} \eta_t^{-1} \cdot \eta_s \ell_s^3 \cdot \Lambda + W^{-D} \\ &\prec W^{C\tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^n \ell_t^2 \eta_t \cdot \Lambda + W^{-D}. \end{aligned}$$

Recalling that $\ell_t^2 \leq \eta_t^{-1}$ (see (49) and (26)) then yields

$$(225) \quad \eta_s \ell_t^{-2} \eta_t^{-1} \cdot \sum_{b_1} Z_{a_2 b_1} Y_{s, b_1} \prec W^{C\tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^n \cdot \Lambda + W^{-D}.$$

Next, we use (230) (using $\tilde{Z}$ instead of Z in order to avoid confusion with the above notation) with

$$\tilde{Z}_{ab_1} := \ell_t^{-2} \eta_t^{-\frac{1}{2}} \cdot \eta_s^{-1} \ell_t^2 \eta_t \cdot (\Xi_1)_{ab_1} \cdot \mathbf{1}[|b_1 - a|_L \leq W^\tau \ell_t]$$

and the same choice of Y_{s, b_1} as above. Note that by (214), this choice of $\tilde{Z}_{ab_1}$ satisfies $\max_b |\tilde{Z}_{ab_1}| \prec \ell_t^{-2} \eta_t^{-1/2}$. Thus, we deduce the following for any $D = O(1)$:

$$\begin{aligned} \eta_s \ell_t^{-2} \eta_t^{-1} \cdot \sum_{b_1} \tilde{Z}_{a_1 b_1} Y_{s, b_1} &\prec W^{C\tau} \ell_t \ell_s \cdot \ell_t^{-2} \eta_t^{-\frac{1}{2}} \cdot \eta_s \ell_t^{-2} \eta_t^{-1} \cdot \eta_s \ell_s^3 \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{n-2} \cdot \Lambda + W^{-D} \\ &\prec W^{C\tau} \cdot \ell_t \eta_t^{\frac{1}{2}} \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^n \cdot \Lambda + W^{-D}. \end{aligned}$$

If we again use $\ell_t \eta_t^{\frac{1}{2}} \leq 1$, then we obtain

$$\eta_s \ell_t^{-2} \eta_t^{-1} \cdot \sum_{b_1} \tilde{Z}_{a_1 b_1} Y_{s, b_1} \prec W^{C\tau} \cdot \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^n \cdot \Lambda + W^{-D}.$$

Finally, we observe that, in the above notation, we have

$$\Gamma = \eta_s \ell_t^{-2} \eta_t^{-1} \sum_{b_1} Z_{a_2 b_1} Y_{s, b_1} + \eta_s \ell_t^{-2} \eta_t^{-1} \sum_{b_1} \tilde{Z}_{a_1 b_1} Y_{s, b_1}.$$

Combining the previous two displays with (225) shows that Γ is controlled by the right-hand side of the desired estimate (206). As noted immediately prior to (224), this completes the proof of case 3, i.e. (206).

Finally, we consider the case 5 of a double sum zero tensor $\mathcal{A}$. Due to (202), we only need to consider the case when σ is alternating. Similarly to the previous cases, we have

$$((\mathcal{U}_{s,t,\sigma} \otimes \mathcal{U}_{s,t,\bar{\sigma}}) \circ \mathcal{A})_{\mathbf{a},\mathbf{a}} = \sum_{\mathbf{b},\mathbf{b}'} \prod_{i=1}^n \psi_i \psi'_i \cdot \mathcal{A}_{\mathbf{b},\mathbf{b}'},$$

where ψ_i are defined in (210) and ψ'_i are defined similarly, up to replacing the b_i with b'_i and ξ_i with $\bar{\xi}_i$. More precisely,

$$(226) \quad \psi'_i = \delta_{a_i b'_i} + \Xi'_i, \quad \Xi'_i := -(s-t)\bar{\xi}_i \cdot \left(S^{(B)} \cdot \Theta_{t\xi_i}^{(B)} \right)_{a_i b'_i}.$$

Using a $2n$ -tensor version of (211), we get that

$$\left| \sum_{\mathbf{b},\mathbf{b}'} \left(\prod_{i=1}^n \psi_i \psi'_i - \prod_{i=1}^n \Xi_i \Xi'_i \right) \cdot \mathcal{A}_{\mathbf{b},\mathbf{b}'} \right| \leq C_n \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{2n} \|\mathcal{A}\|_{\max} + W^{-D+C_n}.$$

Thus, we only need to bound $\sum_{\mathbf{b},\mathbf{b}'} \prod_{i=1}^n \Xi_i \Xi'_i \cdot \mathcal{A}_{\mathbf{b},\mathbf{b}'}$. We decompose $\Xi_i = \Xi_i^0 + \Xi_i^*$ and $\Xi'_i = \Xi_i'^0 + \Xi_i'^*$, where $r_i = b_i - b_1$, $r'_i = b'_i - b_1$ and

$$\begin{aligned} \Xi_i^0 &= \Xi^0(a_i - b_1), & \Xi_i^* &= \Xi^1(a_i - b_1, r_i) + \Xi^2(a_i - b_1, r_i), \\ \Xi_i'^0 &= \Xi^0(a_i - b_1), & \Xi_i'^* &= \Xi^1(a_i - b_1, r'_i) + \Xi^2(a_i - b_1, r'_i). \end{aligned}$$

Due to the double sum zero property of $\mathcal{A}$, we have

$$\sum_{\mathbf{b} \setminus b_1} \prod_{i=1}^n \Xi_i^0 \mathcal{A}_{\mathbf{b},\mathbf{b}'} = 0, \quad \sum_{\mathbf{b}' \setminus b'_1} \prod_{i=1}^n \Xi_i'^0 \mathcal{A}_{\mathbf{b},\mathbf{b}'} = 0.$$

These properties allow us to expand

$$\sum_{\mathbf{b} \setminus b_1, \mathbf{b}' \setminus b'_1} \prod_{i=1}^n \Xi_i \Xi'_i \cdot \mathcal{A}_{\mathbf{b},\mathbf{b}'} = \Xi_1 \Xi_1' \sum_{B, B' \subset \llbracket 2, n \rrbracket, B, B' \neq \emptyset} \prod_{i \in B^c} \Xi_i^0 \prod_{i \in B'^c} \Xi_i'^0 \sum_{\mathbf{b} \setminus b_1, \mathbf{b}' \setminus b'_1} \prod_{i \in B} \Xi_i^* \prod_{i \in B'} \Xi_i'^* \mathcal{A}_{\mathbf{b},\mathbf{b}'}.$$

We bound one of the factors Ξ_i^* with $i \in B$ and one of the factors $\Xi_i'^*$ using (215). We control the rest of the factors $\Xi_i^0, \Xi_i'^0, \Xi_i^*, \Xi_i'^*$ for $i \geq 2$ and Ξ_1' using (214). Then

$$(227) \quad \sum_{\mathbf{b},\mathbf{b}'} \prod_{i=1}^n \Xi_i \Xi'_i \cdot \mathcal{A}_{\mathbf{b},\mathbf{b}'} \leq C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{2n-3} \cdot \max_{\mathbf{b} \setminus b_1, \mathbf{b}' \setminus b'_1} \sum_{b_1} \left[\sum_{i=2}^n \frac{\ell_s^3 \eta_s}{|a_i - b_1|_L + 1} + \frac{\ell_s^3 \eta_s}{\ell_t^2 \eta_t^{1/2}} \right]^2 \cdot \Xi_1 |\mathcal{A}_{\mathbf{b},\mathbf{b}'}| + W^{-D+C_n}.$$

The bound in this case is completely analogous to the bound on the second line of (221) and we get

$$\sum_{\mathbf{b},\mathbf{b}'} \prod_{i=1}^n \Xi_i \Xi'_i \cdot \mathcal{A}_{\mathbf{b},\mathbf{b}'} \leq C_n W^{C_n \tau} \left(\frac{\ell_s^2 \eta_s}{\ell_t^2 \eta_t} \right)^{2n} \cdot \|\mathcal{A}\|_{\max} + W^{-D+C_n},$$

which completes the proof. $\square$

We now record the following lemma, which was used in the proof of (206).

LEMMA 7.4. *Fix $u = O(1)$, and consider a function of $b \in \mathbb{Z}_L^2$ of the following form:*

$$(228) \quad Y_{s,b} = \sum_{\mathbf{x}, \mathbf{y} \in (\mathbb{Z}_{W_L}^2)^u} C_{b,\mathbf{x},\mathbf{y}} \cdot \prod_{i=1}^u G_s(\sigma_i)_{x_i y_i}.$$

Above, $u = O(1)$ is fixed. The coefficients $C_{b,x,y}$ are deterministic, and they satisfy the following conditions. First, for some $C' = O(1)$, we have

$$\|C\|_{\max} \leq N^{C'}.$$

Second, there exists $\tau > 0$ such that

$$\max_i \left(|[x_i] - b|_L + |[y_i] - b|_L \right) \geq W^\tau \ell_s \Rightarrow C_{b,x,y} = 0.$$

Third, there exists a deterministic constant $\Lambda > 0$ such that

$$\max_b |Y_{s,b}| \prec \Lambda.$$

Moreover, suppose that $0 \leq s \leq 1 - N^{-1+\delta}$ for fixed $\delta > 0$, and that (64) and (65) hold at time s .

Now, let $Z = (Z_{ab})_{a,b \in \mathbb{Z}_L^2}$ be a deterministic matrix. We consider two separate cases.

Case 1: the matrix Z satisfies the following constraint:

$$(229) \quad |a - b|_L \geq W^\tau \ell_t \Rightarrow Z_{ab} = 0.$$

In this case, for any $D > 0$ and for some $C = O(1)$, we have

$$(230) \quad \sum_b Z_{ab} (Y_{s,b} - \mathbb{E}Y_{s,b}) \prec W^{C\tau} \ell_t \ell_s \cdot \max_b |Z_{ab}| \cdot \Lambda + W^{-D} \cdot \max_b |Z_{ab}|.$$

Case 2: suppose that we have the bound

$$(231) \quad Z_{ab} \prec (1 + |a - b|_L)^{-1}, \quad a, b \in \mathbb{Z}_L^2.$$

Then, for any $D > 0$ and some $C = O(1)$, we have

$$(232) \quad \sum_b Z_{ab} (Y_{s,b} - \mathbb{E}Y_{s,b}) \prec W^{C\tau} \cdot \ell_s \cdot \Lambda + W^{-D}.$$

Note that the trivial bound on the left-hand side of (230) has ℓ_t instead of ℓ_s on the right-hand side of (230). The key observation leading to the improved bound is that Y_{s,b_1} and Y_{s,b_2} are essentially independent when $|b_1 - b_2|_L \gg \ell_s$. Then the additional factor of $\frac{\ell_s}{\ell_t}$ comes from the CLT-like cancellation of the left-hand side. Here we explain the source of this independence. We claim that, heuristically, the resolvent entries $G_{s,xy}$ in the ℓ_s range around the (b, b) block of the matrix (meaning that $|[x] - b|_L + |[y] - b|_L \leq W^\tau \ell_s$) depend only on the entries of H in the ℓ_s range around b . Indeed, we can compute the correlations of H_{ij} with G_{xy} by integration by parts:

$$(233) \quad \mathbb{E}H_{ij}G_{xy} = S_{ij}\mathbb{E}G_{xj}G_{iy}, \quad \mathbb{E}\overline{H}_{ij}G_{xy} = S_{ij}\mathbb{E}G_{xi}G_{jy}.$$

Due to the decay property of the resolvent entries (65), the correlations (233) are $O(W^{-D})$ for any $D > 0$ when $|[i] - b|_L + |[j] - b|_L \gg W^\tau \ell_s$. This (heuristically) suggests that G_{xy} and $G_{x'y'}$ are almost independent if (x, y) is far from (x', y') . From the definition (228) of $Y_{s,b}$, we see that the random functions Y_{s,b_1} and Y_{s,b_2} depend only on the resolvent entries in the ℓ_s range of b_1 and b_2 blocks, respectively, up to a $O(W^{-D})$ error term.

Thus, the randomness in Y_{s,b_1} and Y_{s,b_2} essentially comes from two disjoint subsets of the entries of H in case $|b_1 - b_2|_L \gg W^\tau \ell_s$. This implies their (almost) independence.

In order to make this argument rigorous, we compute $\mathbb{E}Y_{s,b_1}Y_{s,b_2}$ by interpolating between this correlation and $\mathbb{E}Y'_{s,b_1}Y_{s,b_2}$, where Y'_{s,b_1} is an independent copy of Y_{s,b_1} . Interpolation is achieved by viewing Y_{s,b_1} as a function of entries of H and replacing the entries by their

independent copies one by one. Then on a single step of the interpolation the change in the correlation is bounded in one of the two ways. One option is that the replaced entry H_{ij} is located far away from b_1 ($||[i] - b_1|_L + |[j] - b_1|_L \gg W^\tau \ell_s$), then the change in the function Y_{s,b_1} is $O(W^{-D})$. The other option is that H_{ij} is far away from b_2 ($||[i] - b_2|_L + |[j] - b_2|_L \gg W^\tau \ell_s$). In this case, we Taylor expand Y_{s,b_1} in H_{ij} variable and use the bounds on the correlations of H_{ij} with Y_{s,b_2} explained below (233). A similar argument controls higher moments of the left-hand side of (230). We present the full proof with details below.

PROOF. Throughout this argument, we will let $G_s = (\sqrt{s}H - z_s)^{-1}$; since the final conclusion (230) is a statement about the law of G_s , this convention is okay. We compute $2p$ -th moments. We have

$$(234) \quad \mathbb{E} \left| \sum_b Z_{ab}(Y_{s,b} - \mathbb{E}Y_{s,b}) \right|^{2p} = \sum_{b_1, \dots, b_{2p}} \prod_{k=1}^p Z_{ab_{2k-1}} \prod_{k=p+1}^{2p} \overline{Z}_{ab_k} \mathbb{E} \left[\prod_{k=1}^p (Y_{s,b_{2k-1}} - \mathbb{E}Y_{s,b_{2k-1}}) \prod_{k=p+1}^{2p} (\overline{Y}_{s,b_k} - \mathbb{E}\overline{Y}_{s,b_k}) \right].$$

Suppose that there exists an index, say b_1 , such that $\min_{k=2, \dots, 2p} |b_1 - b_k|_L \geq W^{2\tau} \ell_s$. We claim that

$$(235) \quad \left| \mathbb{E} \left[\prod_{k=1}^p (Y_{s,b_k} - \mathbb{E}Y_{s,b_k}) \prod_{k=p+1}^{2p} (\overline{Y}_{s,b_k} - \mathbb{E}\overline{Y}_{s,b_k}) \right] \right| \leq c_D W^{-D}$$

for any $D > 0$, where c_D depends only on D . To prove this, we first realize $Y_{s,b_k} = Y_{s,b_k}(H)$ as a function of the entries of H . Let H' be an i.i.d. copy of H . For any indices $i \leq j$, define

$$(236) \quad \begin{aligned} \Delta_{ij} Y_{s,b_k} &:= Y_{s,b_k}(H_{11}, H_{12}, \dots, H_{ij}, H'_{i(j+1)}, \dots, H'_{NN}) \\ &\quad - Y_{s,b_k}(H_{11}, H_{12}, \dots, H'_{ij}, H'_{i(j+1)}, \dots, H'_{NN}). \end{aligned}$$

We also set $Y'_{s,b_k} := Y_{s,b_k}(H')$. (Here, and throughout this argument, indices i, j are reserved for elements of $\{1, \dots, N\}$, and they will only be used as indices for H and H' . We will identify these indices with elements in $\mathbb{Z}_{WL}^2$ when we apply the $[\cdot]$ operation, but the exact choice of identification $\{1, \dots, N\} \simeq \mathbb{Z}_{WL}^2$ is unimportant.) By telescoping sum, we have

$$(237) \quad \sum_{i \leq j} \Delta_{ij} Y_{s,b_k} = Y_{s,b_k}(H) - Y_{s,b_k}(H').$$

Now, for the $k = 1$ factor in (235), we rewrite it as $Y_{s,b_1} - \mathbb{E}Y_{s,b_1} = Y_{s,b_1} - Y_{s,b_1}(H') + Y_{s,b_1}(H') - \mathbb{E}Y_{s,b_1}$. Using (237) to rewrite $Y_{s,b_1} - Y_{s,b_1}(H')$ then gives

$$(238) \quad \mathbb{E} \left[\prod_{k=1}^p (Y_{s,b_k} - \mathbb{E}Y_{s,b_k}) \prod_{k=p+1}^{2p} (\overline{Y}_{s,b_k} - \mathbb{E}\overline{Y}_{s,b_k}) \right]$$

$$(239) \quad = \mathbb{E} \left[(Y_{s,b_1}(H') - \mathbb{E}Y_{s,b_1}) \prod_{k=2}^p (Y_{s,b_k} - \mathbb{E}Y_{s,b_k}) \prod_{k=p+1}^{2p} (\overline{Y}_{s,b_k} - \mathbb{E}\overline{Y}_{s,b_k}) \right]$$

$$(240) \quad + \sum_{i \leq j} \mathbb{E} \left[\Delta_{ij} Y_{s,b_1} \prod_{k=2}^p (Y_{s,b_k} - \mathbb{E}Y_{s,b_k}) \prod_{k=p+1}^{2p} (\overline{Y}_{s,b_k} - \mathbb{E}\overline{Y}_{s,b_k}) \right].$$

Of course, we can certainly restrict to i, j such that $S_{ij} \neq 0$ in (240); otherwise, H_{ij} and H'_{ij} are both 0, and thus $\Delta_{ij} Y_{s,b_1} = 0$ in this case. We will implicitly make this assumption for the rest of the proof of (235).

Since Y'_{s,b_1} is independent of all the Y_{s,b_k} , we know that (239) is zero. Since the number of summands in (240) is $O(W^C)$ for some $C = O(1)$, in order to prove (235), it suffices to control each summand in (240). Next, consider the function $\rho \mapsto Y_{s,b_1}(H_{11}, \dots, \rho H_{ij}, H'_{i(j+1)}, \dots, H'_{NN})$. We think of this as a function $f_{s,b_1}(\rho H_{ij}, \rho \bar{H}_{ij})$, so that f_{s,b_1} is smooth in its two inputs. Taylor expanding in ρ gives

$$(241) \quad \begin{aligned} Y_{s,b_1}(H_{11}, \dots, H_{ij}, H'_{i(j+1)}, \dots, H'_{NN}) &= f_{s,b_1}(0) + \sum_{\alpha=1}^{D_1} \frac{1}{\alpha!} \frac{d^\alpha}{d\rho^\alpha} f_{s,b_1}(0, 0) \\ &\quad + \frac{1}{D_1!} \int_0^1 (1 - \rho_{ij})^{D_1} \frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij}) d\rho_{ij}, \end{aligned}$$

where $D_1 > 0$ is any large constant independent of W . (Since f_{s,b_1} is complex-valued, we use the integral form of the remainder.) Using the chain rule, we compute the following for any $\alpha \geq 1$:

$$(242) \quad \frac{d^\alpha}{d\rho^\alpha} f_{s,b_1}(\rho H_{ij}, \rho \bar{H}_{ij}) = \sum_{v=0}^{\alpha} \binom{\alpha}{v} \partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(\rho H_{ij}, \rho \bar{H}_{ij}) \cdot H_{ij}^v \bar{H}_{ij}^{\alpha-v}.$$

Similarly, for the function $\rho \mapsto Y_{s,b_1}(H_{11}, \dots, \rho H'_{ij}, H'_{i(j+1)}, \dots, H'_{NN}) = f_{s,b_1}(\rho H'_{ij}, \rho \bar{H}'_{ij})$, we have

$$(243) \quad \begin{aligned} Y_{s,b_1}(H_{11}, \dots, H'_{ij}, H'_{i(j+1)}, \dots, H'_{NN}) &= f_{s,b_1}(0) + \sum_{\alpha=1}^{D_1} \frac{1}{\alpha!} \frac{d^\alpha}{d\rho^\alpha} f_{s,b_1}(0, 0) \\ &\quad + \frac{1}{D_1!} \int_0^1 (1 - \rho_{ij})^{D_1} \frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} f_{s,b_1}(\rho_{ij} H'_{ij}, \rho_{ij} \bar{H}'_{ij}) d\rho_{ij}. \end{aligned}$$

If we now combine (241), (244), (242), and (236), we deduce

$$(245) \quad \begin{aligned} \Delta_{ij} Y_{s,b_1} &= \sum_{\alpha=1}^{D_1} \frac{1}{\alpha!} \sum_{v=0}^{\alpha} \binom{\alpha}{v} \partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0, 0) \cdot \left[H_{ij}^v \bar{H}_{ij}^{\alpha-v} - (H'_{ij})^v (\bar{H}'_{ij})^{\alpha-v} \right] \\ &\quad + \frac{1}{D_1!} \int_0^1 (1 - \rho_{ij})^{D_1} \frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} \left(f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij}) - f_{s,b_1}(\rho_{ij} H'_{ij}, \rho_{ij} \bar{H}'_{ij}) \right) d\rho_{ij}. \end{aligned}$$

For convenience, let Γ_s denote the product of the two products in (240), so that

$$\Gamma_s := \prod_{k=2}^p (Y_{s,b_k} - \mathbb{E} Y_{s,b_k}) \prod_{k=p+1}^{2p} (\bar{Y}_{s,b_k} - \mathbb{E} \bar{Y}_{s,b_k}).$$

We have

$$(247) \quad \begin{aligned} \mathbb{E}[\Delta_{ij} Y_{s,b_1} \cdot \Gamma_s] &= \sum_{\alpha=1}^{D_1} \sum_{v=0}^{\alpha} \frac{1}{\alpha!} \binom{\alpha}{v} \mathbb{E} \left[\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0, 0) \cdot \Gamma_s \cdot \left\{ H_{ij}^v \bar{H}_{ij}^{\alpha-v} - (H'_{ij})^v (\bar{H}'_{ij})^{\alpha-v} \right\} \right] \\ &\quad + \frac{1}{D_1!} \int_0^1 (1 - \rho_{ij})^{D_1} \mathbb{E} \left[\frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij}) \cdot \Gamma_s \right] d\rho_{ij} \\ &\quad - \frac{1}{D_1!} \int_0^1 (1 - \rho_{ij})^{D_1} \mathbb{E} \left[\frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} f_{s,b_1}(\rho_{ij} H'_{ij}, \rho_{ij} \bar{H}'_{ij}) \cdot \Gamma_s \right] d\rho_{ij}. \end{aligned}$$

We will estimate each term on the right-hand side of (247). For this, we need some auxiliary estimates. Let $G^{i,j} = (\sqrt{s}H^{i,j} - z_s)^{-1}$, where $H^{i,j} = H - (1 - \gamma_{ij})H_{ij}e_i e_j^\dagger - (1 - \gamma_{ij})H_{ji}e_j e_i^\dagger$, be the resolvent G_s except we replace the entries H_{ij}, H_{ji} by $\gamma_{ij}H_{ij}, \gamma_{ij}H_{ji}$, respectively. We will only assume that $\gamma_{ij} = O(1)$; they can be random and depend on H_{uv}, H'_{uv} entries, and we will be interested in either $\gamma_{ij} = 0$ or $\gamma_{ij} = \rho_{ij} \in [0, 1]$ from (241).

- By resolvent expansion, Gaussianity of H entries, and the assumption $\|G_s\|_{\max} \prec 1$ from (64), we have $G_{xy}^{i,j} = (G_s)_{xy} + O_{\prec}(W^{-1}\|G^{i,j}\|_{\max})$. Taking a maximum over x, y gives

$$(248) \quad \|G^{i,j}\|_{\max} \prec \|G_s\|_{\max} \prec 1.$$

- By the same resolvent expansion, for any $K_0 > 0$ large and any indices $x, y = 1, \dots, N$, we have

$$G_{xy}^{i,j} = (G_s)_{xy} + \sum_{k=1}^{K_0} (G_s(X^{i,j}G_s)^k)_{xy} + (G^{i,j}(X^{i,j}G_s)^{K_0+1})_{xy},$$

where $X^{i,j}$ is supported at (i, j) and (j, i) entries, and its entries are all $O_{\prec}(S_{ij}^{1/2})$. We now compute

$$(249) \quad (G_s(X^{i,j}G_s)^k)_{xy} = \sum_{\substack{\{i_1, i_2\}=\{i,j\} \\ \{i_{2k-1}, i_{2k}\}=\{i,j\}}} (G_s)_{xi_1} X_{i_1 i_2}^{i,j} (G_s)_{i_2 i_3} \dots X_{i_{2k-1} i_{2k}}^{i,j} (G_s)_{i_{2k} y},$$

$$(250) \quad (G^{i,j}(X^{i,j}G_s)^k)_{xy} = \sum_{\substack{\{i_1, i_2\}=\{i,j\} \\ \{i_{2k-1}, i_{2k}\}=\{i,j\}}} G_{xi_1}^{i,j} X_{i_1 i_2}^{i,j} (G_s)_{i_2 i_3} \dots X_{i_{2k-1} i_{2k}}^{i,j} (G_s)_{i_{2k} y}.$$

In each identity, the number of summands is bounded by a constant depending only on k . Thus, as long as K_0 is finite, we can use $\|X^{i,j}\|_{\max} \prec W^{-1}$ and $\|G^{i,j}\|_{\max} + \|G_s\|_{\max} \prec 1$ to deduce that $(G^{i,j}(X^{i,j}G_s)^{K_0+1})_{xy} \prec W^{-(K_0+1)}$.

Next, fix any $\tau', D' > 0$. We also fix $a, b \in \mathbb{Z}_L^2$ so that $x \in \mathcal{I}_a^{(2)}$ and $y \in \mathcal{I}_b^{(2)}$. We claim that if $|a - b|_L > W^{2\tau'} \ell_s$, where $|\cdot|_L$ is periodic distance on $\mathbb{Z}_L^2$, then $(G_s(X^{i,j}G_s)^k)_{xy} \prec W^{-D'}$. Indeed, take (249). If $|a - b|_L > W^{2\tau'} \ell_s$ and $k = O(1)$, then at least one of the following must be true:

- There exists some i_n, i_{n+1} indices such that $|i_n - i_{n+1}|_L \geq W^{1+\tau'} \ell_s$; indeed, if $|a - b|_L > W^{2\tau'} \ell_s$, then $|x - y| \geq W^{1+2\tau'} \ell_s$. If this pair of indices corresponds to the factor $X_{i_n i_{n+1}}^{i,j}$, then this factor is equal to 0. If this pair of indices corresponds to the factor $(G_s)_{i_n i_{n+1}}$, then by the assumption (65) and (84), we have $|(G_s)_{i_n i_{n+1}}| \prec W^{-D'}$. Moreover, since $\|X^{i,j}\|_{\max} + \|G_s\|_{\max} \prec 1$, this implies that the corresponding summand on the right-hand side of (249) is $O_{\prec}(W^{-D'})$.
 - We have either $|a - i_1|_L \geq W^{1+\tau'} \ell_s$ or $|i_{2k} - b|_L \geq W^{1+\tau'} \ell_s$. The same reasoning in the previous paragraph implies that the corresponding summand in (249) is $O_{\prec}(W^{-D'})$.
- Ultimately, we deduce that for any $\tau', D' > 0$, we have $(G_s(X^{i,j}G_s)^k)_{xy} \prec W^{-D'}$ if $|a - b|_L \geq W^{\tau'} \ell_s$. Finally, since $|(G_s)_{xy}| \prec W^{-D'}$ if $|a - b|_L \geq W^{\tau'} \ell_s$, we conclude that for any $\tau', D' > 0$, we have

$$(251) \quad \mathbf{1}[|a - b|_L \geq W^{\tau'} \ell_s] \sup_{x \in \mathcal{I}_a^{(2)}} \sup_{y \in \mathcal{I}_b^{(2)}} |G_{xy}^{i,j}| \prec W^{-D'}.$$

- Our last auxiliary estimate is a bound on $\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}$ in (242). Recall that

$$f_{s,b_1}(\rho H_{ij}, \rho \bar{H}_{ij}) = Y_{s,b_1}(H_{11}, \dots, \rho H_{ij}, H'_{i(j+1)}, \dots, H'_{NN}),$$

and recall that Y_{s,b_1} has the form (228). In particular, $\partial_1 f_{s,b_1}$ is the derivative of Y_{s,b_1} from (228) with respect to H_{ij} (treating H_{ij} and $\bar{H}_{ij}$ as separate variables), and $\partial_2 f_{s,b_1}$ is the derivative of Y_{s,b_1} in $\bar{H}_{ij}$. Each such derivative can only act on entries of G_s or $G_s^\dagger$ in (228); when it hits one such entry, it returns two entries of G_s or $G_s^\dagger$. More precisely, we have $\partial_{H_{ij}}(G_s)_{xy} = -(G_s)_{xi}(G_s)_{jy}$, and similar standard formulas hold for $\bar{H}_{ij}$ instead of H_{ij} and/or $G_s^\dagger$ instead of G_s . Therefore, since $\|C\|_{\max} \leq N^{C'}$ in (228), and since $W \geq N^c$ gives $\|C\|_{\max} = O(W^C)$ for some $C = O(1)$, we deduce from (248) that for any $\alpha = O(1)$ and $0 \leq v \leq \alpha$ and $\gamma_{ij} = O(1)$, we have

$$(252) \quad |\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(\gamma_{ij} H_{ij}, \gamma_{ij} \bar{H}_{ij})| \prec W^C.$$

If we instead bound entries of Green's functions that appear in $\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}$ using the deterministic bound $O(W^C)$ for some $C = O(1)$, then we instead have the following deterministic bound for any $\alpha = O(1)$ and $0 \leq v \leq \alpha$ and $\gamma_{ij} = O(1)$, where now C_{D_1} depends on $D_1 > 0$:

$$(253) \quad |\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(\gamma_{ij} H_{ij}, \gamma_{ij} \bar{H}_{ij})| = O(W^{C_{D_1}}).$$

- We note that (248), (251), (252), and (253) all hold if we replace $H_{ij}, \bar{H}_{ij}$ by $H'_{ij}, \bar{H}'_{ij}$, respectively. Indeed, the proofs of (248), (251) use only $|H_{ij}| \prec S_{ij}^{1/2}$. The proof of (252) uses only the distribution of H_{ij} . The proof of (253) is deterministic in H_{ij} .

We will now use the bounds (248), (251), (252), and (253) to control the second term on the right-hand side of (247). To this end, we fix $\rho_{ij} \in [0, 1]$ and bound the expectation in the integrand, uniformly in ρ_{ij} . First, we recall that Γ_s is the product of the $k \geq 2$ factors in (240). By (228), we know that $|\Gamma_s| = O(W^C)$ deterministically for some $C = O(1)$ depending only on p . This gives

$$(254) \quad \begin{aligned} & \left| \mathbb{E} \left[\frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij}) \cdot \Gamma_s \right] \right| \\ & \leq O(W^C) \mathbb{E} \left[\left| \frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij}) \right| \right] \\ & \leq O(W^C) \sum_{v=0}^{D_1+1} \mathbb{E} \left| \partial_1^v \partial_2^{D_1+1-v} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij}) \cdot H_{ij}^v \bar{H}_{ij}^{D_1+1-v} \right|, \end{aligned}$$

where the last line follows by (242) and the triangle inequality. Now, fix any $D_2 > 0$ and any $0 \leq v \leq D_1 + 1$. By (252) for $\alpha = D_1 + 1$, there exists a good event $\mathcal{E}$ such that $\mathbb{P}(\mathcal{E}) = 1 - O(W^{-D_2})$, and such that on $\mathcal{E}$, we have $|\partial_1^v \partial_2^{D_1+1-v} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij})| = O(W^{2C})$ for some $C = O(1)$. We now estimate the expectation in the last line above by separating into $\mathcal{E}$ and its complement $\mathcal{E}^c$. On $\mathcal{E}$, we use the bound $|\partial_1^v \partial_2^{D_1+1-v} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij})| = O(W^{2C})$. On the bad event $\mathcal{E}^c$, we use the deterministic bound (253) for $\alpha = D_1 + 1$. This implies that

$$(255) \quad \mathbb{E} \left| \partial_1^v \partial_2^{D_1+1-v} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij}) \cdot H_{ij}^v \bar{H}_{ij}^{D_1+1-v} \right|$$

$$(256) \quad \leq O(W^{2C}) \mathbb{E} |H_{ij}^v \bar{H}_{ij}^{D_1+1-v}| + O(W^{2C_{D_1}}) \mathbb{E} [\mathbf{1}_{\mathcal{E}^c} |H_{ij}^v \bar{H}_{ij}^{D_1+1-v}|]$$

$$(257) \quad \leq O(W^{2C} W^{-(D_1+1)}) + O(W^{2C_{D_1}}) \mathbb{P}(\mathcal{E}^c)^{1/2}$$

$$(258) \quad \leq O(W^{2C} W^{-(D_1+1)}) + O(W^{2C_{D_1}} W^{-D_2/2}).$$

Fix any $D' > 0$. We can choose $D_1 > 0$ large enough and then $D_2 > 0$ large enough depending on D_1 such that the last line above is $O(W^{-D'})$. In particular, combining this with (254) implies the following. For any $D_3 > 0$, there exists $D_1 > 0$ such that

$$(259) \quad \mathbb{E} \left[\frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} f_{s,b_1}(\rho_{ij} H_{ij}, \rho_{ij} \bar{H}_{ij}) \cdot \Gamma_s \right] = O(W^{-D_3}) \quad \text{uniformly in } \rho_{ij} \in [0, 1].$$

The same argument provides the same bound on the last term in (247), i.e. for any $D_3 > 0$, there exists $D_1 > 0$ such that

$$(260) \quad \mathbb{E} \left[\frac{d^{D_1+1}}{d\rho_{ij}^{D_1+1}} f_{s,b_1}(\rho_{ij} H'_{ij}, \rho_{ij} \bar{H}'_{ij}) \cdot \Gamma_s \right] = O(W^{-D_3}) \quad \text{uniformly in } \rho_{ij} \in [0, 1].$$

We are left to control the first term on the right-hand side of (247), which is a sum over α, v ; fix any such α, v . We first expand $H_{ij}^v \bar{H}_{ij}^{\alpha-v}$ as a linear combination of Hermite polynomials in $H_{ij}, \bar{H}_{ij}$ of degree at most α . We do the same to $(H'_{ij})^v (\bar{H}'_{ij})^{\alpha-v}$. In these expansions, the zero-th degree (constant) terms cancel each other out. Then, by Gaussian integration-by-parts, we get

$$(261) \quad \mathbb{E} \left[\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) \cdot \Gamma_s \cdot \left\{ H_{ij}^v \bar{H}_{ij}^{\alpha-v} - (H'_{ij})^v (\bar{H}'_{ij})^{\alpha-v} \right\} \right]$$

$$(262) \quad = \sum_{\substack{q_1, q_2=0, \dots, \alpha \\ q_1+q_2>0}} c_{q_1, q_2} \mathbb{E} \left[\left\{ S_{ij}^{q_1+q_2} \partial_{H_{ij}}^{q_1} \partial_{\bar{H}_{ij}}^{q_2} \right\} (\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) \cdot \Gamma_s) \right]$$

$$(263) \quad - \sum_{\substack{q_1, q_2=0, \dots, \alpha \\ q_1+q_2>0}} c_{q_1, q_2} \mathbb{E} \left[\left\{ S_{ij}^{q_1+q_2} \partial_{H'_{ij}}^{q_1} \partial_{\bar{H}'_{ij}}^{q_2} \right\} (\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) \cdot \Gamma_s) \right].$$

Above, $c_{q_1, q_2} = O(1)$ are deterministic constants. The last line vanishes, since nothing inside the expectation depends on $H'_{ij}, \bar{H}'_{ij}$. For the first term on the right-hand side above, we note that because we evaluate the f_{s,b_1} derivatives at $(0,0)$, they have no dependence on $H_{ij}, \bar{H}_{ij}$, so the derivatives act only on Γ_s (which we recall to be the product of the $k \geq 2$ factors in (240)). Thus, we have

$$(264) \quad \mathbb{E} \left[\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) \cdot \Gamma_s \cdot \left\{ H_{ij}^v \bar{H}_{ij}^{\alpha-v} - (H'_{ij})^v (\bar{H}'_{ij})^{\alpha-v} \right\} \right] \\ = \sum_{\substack{q_1, q_2=0, \dots, \alpha \\ q_1+q_2>0}} c_{q_1, q_2} \mathbb{E} \left[\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) \cdot \left\{ S_{ij}^{q_1+q_2} \partial_{H_{ij}}^{q_1} \partial_{\bar{H}_{ij}}^{q_2} \right\} \Gamma_s \right].$$

We now study further the derivatives of Γ_s in (264); again, recall that Γ_s is the product of $k \geq 2$ factors in (240). Thus, by the Leibniz rule, we have the following for some deterministic constants $c_{\mathbf{w}_1, \mathbf{w}_2} = O(1)$:

$$(265)$$

$$\left\{ \partial_{H_{ij}}^{q_1} \partial_{\bar{H}_{ij}}^{q_2} \right\} \Gamma_s \\ = \sum_{\substack{\mathbf{w}_1 := (w_{1,2}, \dots, w_{1,2p}) \in \mathbb{Z}_{\geq 0}^{2p-1} \\ w_{1,2} + \dots + w_{1,2p} = q_1 \\ \mathbf{w}_2 := (w_{2,2}, \dots, w_{2,2p}) \in \mathbb{Z}_{\geq 0}^{2p-1} \\ w_{2,2} + \dots + w_{2,2p} = q_2}} c_{\mathbf{w}_1, \mathbf{w}_2} \prod_{k=2}^p \partial_{H_{ij}}^{w_{1,k}} \partial_{\bar{H}_{ij}}^{w_{2,k}} (Y_{s,b_k} - \mathbb{E} Y_{s,b_k}) \cdot \prod_{k=p+1}^{2p} \partial_{H_{ij}}^{w_{1,k}} \partial_{\bar{H}_{ij}}^{w_{2,k}} (\bar{Y}_{s,b_k} - \mathbb{E} \bar{Y}_{s,b_k}).$$

Now, we use the form (228) for Y_{s,b_k} , and suppose that $w_{1,k} + w_{2,k} > 0$. Since the $C_{b,\mathbf{x},\mathbf{y}}$ coefficients therein are deterministic, we have

(266)

$$\partial_{H_{ij}}^{w_{1,k}} \partial_{\bar{H}_{ij}}^{w_{2,k}} (Y_{s,b_k} - \mathbb{E}Y_{s,b_k}) = \sum_{\mathbf{x}, \mathbf{y} \in (\mathbb{Z}_{WL}^2)^u} C_{b_k, \mathbf{x}, \mathbf{y}} \sum_{\substack{\gamma_1 = (\gamma_{11}, \dots, \gamma_{1u}) \\ \gamma_{11} + \dots + \gamma_{1u} = w_{1,k} \\ \gamma_2 = (\gamma_{21}, \dots, \gamma_{2u}) \\ \gamma_{21} + \dots + \gamma_{2u} = w_{2,k}}} \prod_{r=1}^u \partial_{H_{ij}}^{\gamma_{1r}} \partial_{\bar{H}_{ij}}^{\gamma_{2r}} G_s(\sigma_r)_{x_r y_r}.$$

(The expectation on the left-hand side above vanishes under the derivatives.) Now, recall $w_{1,k} + w_{2,k} > 0$. In this case, there must exist a factor of the form $G_s(\sigma)_{\beta\iota}$ or $G_s(\sigma)_{\iota\beta}$ with $\iota \in \{i, j\}$ (for some $\sigma \in \{+, -\}$), where $\beta = x_r$ or $\beta = y_r$ for some r . This follows by differentiation formulas for the resolvent, e.g. $\partial_{H_{ij}}(G_s)_{xy} = -(G_s)_{xi}(G_s)_{jy}$. (For mixed derivatives in H_{ij} and $\bar{H}_{ij}$, all such factors may carry the same index ι , e.g. $(G_s)_{xi}(G_s)_{jj}(G_s)_{iy}$.) Given this information, suppose now that $||[i] - b_k|_L + |[j] - b_k|_L \geq 3W^\tau \ell_s$. Since $S_{ij} \neq 0$, we have $||[i] - [j]|_L \leq 1$, and hence $||[i] - b_k|_L \geq 3W^\tau \ell_s/2 - 1$ for both $\iota \in \{i, j\}$. Take the factor $G_s(\sigma)_{\beta\iota}$ (or $G_s(\sigma)_{\iota\beta}$), where $\beta = x_r$ or $\beta = y_r$ for some r . If $||[\beta] - b_k| \geq W^\tau \ell_s$, then the corresponding coefficient $C_{b_k, \mathbf{x}, \mathbf{y}}$ vanishes by assumptions. If $||[\beta] - b_k|_L \leq W^\tau \ell_s$, then we have $||[\iota] - [\beta]|_L \geq W^\tau \ell_s/2 - 1 \geq W^\tau \ell_s/4$. In this case, we can use the fast decay property for G_s (see (65) and (84)) to deduce the bound $G_s(\sigma)_{\beta\iota} \prec W^{-D_0}$ for any $D_0 > 0$. Finally, we can bound the remaining factors in (266) by $O_{\prec}(N^{C'})$ for some $C' = O(1)$ (this uses the assumption (64)). We deduce

(267)

$$\mathbf{1}[w_{1,k} + w_{2,k} > 0] \mathbf{1}[|[i] - b_k|_L + |[j] - b_k|_L \geq 3W^\tau \ell_s] |\partial_{H_{ij}}^{w_{1,k}} \partial_{\bar{H}_{ij}}^{w_{2,k}} (Y_{s,b_k} - \mathbb{E}Y_{s,b_k})| \prec W^{-D'}$$

for any $D' = O(1)$. On the other hand, even if $w_{1,k} + w_{2,k} = 0$, we can use the bounds $\|C'\|_{\max} = O(W^{C'})$ and $\|G_s\|_{\max} = O(W^{C'})$ for some $C' = O(1)$ to get

(268)

$$|\partial_{H_{ij}}^{w_{1,k}} \partial_{\bar{H}_{ij}}^{w_{2,k}} (Y_{s,b_k} - \mathbb{E}Y_{s,b_k})| \prec W^{C'}$$

for some (possibly different) $C' = O(1)$. We emphasize that the previous two bounds are true if we replace Y_{s,b_k} by its complex conjugate as well. We also clarify that the bound $\|G_s\|_{\max} = O(W^{C'})$ comes from the assumption that $s \leq 1 - N^{-1+\delta}$ for some $\delta > 0$, and thus $\eta_s^{-1} = O(N^{1-\delta})$.

Now, take (265), and recall that $q_1 + q_2 > 0$ therein; see (264) for this constraint. Thus, at least one pair $(w_{1,k}, w_{2,k})$ must satisfy $w_{1,k} + w_{2,k} > 0$. For this pair, we bound the corresponding factor using (267). The other factors in (265) can be bounded using (267) and (268). Ultimately, we deduce

$$(269) \quad \mathbf{1} \left[\min_{k=2, \dots, 2p} (|[i] - b_k|_L + |[j] - b_k|_L) \geq 3W^\tau \ell_s \right] \cdot \left\{ \partial_{H_{ij}}^{q_1} \partial_{\bar{H}_{ij}}^{q_2} \right\} \Gamma_s \prec W^{-D'},$$

(270)

$$\left\{ \partial_{H_{ij}}^{q_1} \partial_{\bar{H}_{ij}}^{q_2} \right\} \Gamma_s = O(W^C),$$

where $D' = O(1)$ is arbitrary and $C = O(1)$ is fixed. This bounds the last factor in the expectation in (264).

To control $\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0)$, we note that $\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0)$ has the same form as the summand on the right-hand side of (265) (or its complex conjugate) with $p = 1$ and $w_{1,2} + w_{2,2} > 1$, except we evaluate the Green's functions at $H_{ij}, \bar{H}_{ij} = 0$. (Indeed, recall from (241) that $f_{s,b_1}(H_{ij}, \bar{H}_{ij}) = Y_{s,b_1}$.) Therefore, if we instead use the Green's function

estimates (248) and (251) to account for setting $H_{ij}, \bar{H}_{ij} = 0$, then for any $\tau', D' > 0$ and for some $C = O(1)$, we have the same bounds

$$(271) \quad \mathbf{1}[|[i] - b_1|_L + |[j] - b_1|_L \geq 3W^\tau \ell_s] \cdot \partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) \prec W^{-D'},$$

$$(272) \quad \partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) = O(W^C).$$

Now, consider (264), and recall the assumption $\min_{k=2,\dots,2p} |b_k - b_1|_L \geq W^{2\tau} \ell_s$ that was stated before (235). This implies that for any i, j indices, we either have $|[i] - b_1|_L + |[j] - b_1|_L \geq 3W^\tau \ell_s$, or $|[i] - b_k|_L + |[j] - b_k|_L \geq 3W^\tau \ell_s$ for all $k = 2, \dots, 2p$. So, by (269), (270), (271), and (272), we get the following. For any $D_4, D_5 > 0$ and any i, j indices, there exists another good event $\mathcal{E}_{good}$ such that $\mathbb{P}(\mathcal{E}_{good}) = 1 - O(W^{-D_4})$ and

$$(273) \quad \mathbf{1}_{\mathcal{E}_{good}} \partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) \cdot \left\{ S_{ij}^{q_1+q_2} \partial_{H_{ij}}^{q_1} \partial_{\bar{H}_{ij}}^{q_2} \right\} \Gamma_s = O(W^{-D_5}).$$

When we estimate the expectation in (264), we separate into the good event $\mathcal{E}_{good}$ and the bad event $\mathcal{E}_{good}^c$. On the former, we use (273). On the bad event, we use (270) and (272). This gives

$$(274) \quad \mathbb{E} \left[\partial_1^v \partial_2^{\alpha-v} f_{s,b_1}(0,0) \cdot \left\{ S_{ij}^{q_1+q_2} \partial_{H_{ij}}^{q_1} \partial_{\bar{H}_{ij}}^{q_2} \right\} \Gamma_s \right]$$

$$(275) \quad = O(W^{-D_5}) + O(W^{2C}) \mathbb{P}(\mathcal{E}_{good}^c) \leq O(W^{-D_5}) + O(W^{2C-D_4}).$$

We plug this bound into (264), which we then plug into the first term on the right-hand side of (247). Since $D_4, D_5 > 0$ are any large constants independent of W in the previous display, this shows that for any $D > 0$ independent of W , the first term on the right-hand side of (247) is $O(W^{-D})$. Combining this with (259) and (260) and (247) ultimately gives $\mathbb{E}[\Delta_{ij} Y_{s,b_1} \cdot \Gamma_s] = O(W^{-D})$ for any $D > 0$ independent of W . By (240), this completes the proof of (235).

We now finish the proof. Take any large $D_0 > 0$. By (235), if we give up an error term of $O(W^{-D_0})$, we can restrict to indices $b_1, \dots, b_{2p}$ so that for every i , we have $\min_{k \neq i} |b_i - b_k|_L \leq W^{2\tau} \ell_s$. In particular, for any $j = 1, \dots, p$, choose j many “free indices”, and label them $b_1, \dots, b_j$. The remaining indices $b_{j+1}, \dots, b_{2p}$ must be assigned to one of the $b_1, \dots, b_j$ in the following sense. If b_{j+1} is assigned to b_1 , then $|b_{j+1} - b_1|_L \leq W^{3\tau} \ell_s$. Thus, from (234) and (235), we have

$$(276) \quad \mathbb{E} \left| \sum_b Z_{ab}(Y_{s,b} - \mathbb{E} Y_{s,b}) \right|^{2p} \prec \sum_{j=1}^p \sum_{\substack{k_1, \dots, k_j \geq 1 \\ k_1 + \dots + k_j = 2p-j}} \prod_{i=1}^j \sum_{b_i} |Z_{ab_i}| \left(\sum_{|b_{i,2} - b_i|_L \leq W^{3\tau} \ell_s} |Z_{ab_{i,2}}| \right)^{k_i} \cdot \mathbb{E}(\|Y_s\|_{\max}^{2p}) \\ + O(W^{-D} \max_b |Z_{ab}|^{2p}).$$

We now focus on case 1, i.e. the proof of (230). By our assumption on the range of Z_{ab} , we can restrict each b_i summation to $O(W^{C\tau} \ell_t^2)$ many terms. Thus, the total number of summands in the first term on the right-hand side of (276) is $O(W^{2pC\tau} \ell_t^{2p} \ell_s^{2p})$. (There are at most p -many b_i -sums, and a total of $2p$ -sums overall.) Then we bound entries of Z by its max-norm. Also, recall the assumption $\|Y_s\|_{\max} \prec \Lambda$, and note that $\|Y_s\|_{\max} = O(W^C)$ for some $C = O(1)$, which follows from (228) and $s \leq 1 - N^{-1+\delta}$ for some $\delta > 0$. This implies that $\mathbb{E}(\|Y_s\|_{\max}^{2p}) \prec \Lambda^{2p} + W^{-D'}$ for any $D' = O(1)$. Ultimately, we deduce (for any $D = O(1)$) that

$$\mathbb{E} \left| \sum_b Z_{ab}(Y_{s,b} - \mathbb{E} Y_{s,b}) \right|^{2p} \prec W^{C\tau} \ell_t^{2p} \ell_s^{2p} \max_b |Z_{ab}|^{2p} \cdot \Lambda^{2p} + W^{-D} \max_b |Z_{ab}|^{2p}.$$

Since $D, p = O(1)$ are arbitrary, we obtain the desired result (230) in case 1.

We now move to case 2. On the right-hand side of (276), we fix some sequence of $k_1, \dots, k_p$ and some $i \in \{1, \dots, p\}$. Consider the object

$$(277) \quad \sum_{b_i} |Z_{ab_i}| \cdot \left(\sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} |Z_{ab_{i,2}}| \right)^{k_i}.$$

We emphasize that $k_i \geq 1$. We now rewrite (277) as follows:

$$\sum_{b_i} |Z_{ab_i}| \sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} |Z_{ab_{i,2}}| \cdot \left(\sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} |Z_{ab_{i,2}}| \right)^{k_i-1}.$$

Since $k_i \geq 1$, we can use the a priori estimate of $|Z_{ab}| \prec (1 + |a - b|_L)^{-1}$ to show that the last factor in the previous display is $O_{\prec}(W^{3\tau(k_i-1)}\ell_s^{k_i-1})$. On the other hand, we can use this same bound on $|Z_{ab}|$ to get

$$\begin{aligned} \sum_{b_i} |Z_{ab_i}| \sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} |Z_{ab_{i,2}}| &\prec \sum_{b_i} \frac{1}{|a - b_i|^2 + 1} \sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} \frac{|a - b_i|_L + 1}{|a - b_{i,2}|_L + 1} \\ &\leq \sum_{b_i} \frac{1}{|a - b_i|^2 + 1} \sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} \left(1 + \frac{|b_i - b_{i,2}|_L}{1 + |a - b_{i,2}|_L} \right) \\ &\prec W^{6\tau}\ell_s^2 \cdot \sum_{b_i} \frac{1}{|a - b_i|^2 + 1} \prec W^{6\tau}\ell_s^2. \end{aligned}$$

Therefore, we deduce

$$(278) \quad \sum_{b_i} |Z_{ab_i}| \sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} |Z_{ab_{i,2}}| \cdot \left(\sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} |Z_{ab_{i,2}}| \right)^{k_i-1} \prec W^{6\tau} W^{3\tau(k_i-1)} \ell_s^{k_i+1}.$$

The benefit of this bound is that there is no dependence on the range of Z . Now, when we take a product over all $i = 1, \dots, j$, we deduce

$$\prod_{i=1}^j \sum_{b_i} |Z_{ab_i}| \cdot \left(\sum_{|b_{i,2}-b_i|_L \leq W^{3\tau}\ell_s} |Z_{ab_{i,2}}| \right)^{k_i} \prec \prod_{i=1}^j W^{C\tau k_i} \ell_s^{k_i+1} \prec W^{(2p-j)C\tau} \ell_s^{2p-j} \cdot \ell_s^j \prec W^{2pC\tau} \ell_s^{2p}$$

since the k_i sum to $2p - j$ over $i = 1, \dots, j$. In particular, if we plug the previous display into the first term on the right-hand side of (276) and use $|Z_{ab}| \prec 1$, we obtain the following for any $D = O(1)$:

$$\mathbb{E} \left| \sum_b Z_{ab}(Y_{s,b} - \mathbb{E}Y_{s,b}) \right|^{2p} \prec W^{2pC\tau} \ell_s^{2p} \cdot \mathbb{E}(\|Y_s\|_{\max}^{2p}) + W^{-D}.$$

If we again use $\|Y_s\|_{\max} \prec \Lambda$ and $\|Y_s\|_{\max} = O(W^C)$ for some $C = O(1)$, then the previous display yields the desired estimate (232) (since $D, p = O(1)$ are arbitrary). This completes the proof. $\square$

8. Proof of Lemma 2.14. Throughout this section we fix $\xi \in \mathbb{C}$ with $|\xi| < 1$, and we abbreviate

$$(279) \quad \kappa := |1 - \xi|^{1/2}, \quad \hat{\ell} = \hat{\ell}(\xi) = \min(\kappa^{-1}, L), \quad x := a - b \in \mathbb{Z}_L^2.$$

Recall that $S_{ab}^{(B)} = \frac{1}{5} \mathbf{1}(|a - b|_L \leq 1)$ is the transition matrix of the lazy nearest-neighbor random walk on $\mathbb{Z}_L^2$. Recall also that $|\cdot|_L$ denotes the periodic L^1 distance on $\mathbb{Z}_L^2$; since the periodic L^1 and L^2 distances on $\mathbb{Z}_L^2$ differ by at most a factor of $\sqrt{2}$, and since all the estimates of Lemma 2.14 are stated up to constants in the exponents and up to $\prec$, we may and do use the two interchangeably in this section. In particular, we write

$$|x|_L = \min_{n \in \mathbb{Z}^2} |x + nL|$$

for the periodic Euclidean norm of $x \in \mathbb{Z}_L^2$, where $|\cdot|$ is the Euclidean norm on $\mathbb{Z}^2$. Finally, we recall that $A \prec B$ means that for every fixed $\varepsilon > 0$ we have $A \leq C_\varepsilon L^\varepsilon B$; in particular, powers of $\log L$ are harmless.

The first four assertions of Lemma 2.14 are immediate. Indeed, $S^{(B)}$ is a real symmetric circulant matrix and $\Theta_\xi^{(B)} = (1 - \xi S^{(B)})^{-1}$ is a function of $S^{(B)}$; this gives the symmetry, the translation invariance, and the commutation relations in properties 1–3. Moreover, $S^{(B)}$ is stochastic and symmetric, so $\|S^{(B)}\| = 1$, and therefore the Neumann series

$$(280) \quad \Theta_\xi^{(B)} = (1 - \xi S^{(B)})^{-1} = \sum_{k=0}^{\infty} \xi^k (S^{(B)})^k$$

converges in operator norm for every $|\xi| < 1$. For $0 \leq \xi < 1$ all the terms in (280) are non-negative and $[(S^{(B)})^k]_{ab}$ is the k -step transition probability of the lazy random walk; this is property 4.

We prove the remaining entrywise estimates, i.e. properties 5 and 6, by Fourier analysis. We emphasize that the argument below only uses $|\xi| < 1$; in particular it covers the three cases $\xi \in \{tm^2, t\overline{m}^2, t|m|^2\}$ appearing in properties 5 and 6 of Lemma 2.14.

8.1. *Fourier setup and an ellipticity estimate.* Set

$$\mathbb{T}_L^2 := \left(\frac{2\pi}{L} \mathbb{Z}_L\right)^2.$$

The Fourier symbol of $S^{(B)}$ is

$$(281) \quad \widehat{S}(p) = \frac{1 + 2 \cos p_1 + 2 \cos p_2}{5}, \quad p = (p_1, p_2) \in \mathbb{T}_L^2.$$

Consequently, with $x = a - b$,

$$(282) \quad (\Theta_\xi^{(B)})_{ab} = K_{\xi, L}(x) := \frac{1}{L^2} \sum_{p \in \mathbb{T}_L^2} \frac{e^{ip \cdot x}}{1 - \xi \widehat{S}(p)}.$$

Let

$$(283) \quad q(p) := 1 - \widehat{S}(p) = \frac{2}{5} \left[(1 - \cos p_1) + (1 - \cos p_2) \right],$$

and write $|p|_*^2 := \text{dist}(p_1, 2\pi\mathbb{Z})^2 + \text{dist}(p_2, 2\pi\mathbb{Z})^2$, so that

$$(284) \quad q(p) \sim |p|_*^2.$$

We shall use the following elementary ellipticity estimate:

$$(285) \quad |1 - \xi \widehat{S}(p)| \sim \kappa^2 + q(p) \sim \kappa^2 + |p|_*^2.$$

To verify (285), write $\lambda = \widehat{S}(p) \in [-3/5, 1]$. If $\lambda \leq 1/2$, then $|\lambda| \leq 3/5$ and hence $|1 - \lambda\xi| \geq 1 - |\lambda||\xi| \geq 2/5$, while $\kappa^2 + q(p) \leq |1 - \xi| + 2 \leq C$; this gives (285) in this regime. If $\lambda \geq 1/2$, then

$$1 - \lambda\xi = (1 - \lambda) + \lambda(1 - \xi).$$

Since $|\xi| < 1$ we have $\operatorname{Re}(1 - \xi) > 0$, so the real parts of the two summands above cannot cancel, and therefore

$$\begin{aligned} |1 - \lambda\xi| &\geq \max \left\{ (1 - \lambda) + \lambda \operatorname{Re}(1 - \xi), \lambda |\operatorname{Im}(1 - \xi)| \right\} \\ &\geq \frac{1}{2} \left[(1 - \lambda) + \lambda |1 - \xi| \right] \geq \frac{1}{4} \left[(1 - \lambda) + |1 - \xi| \right], \end{aligned}$$

where we used $\lambda \geq 1/2$ in the last step. The matching upper bound $|1 - \lambda\xi| \leq (1 - \lambda) + |1 - \xi|$ is clear. Since $1 - \lambda = q(p)$ and $|1 - \xi| = \kappa^2$, this proves (285).

8.2. *Proof of the decay estimate (42).* Consider first the infinite-volume kernel

$$(286) \quad K_{\xi, \infty}(x) := \frac{1}{(2\pi)^2} \int_{[-\pi, \pi]^2} \frac{e^{ip \cdot x}}{1 - \xi \widehat{S}(p)} dp, \quad x \in \mathbb{Z}^2.$$

We give the contour-shift argument in some detail. Write

$$D_\xi(p) := 1 - \xi \widehat{S}(p) = 1 - \xi + \xi q(p).$$

For real $p \in [-\pi, \pi]^2$ and a real vector η with $|\eta| \leq 1$, the explicit expression (283) for q gives

$$|q(p + i\eta) - q(p)| \leq C(|\eta||p|_* + |\eta|^2).$$

Indeed, this follows directly from $\cos(u + iv) = \cos u \cosh v - i \sin u \sinh v$ together with $|\sin u| \leq \operatorname{dist}(u, 2\pi\mathbb{Z})$. Hence

$$|D_\xi(p + i\eta) - D_\xi(p)| \leq C(|\eta||p|_* + |\eta|^2).$$

Combining this estimate with (285) and choosing $c_0 > 0$ sufficiently small, we obtain

$$(287) \quad |D_\xi(p + i\eta)| \geq c(\kappa^2 + |p|_*^2), \quad |\eta| \leq c_0 \kappa.$$

To see the last step, use $\kappa|p|_* \leq (\kappa^2 + |p|_*^2)/2$, so that the above perturbation can be absorbed into the lower bound in (285). In particular, the denominator in (286) has no zeros in this strip.

Choose $j \in \{1, 2\}$ with $|x_j| = \max(|x_1|, |x_2|)$. If $x \neq 0$, we shift the p_j -contour from $[-\pi, \pi]$ to $[-\pi, \pi] + i \operatorname{sgn}(x_j) c_0 \kappa$. The integrand is 2π -periodic in p_j because $x_j \in \mathbb{Z}$, so the integrals over the two vertical sides of the corresponding rectangle cancel. Cauchy's theorem and (287) thus give

$$|K_{\xi, \infty}(x)| \leq e^{-c_0 \kappa |x_j|} \cdot \frac{C}{(2\pi)^2} \int_{[-\pi, \pi]^2} \frac{dp}{\kappa^2 + |p|_*^2}.$$

Since $|x_j| \geq |x|/\sqrt{2}$, and since the same integral bound holds directly when $x = 0$, we conclude that

$$(288) \quad |K_{\xi, \infty}(x)| \leq C \log(2 + \kappa^{-1}) e^{-c\kappa|x|}.$$

Here the logarithm comes from the familiar two-dimensional integral

$$(289) \quad \int_{[-\pi, \pi]^2} \frac{dp}{\kappa^2 + |p|_*^2} \leq C \log(2 + \kappa^{-1}).$$

Suppose first that $\kappa L \geq 1$, so that $\hat{\ell} = \kappa^{-1}$. By Poisson summation, or simply by comparing Fourier coefficients in (282) and (286), the torus kernel is the periodization

$$K_{\xi,L}(x) = \sum_{n \in \mathbb{Z}^2} K_{\xi,\infty}(x + nL).$$

Using (288) and $\kappa L \geq 1$ to sum the geometric series over n , we obtain

$$|K_{\xi,L}(x)| \leq C \log(2 + \kappa^{-1}) e^{-c\kappa|x|_L} \prec e^{-c\kappa|x|_L}.$$

Since $\kappa^2 \hat{\ell}^2 = 1$ in this regime, this is exactly (42).

Suppose next that $\kappa L < 1$, so that $\hat{\ell} = L$. We separate the zero Fourier mode in (282) and use (285) for the remaining modes:

$$|K_{\xi,L}(x)| \leq \frac{1}{L^2|1-\xi|} + \frac{C}{L^2} \sum_{p \in \mathbb{T}_L^2 \setminus \{0\}} \frac{1}{|p|_*^2} \leq \frac{1}{\kappa^2 L^2} + C \log L \prec \frac{1}{\kappa^2 L^2},$$

where in the last step we used $\kappa^2 L^2 < 1$. Since $|x|_L \leq CL = C\hat{\ell}$, the factor $\exp(-c|x|_L/\hat{\ell})$ is bounded below by a positive constant, and $|1-\xi|\hat{\ell}^2 = \kappa^2 L^2$. This again yields (42), and completes the proof of property 5.

8.3. Proof of the derivative bounds (43)–(44). We first record the standard dyadic Fourier estimate needed below. Separate the zero mode in (282) and decompose the remaining momenta into dyadic annuli $|p|_* \sim r$, where $L^{-1} \lesssim r \lesssim 1$, using a smooth partition of unity. If K_r denotes the contribution of one such annulus to (282), then discrete summation by parts, (285), and the corresponding derivative bounds on the multiplier give, for $m = 0, 1, 2$,

$$(290) \quad |\nabla^m K_r(x)| \leq C_M \frac{r^{m+2}}{\kappa^2 + r^2} (1 + r|x|_L)^{-M}.$$

Here K_r is viewed as a trigonometric polynomial on the continuous torus, and M can be chosen arbitrarily large. For completeness, we recall the mechanism: each summation by parts produces a factor $(1 + r|x|_L)^{-1}$, while each differentiation of the multiplier or of the cutoff costs one power of r^{-1} ; the annulus contains $O(L^2 r^2)$ momenta, its multiplier is $O((\kappa^2 + r^2)^{-1})$ by (285), and each of the m derivatives in x produces a factor $O(r)$. Together these give (290).

Let $d := |a - b|_L$, and choose the representatives of $x = a - b$ and of s in $\mathbb{Z}^2$ with minimal Euclidean norm.

Case 1: $|s|_L \leq d/2$. In this case the line segments joining x to $x - s$, and $x - s$ to $x + s$, remain at distance comparable to d from the origin. The fundamental theorem of calculus and (290) therefore imply

$$(291) \quad |K_{\xi,L}(x) - K_{\xi,L}(x - s)| \leq C|s|_L \sum_r \frac{r^3}{\kappa^2 + r^2} (1 + rd)^{-M},$$

$$(292) \quad |2K_{\xi,L}(x) - K_{\xi,L}(x - s) - K_{\xi,L}(x + s)| \leq C|s|_L^2 \sum_r \frac{r^4}{\kappa^2 + r^2} (1 + rd)^{-M},$$

where the sums run over dyadic $L^{-1} \lesssim r \lesssim 1$ (the zero mode is constant and cancels from both differences). Splitting each sum at $r \sim \kappa$ and at $r \sim (d+1)^{-1}$ gives

$$(293) \quad \sum_r \frac{r^3}{\kappa^2 + r^2} (1 + rd)^{-M} \leq C \left(\frac{1}{d+1} + \frac{1}{\hat{\ell}} \right),$$

$$(294) \quad \sum_r \frac{r^4}{\kappa^2 + r^2} (1 + rd)^{-M} \leq C \left(\frac{1}{d^2+1} + \frac{1}{\hat{\ell}^2} \right).$$

Since $\kappa\hat{\ell} \leq 1$ by (279), we have

$$\frac{1}{\hat{\ell}} \leq \frac{1}{\kappa\hat{\ell}^2} = \frac{1}{[\hat{\ell}(\xi)]^2 |1 - \xi|^{1/2}}.$$

Combining this with (291)–(294) proves (43) and (44) when $|s|_L \leq d/2$.

Case 2: $|s|_L > d/2$. The zero Fourier mode is constant and cancels from both differences, so by (285),

$$\begin{aligned} & |K_{\xi,L}(x) - K_{\xi,L}(x-s)| + |2K_{\xi,L}(x) - K_{\xi,L}(x-s) - K_{\xi,L}(x+s)| \\ & \leq \frac{C}{L^2} \sum_{p \in \mathbb{T}_L^2 \setminus \{0\}} \frac{1}{\kappa^2 + |p|_*^2} \leq C \log L. \end{aligned}$$

Thus both differences are $O(\log L)$, hence $\prec 1$. On the other hand, unless $s = 0$ we have $|s|_L \geq 1$ and $d < 2|s|_L$, so that

$$\frac{|s|_L}{d+1} \geq c, \quad \frac{|s|_L^2}{d^2+1} \geq c$$

for some constant $c > 0$. The case $s = 0$ is trivial. This proves (43) and (44) in all cases, and completes the proof of Lemma 2.14. $\square$

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